

Review

A Review of Sensing Technologies for Indoor Autonomous Mobile Robots

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Abstract: As a fundamental issue in robotics academia and industry, indoor autonomous mobile robots (AMRs) have been extensively studied. For AMRs, it is crucial to obtain information about their working environment and themselves, which can be realized through sensors and the extraction of corresponding information from the measurements of these sensors. The application of sensing technologies can enable mobile robots to perform localization, mapping, target or obstacle recognition, and motion tasks, etc. This paper reviews sensing technologies for autonomous mobile robots in indoor scenes. The benefits and potential problems of using a single sensor in application are analyzed and compared, and the basic principles and popular algorithms used in processing these sensor data are introduced. In addition, some mainstream technologies of multi-sensor fusion are introduced. Finally, this paper discusses the future development trends in the sensing technology for autonomous mobile robots in indoor scenes, as well as the challenges in the practical application environments.

Keywords: sensing technology; SLAM; obstacle avoidance; sensor fusion



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1. Introduction

In recent times, there has been a rapid development in autonomous mobile robots (AMRs). Due to their excellent work ability, AMRs are being used to replace humans in some indoor scenarios. Mobile robots can help reduce the burden of human work and improve production efficiency. They have been widely used in many industries and services, such as warehousing, logistics, healthcare, restaurant service, and personal services [1–3].

The basics of mobile robotics consist of the fields of locomotion, perception, cognition, and navigation [4]. The locomotion problem is mainly concerned with the motion system of the mobile robot. The motion system design is based on the requirements of the provided services, such as the motion environment, controllability, efficiency, stability, and other relevant indicators. Perception involves acquiring information about the mobile robot's working environment and itself. Cognition involves analyzing and processing data from the perception system and providing control solutions to the motion system to accomplish the mobile robot's tasks. Mobile robot navigation relies on perception, cognition, and motion control to move from a starting point to a task goal point in a work environment, whether known or unknown. Perception is a crucial part of mobile robotics research. For mobile robots to safely and efficiently perform mobile tasks, they must sense their environment and make decisions based on that information. If a mobile robot cannot perceive its environment accurately and efficiently, then it will not be able to perform even simple tasks [5].

The perception system of the mobile robot utilizes relevant sensing techniques to provide information about the environment, the robot itself, and the relationship between the two. This information is then used for path planning, controlling the robot's movement, and ultimately completing the navigation task. The main tasks of the perceptual system include localization, map building, object detection, etc.

Review

室内自主移动机器人传感技术综述

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摘要: 作为机器人学术界和工业界的一个基本问题, 室内自主移动机器人 (AMR) 得到了广泛的研究。对于AMR来说, 获取其工作环境和自身的信息至关重要, 这可以通过传感器并从这些传感器的测量中提取相应的信息来实现。传感技术的应用可以使移动机器人执行定位、建图、目标或障碍物识别以及运动任务等。本文综述了室内场景中自主移动机器人的传感技术。分析和比较了应用中使用单个传感器的好处和潜在问题, 并介绍了处理这些传感器数据的基本原理和流行算法。此外, 还介绍了多传感器融合的一些主流技术。最后, 本文讨论了室内场景自主移动机器人传感技术的未来发展趋势, 以及实际应用环境中的挑战。

关键词: 传感技术; SLAM; 避障; 传感器融合

一、简介

近年来, 自主移动机器人 (AMR) 发展迅速。由于其出色的工作能力, AMR正在一些室内场景中替代人类。移动机器人可以帮助减轻人类工作负担, 提高生产效率。它们已广泛应用于许多行业和服务中, 例如仓储、物流、医疗保健、餐饮服务和个人服务[1-3]。

移动机器人技术的基础包括运动、感知、认知和导航领域[4]。运动问题主要涉及移动机器人的运动系统。运动系统设计基于所提供服务的要求, 如运动环境、可控性、效率、稳定性等相关指标。感知涉及获取有关移动机器人工作环境及其自身的信息。认知涉及分析和处理来自感知系统的数据, 并向运动系统提供控制解决方案以完成移动机器人的任务。移动机器人导航依赖于感知、认知和运动控制, 在工作环境 (无论是已知还是未知) 中从起点移动到任务目标点。感知是移动机器人研究的重要组成部分。为了使移动机器人安全有效地执行移动任务, 它们必须感知环境并根据该信息做出决策。如果移动机器人不能准确有效地感知其环境, 那么它甚至无法执行简单的任务[5]。

移动机器人的感知系统利用相关的传感技术来提供有关环境、机器人本身以及两者之间关系的信息。这些信息随后用于路径规划、控制机器人的运动, 并最终完成导航任务。感知系统的主要任务包括定位、地图构建、物体检测等。



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Localization is the process of estimating a mobile robot's position relative to the surrounding work area during its movement. This information serves as the foundation for the mobile robot's navigation [6–8]. Mobile robot localization techniques include inertial navigation, global positioning systems (GPS), active beacon-based navigation, landmark navigation, and model matching. Inertial navigation is a navigation method that utilizes data from sensors responding to inertial forces. The global positioning system (GPS) is a satellite-based technology that provides accurate information on longitude, latitude, and altitude for any location on Earth. It is a highly effective technique for locating mobile robots in outdoor environments. However, GPS is of limited use for localizing in indoor environments, due to non-line-of-sight and signal-blocking problems in complex environments [9]. Active beacon-based navigation and landmark navigation use relative positional relationships to beacons or landmarks with known locations in the environment for localization, as shown in Figure 1a. Model matching is a method of positioning by matching sensor data with specific map model data. For instance, in 2D LiDAR localization, the LiDAR measurement obtains 2D laser point cloud data, and the map model is an occupancy grid map. The mobile robot localization is performed by matching the point cloud data with the occupancy grid map, as shown in Figure 1b.

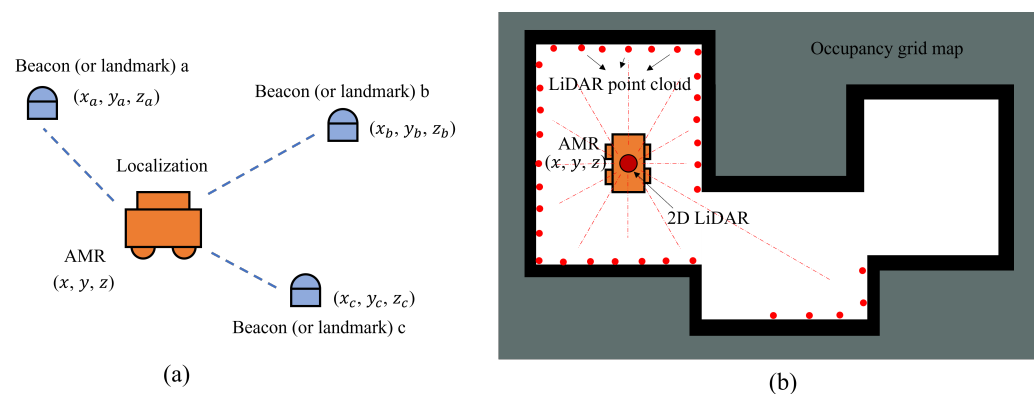


Figure 1. Schematic diagram of the localization method: (a) the localization method base on active beacon or landmark, and (b) the localization method base on model matching of 2D LiDAR.

Map building typically involves Simultaneous Localization and Mapping (SLAM), which is a crucial research field for mobile robots. SLAM is a technique used by mobile robots to determine their location in an unknown environment while simultaneously constructing an accurate map of the environment. The localization component estimates the robot's position on an existing map, while the mapping component constructs the environment's map [10–12]. The two components are interdependent in completing the map building process and enhancing accuracy through continuous iteration. The two most frequently used sensors in SLAM are LIDAR and vision cameras. SLAM techniques can be classified into two main categories based on short- and long-time optimization processing: filter-based methods and graph optimization-based methods. Filter-based SLAM algorithms typically use various types of filters to estimate and optimize the robot's trajectory and map, as shown in Figure 2. The two most common filter-based algorithms are the Kalman filter SLAM algorithm and the particle filter SLAM algorithm.

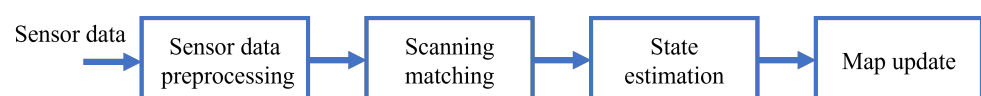


Figure 2. Schematic diagram of the SLAM method based on filter.

The graph optimization algorithm mainly consists of two processes, front-end and back-end, as shown in Figure 3a. The graph optimization algorithm models the robot's

定位是估计移动机器人在移动过程中相对于周围工作区域的位置的过程。该信息作为移动机器人导航的基础[6-8]。移动机器人定位技术包括惯性导航、全球定位系统（GPS）、基于主动信标的导航、地标导航和模型匹配。惯性导航是一种利用传感器响应惯性力的数据的导航方法。全球定位系统 (GPS) 是一种基于卫星的技术，可提供地球上任何位置的经度、纬度和高度的准确信息。这是一种在室外环境中定位移动机器人的高效技术。然而，由于复杂环境中的非视距和信号阻塞问题，GPS 在室内环境中的定位用途有限[9]。基于主动信标的导航和地标导航使用环境中已知位置的信标或地标的相对位置关系来进行定位，如图 1a 所示。模型匹配是一种通过传感器数据与特定地图模型数据进行匹配来进行定位的方法。例如，在2D LiDAR定位中，LiDAR测量获得2D激光点云数据，地图模型是占用网格地图。移动机器人定位是通过将点云数据与占用网格图匹配来执行的，如图1b所示。

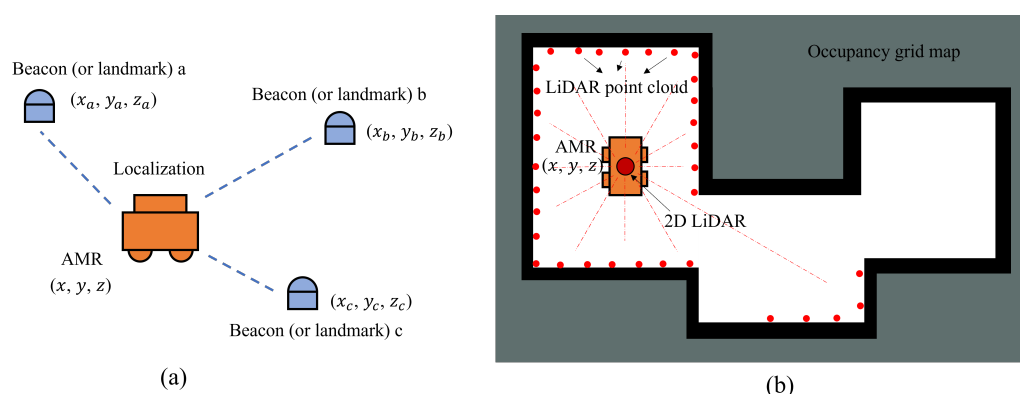


图1 定位方法示意图：（a）基于主动信标或地标的定位方法，（b）基于2D LiDAR模型匹配的定位方法。

地图构建通常涉及同步定位与建图（SLAM），这是移动机器人的一个重要研究领域。SLAM 是移动机器人用来确定其在未知环境中的位置，同时构建准确的环境地图的技术。定位组件估计机器人在现有地图上的位置，而映射组件则构建环境地图[10-12]。这两个组件在完成地图构建过程并通过不断迭代提高准确性方面相互依赖。SLAM 中最常用的两种传感器是激光雷达和视觉相机。根据短期和长期优化处理，SLAM技术可以分为两大类：基于过滤器的方法和基于图优化的方法。基于滤波器的 SLAM 算法通常使用各种类型的滤波器来估计和优化机器人的轨迹和地图，如图 2 所示。两种最常见的基于滤波器的算法是卡尔曼滤波器 SLAM 算法和粒子滤波器 SLAM 算法。

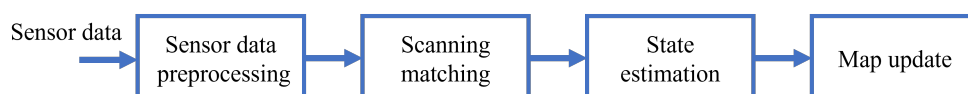


图2 基于滤波器的SLAM方法示意图。

图优化算法主要由前端和后端两个流程组成，如图3a所示。图优化算法对机器人进行建模

trajectory and the environment's topology as a graph, and minimizes the error by optimizing the nodes and edges in the graph to estimate the robot's position and obtain a map of the environment, as shown in Figure 3b. Nodes generally represent the variables to be optimized in the SLAM process of the sensor, such as sensor position, feature point location, estimated trajectory, etc. Edges represent the errors or constraints between the variables. The optimization algorithm is iterated to minimize the error of the edges to find the optimal solution [13].

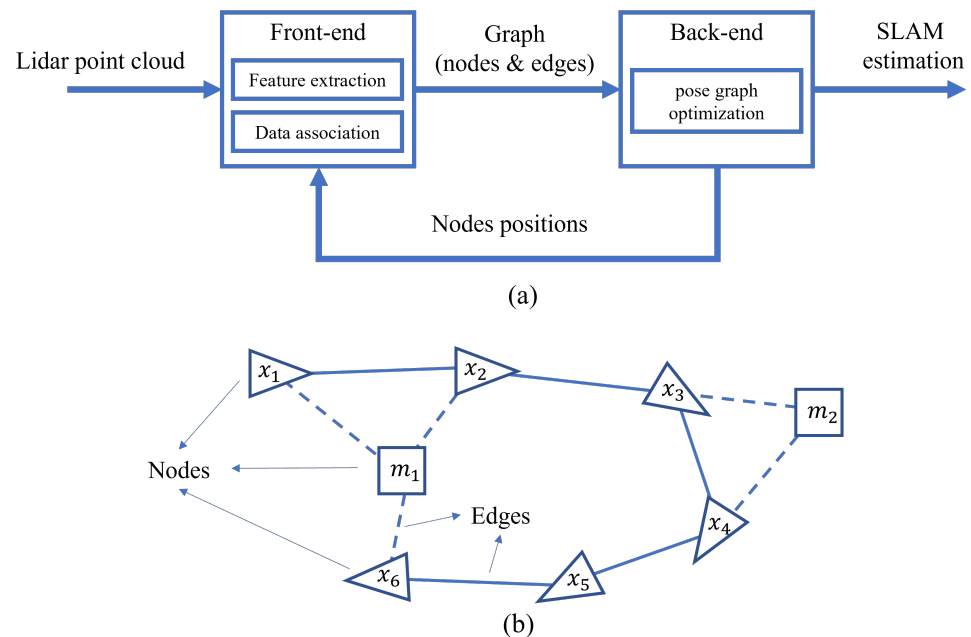


Figure 3. Schematic diagram of the SLAM method based on graph optimization: (a) representation of a graph optimization SLAM process, and (b) schematic representation of the graph optimization algorithm.

With the development of artificial intelligence, the emergence of reinforcement learning gradually allows mobile robots to gradually discard the limitations associated with a priori maps, and achieve the judgement of navigation actions only through image ingestion [14,15]. The main logic is: to set up a series of exploration reward mechanisms around the mobile robot exploration problem, such as: arrival reward, collision reward, curiosity reward, etc., and to establish the relevant neural network through reinforcement learning methods, and to train the network using the data set. Finally, the navigation decision of the autonomous mobile robot is realised.

To accomplish the navigation task efficiently and accurately, the mobile robot must avoid colliding with obstacles. Obstacle avoidance is a crucial task for autonomous mobile robots moving in uncertain environments [16,17]. Obstacle avoidance focuses on the problem by obtaining the global position information of the robot relative to the surrounding environment to the mobile robot only needs to obtain the local position information of the close objects around the robot relative to itself. Obstacle detection is crucial for perception systems as well. The mobile robot's cognition system performs path planning to avoid obstacles based on the distance and direction information about obstacles obtained from the perception system. In principle, range sensors (ultrasonic sensors and LiDAR) are highly suitable for obstacle detection tasks. In addition, vision cameras can recognize obstacles in the environment and even predict the trajectory of dynamic obstacles.

The perception system achieves these functions through the use of sensors and extracting relevant information from their measurements. Sensors used for data collection are categorized into two major aspects: proprioceptive/exteroceptive sensors and active/passive sensors [3]. Proprioceptive sensors gather data about the interior of the mobile

将轨迹和环境的拓扑绘制为图，并通过优化图中的节点和边来最小化误差，以估计机器人的位置并获得环境的map，如图3b所示。节点一般表示传感器SLAM过程中需要优化的变量，如传感器位置、特征点位置、估计轨迹等。边表示变量之间的误差或约束。迭代优化算法以最小化边缘的误差以找到最优解[13]。

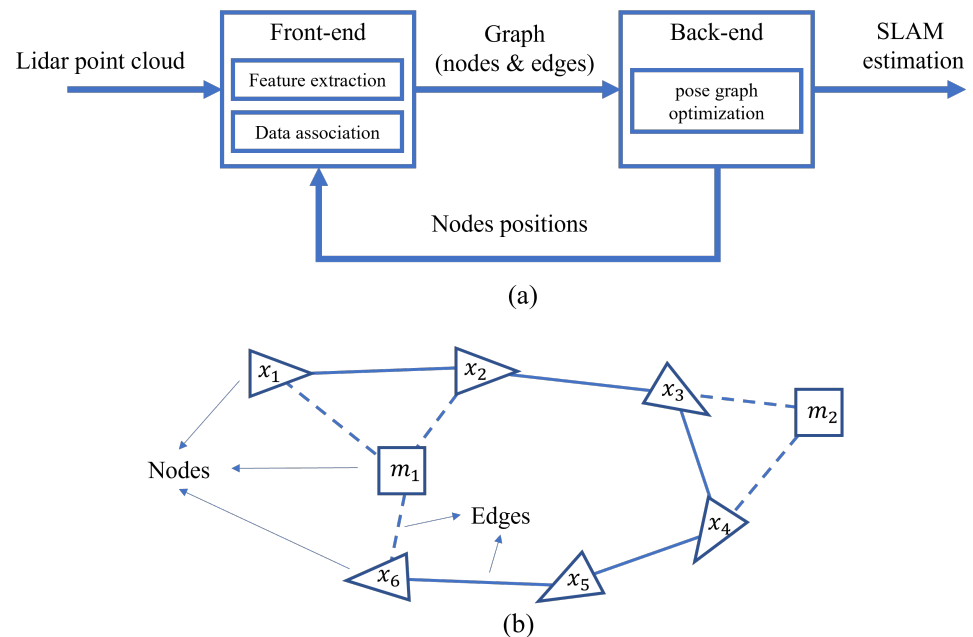


图3 基于图优化的SLAM方法示意图：(a)图优化SLAM流程示意图，(b)图优化算法示意图

随着人工智能的发展，强化学习的出现逐渐让移动机器人逐渐摆脱与先验地图相关的限制，仅通过图像摄取来实现导航动作的判断[14,15]。主要逻辑是：围绕移动机器人探索问题建立一系列探索奖励机制，如：到达奖励、碰撞奖励、好奇心奖励等，并通过强化学习方法建立相关神经网络，并利用数据集训练网络。最终实现自主移动机器人的导航决策。

为了高效、准确地完成导航任务，移动机器人必须避免与障碍物发生碰撞。避障是自主移动机器人在不确定环境中移动的一项关键任务[16,17]。避障通过获取机器人相对于周围环境的全局位置信息来解决问题，而移动机器人只需要获取机器人周围较近物体相对于自身的局部位置信息。障碍物检测对于感知系统也至关重要。移动机器人的认知系统根据感知系统获得的障碍物距离和方向信息进行路径规划以避免障碍物。原则上，距离传感器（超声波传感器和激光雷达）非常适合障碍物检测任务。此外，视觉相机可以识别环境中的障碍物，甚至可以预测动态障碍物的轨迹。

感知系统通过使用传感器并从测量结果中提取相关信息来实现这些功能。用于数据收集的传感器分为两个主要方面：本体/外感受传感器和主动/被动传感器[3]。本体感受传感器收集有关手机内部的数据

robot, such as motor speed, wheel rotation angle, etc. Currently, encoders [18], accelerometers, and gyroscopes [19] are widely used in this type of sensor. Exteroceptive sensors acquire information about the environment around the mobile robot, such as images, sound waves, and wireless signals. Examples of exteroceptive sensors include vision sensors [20], ultrasonic sensors [21], and LiDAR [22]. Active sensors send energy to the external environment and then measure the environmental feedback, such as sonar sensors, LIDAR, etc. Passive sensors measure the environmental energy coming into the sensor, such as vision sensors (Charge Coupled Device (CCD) or Complementary Metal Oxide Semiconductor (CMOS) cameras).

This paper describes the functions and principles related to the perception system of mobile robots. Further, this paper analyzes the application of various types of sensors in sensing technology. Additionally, the research results for multi-sensor fusion are presented here. The rest of this article is organized as follows: Section 2 describes some extensively adopted sensors; Section 3 presents the mainstream multi-sensor fusion schemes; and Section 4 discusses perception techniques and potential technology trends.

2. Overview of Single Sensor Sensing Technologies

2.1. Inertial Measurement Unit (IMU)

An IMU is an electronic device that integrates an accelerometer, gyroscope, and magnetometer to measure an object's acceleration, angular velocity, and the direction of the geomagnetic field [23]. The IMU is the proprioceptive sensor. The Inertial Navigation System (INS) is a dead-reckoning navigation system that uses the Inertial Measurement Unit (IMU) and is commonly used for mobile robot navigation. Magnetometer-measured orientation parameters have low accuracy and are challenging to use for precise navigation tasks. Therefore, the most frequently used parameters in IMUs are acceleration and angular velocity. The principle of dead reckoning is demonstrated in Figure 4. By using the considered IMU, the trajectory of an indoor AMR is determined by integrating the acceleration and angular velocity based on the initial position, direction, and velocity.

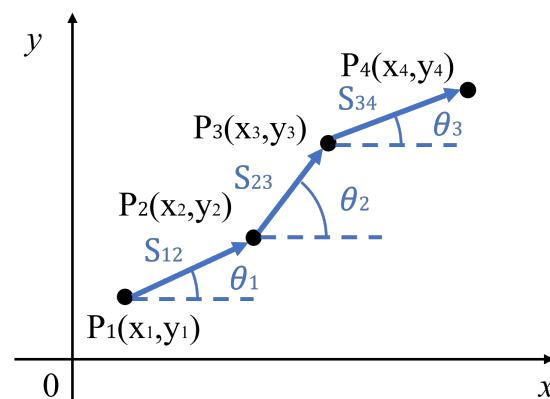


Figure 4. IMU dead reckoning.

An IMU has two typical advantages: a high output update rate and immunity to external interference. These advantages are irreplaceable when a high data update rate or high reliability are required, or when the external signal acquired by the mobile robot system is unreliable.

However, IMUs have limitations when applied to mobile robots. IMUs have two typical disadvantages:

1. The calculation process must depend on the initial conditions.
2. The navigation error increases with time, which is due to the need for double integration of its measurements.

机器人，如电机速度、车轮旋转角度等。目前，编码器[18]、加速度计和陀螺仪[19]广泛用于此类传感器。外感受传感器获取移动机器人周围环境的信息，例如图像、声波和无线信号。外感受传感器的例子包括视觉传感器[20]、超声波传感器[21]和LiDAR[22]。有源传感器向外部环境发送能量，然后测量环境反馈，例如声纳传感器、激光雷达等。无源传感器测量进入传感器的环境能量，例如视觉传感器（电荷耦合器件（CCD）或互补金属氧化物半导体（CMOS）相机）。

本文介绍了移动机器人感知系统的相关功能和原理。进一步分析了各类传感器在传感技术中的应用。此外，这里还介绍了多传感器融合的研究成果。本文其余部分的结构如下：第2节介绍了一些广泛采用的传感器；第3节介绍了主流的多传感器融合方案；第4节讨论感知技术和潜在的技术趋势。

2. 单传感器传感技术概述

2.1. Inertial Measurement Unit (IMU)

IMU是一种集成了加速度计、陀螺仪和磁力计的电子设备，用于测量物体的加速度、角速度和地磁场方向[23]。IMU是本体感觉传感器。惯性导航系统（INS）是一种使用惯性测量单元（IMU）的航位推算导航系统，通常用于移动机器人导航。磁力计测量的方向参数精度较低，难以用于精确导航任务。因此，IMU中最常用的参数是加速度和角速度。航位推算原理如图4所示。通过使用所考虑的IMU，室内AMR的轨迹是通过基于初始位置、方向和速度对加速度和角速度进行积分来确定的。

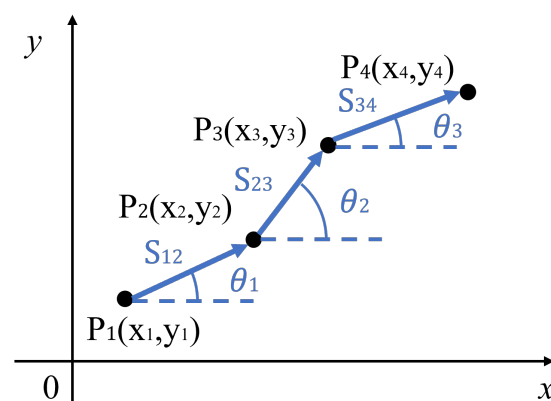


图 4.IMU 航位推算。

IMU 有两个典型的优点：高输出更新率和抗外部干扰。当需要高数据更新率或高可靠性时，或者当移动机器人系统获取的外部信号不可靠时，这些优势是不可替代的。

然而，IMU 在应用于移动机器人时存在局限性。IMU 有两个典型的缺点：1. 计算过程必须取决于初始条件。2. 导航误差随着时间的推移而增加，这是由于需要对其测量进行双重积分。

It should be mentioned that its duration and sensor accuracy have a significant impact on the accuracy of IMU-based navigation [23]. To overcome these disadvantages, an IMU is frequently used in fusion with other sensors and as auxiliary measurements in other navigation methods. L. Luo et al. [24] designed a positioning method that combines an IMU, Bluetooth, and a magnetometer for the mobile robot positioning problem. In the above-mentioned method, the inertial navigation system (INS) serves as the core, and the Bluetooth AOA positioning base station observes the position, while the magnetometer observes the heading angle. Alatis M. B. et al. [25] fused IMU and vision sensors in order to achieve accurate positioning of a mobile robot. The accelerometer data are used as inputs to the control, while gyroscope and image data are used as observations for position pose estimation using extended Kalman filtering. Zhang et al. [26] developed an algorithmic framework for the indoor localization of mobile robots by combining a wheel odometer, an inertial navigation unit (IMU), and Ultra-Wideband (UWB). The framework uses a fusion method with extended Kalman filtering, where the wheel odometer provides localization information as a prediction and the IMU and UWB provide attitude angle and position information as observations.

2.2. Ultrasonic Sensor/Sonar

Ultrasound refers to sound waves that vibrate at frequencies above 20 kHz, which are beyond the upper limit of human hearing. Sonar also employs ultrasound. Ultrasonic sensors are typically distance measurement sensors. Distance measurement sensors can be classified into two types of measurement methods: the reflection-type distance measuring method and the unidirectional distance measuring method, as shown in Figure 5. In the reflection-type method, the sensor sends a signal to the surrounding environment, which is then reflected when it encounters an obstacle. The sensor then receives the reflected signal to measure the distance between the sensor and the obstacle. The unidirectional distance-measuring method involves placing the transmitter and receiver in different locations. The receiver measures the distance between the two by accepting the signal from the transmitter. Ultrasonic ranging measures distance using three main criteria: time, phase, and acoustic vibration.

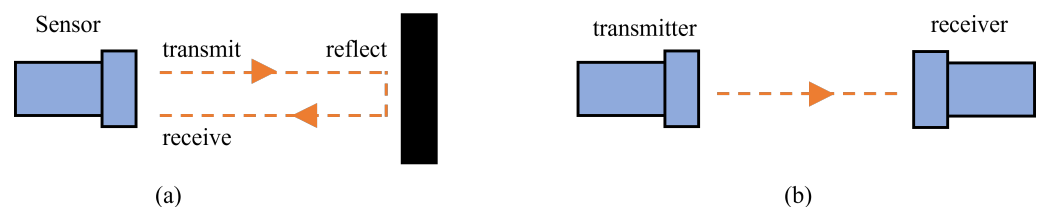


Figure 5. Sensor distance measuring method: (a) reflection-type distance measuring method. (b) unidirectional distance measuring method.

Ultrasonic sensors typically use a reflection-type distance measuring method to determine the distance to an object, which is used for mobile robot positioning and navigation. However, the sonar sensor's ultrasound beam width is too wide to precisely determine the object's direction. Additionally, specular reflection on object surfaces often results in inaccurate distance calculations, reducing the reliability of distance measurements from sonar sensors. It is not possible to construct a complete grid map using erroneously measured sonar data when the frequency of erroneous distance measurements is too high. To overcome this problem, S.-J. Lee et al. [27] designed a method for constructing occupancy grid maps for mobile robots using sonar data. This method assigns weights to each sonar data point based on its geometric reliability to minimize the impact of incorrect sonar data on the grid map construction. Anomalous sonar data are identified and assigned a lower geometric reliability value through a morphological comparison with neighboring sonar data. H. Liu et al. [28] used sonar sensors to overcome the limitations of sensor performance by accumulating sonar data to achieve room-level localization. Y. Liu et al. [29]

应该提到的是，其持续时间和传感器精度对基于 IMU 的导航精度有重大影响[23]。为了克服这些缺点，IMU 经常与其他传感器融合使用，并作为其他导航方法中的辅助测量。L.罗等人。[24]针对移动机器人定位问题，设计了一种结合IMU、蓝牙和磁力计的定位方法。上述方法以惯性导航系统（INS）为核心，蓝牙AOA定位基站观测位置，磁力计观测航向角。阿拉蒂斯 M. B. 等人。[25]融合IMU和视觉传感器以实现移动机器人的精确定位。加速度计数据用作控制的输入，而陀螺仪和图像数据用作使用扩展卡尔曼滤波进行位置姿态估计的观测值。张等人。[26]通过结合车轮里程计、惯性导航单元（IMU）和超宽带（UWB），开发了一种用于移动机器人室内定位的算法框架。该框架采用扩展卡尔曼滤波的融合方法，其中车轮里程计提供定位信息作为预测，IMU 和 UWB 提供姿态角和位置信息作为观测。

2.2. Ultrasonic Sensor/Sonar

超声波是指频率在20kHz以上振动的声波，该频率超出了人类听觉的上限。声纳也采用超声波。超声波传感器通常是距离测量传感器。测距传感器可分为两种测量方式：反射式测距法和单向测距法，如图5所示。反射式测距传感器向周围环境发送信号，遇到障碍物时信号被反射。然后传感器接收反射信号以测量传感器与障碍物之间的距离。单向距离测量方法涉及将发射器和接收器放置在不同的位置。接收器通过接受来自发射器的信号来测量两者之间的距离。超声波测距使用三个主要标准来测量距离：时间、相位和声振动。

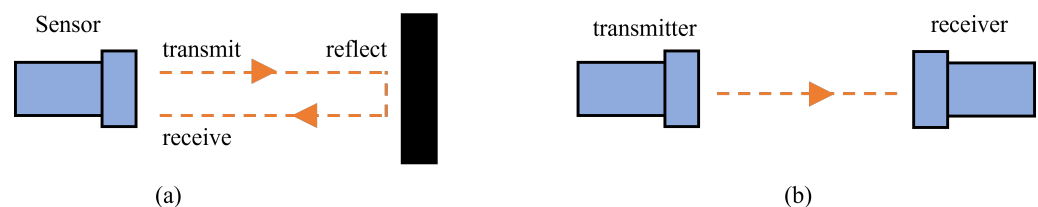


图5.传感器测距方法：(a)反射式测距方法。(b)单向测距法。

超声波传感器通常使用反射式距离测量方法来确定到物体的距离，用于移动机器人定位和导航。然而，声纳传感器的超声波束宽度太宽，无法精确确定物体的方向。此外，物体表面的镜面反射通常会导致距离计算不准确，从而降低声纳传感器距离测量的可靠性。当错误距离测量的频率太高时，不可能使用错误测量的声纳数据构建完整的网格地图。为了克服这个问题，S.-J.李等人。[27]设计了一种使用声纳数据构建移动机器人占用网格地图的方法。该方法根据每个声纳数据点的几何可靠性为其分配权重，以尽量减少不正确的声纳数据对网格地图构建的影响。通过与邻近声纳数据的形态比较，异常声纳数据被识别并分配较低的几何可靠性值。H.刘等人。[28]使用声纳传感器通过积累声纳数据来克服传感器性能的限制，以实现房间级定位。Y.刘等人。[29]

presented a mobile robot on the basis of LEGO MINDSTORM NXT, in which an ultrasonic sensor was mounted on a 360° rotating motor and a particle-filtering approach was used to implement its localization function. Mingqi Shen et al. [30] provided a localization method for autonomous mobile robots based on multiple ultrasonic sensors. The method calculates the target's coordinates based on the ratio of three flight times measured by three ultrasonic sensors. This improves the localization accuracy by avoiding the effect of ambient temperature on localization.

In addition, ultrasonic sensors can also be used for the unidirectional distance measuring method, which has been applied in the localization of indoor mobile robots. R. Li et al. [21] set up an ultrasonic receiver on the robot and had multiple ultrasonic generators in the scene. The three-sided localization principle is used to achieve the localization of the mobile robot. Chen-Chien Hsu et al. [31] presented a localization method based on omnidirectional ultrasonic sensing. The proposed method utilizes reflective cones to generate a 360° propagating ultrasonic beam to address the detection angle limitation of conventional ultrasonic sensors. Ultrasonic waves are emitted directly onto a cone, which reflects them in all directions. The ultrasonic signal is received by the installed receiver in the environment. The mobile robot's position is determined through the ToF (time-of-flight) method and the trilateration localization method.

Ultrasonic sensors are susceptible to environmental noise and long-range strength degradation [32] and are not accurate enough to measure detailed features of the environment. There are relatively few cases where ultrasonic sensors are used for positioning in high-precision indoor navigation applications. However, ultrasonic sensors have a noticeable advantage in obstacle detection. They are highly accurate in detecting highly transmissive or reflective objects, such as glass and mirrors, which can be challenging for many optical sensors. H. Takai et al. [33] used ultrasonic stereo sonar and a single image sensor to detect obstacles in the workspace. J.-H. Jean et al. [34] mainly used the environmental distance information collected by ultrasonic sensors to guide the robot along the wall baseline using a steering controller based on the potential field method. Grami T. et al. [35] designed a mobile robot equipped with seven ultrasonic sensors and used a particle-filtering approach to implement its localization function. The mobile robot can move freely while avoiding obstacles. Maryna Derkach et al. [36] utilized four ultrasonic sensors for obstacle avoidance in mobile robots. Their proposed algorithm uses a linear recursive Kalman filter to process the sensor data, allowing the robot to avoid additional obstacles.

2.3. Infrared Sensor

Infrared is an electromagnetic wave with a wavelength between visible light and microwaves. Infrared sensors use an emitter to emit an infrared beam and a receiver to measure the intensity of the beam. The intensity of the infrared beam decreases as the distance traveled increases. Infrared sensors can be used to measure distance by assessing the intensity of the beam, in addition to the ToF (time-of-flight) method. Similar to ultrasonic sensors, infrared sensors use reflection-type and unidirectional distance measuring methods.

There are many early studies applying infrared sensors to mobile robot localization. Eric Brassart et al. [37] designed a mobile robot localization system based on infrared beacons. In this system, a ceiling-mounted infrared beacon sends an encoded infrared signal, which is then received by a CCD camera mounted on the mobile robot. The robot's location is determined using triangulation and trilateration methods. J. Krejsa et al. [38] presented an interior localization method for mobile robots that uses infrared beacons. The beacons are placed at known locations in the environment, and a ring beacon scanning device with 16 receivers is used to receive the infrared signals. The scanner is mounted on the robot to measure the relative angle between the robot and the beacon. The position of the mobile robot is estimated by fusing orientation information with motion controller commands or an odometer through an extended Kalman filter. Infrared sensors can be

提出了一种基于乐高 MINDSTORM NXT 的移动机器人，其中超声波传感器安装在 360° 旋转电机上，并采用粒子过滤方法来实现其定位功能。沉明琪等.[30]提供了一种基于多个超声波传感器的自主移动机器人的定位方法。该方法根据三个超声波传感器测量的三个飞行时间的比值计算目标的坐标。这样可以避免环境温度对定位的影响，从而提高定位精度。

此外，超声波传感器还可用于单向测距方法，该方法已应用于室内移动机器人的定位。R.李等人。[21]在机器人上设置了超声波接收器，并在场景中设置了多个超声波发生器。利用三边定位原理实现移动机器人的定位。Chen-Chien Hsu 等人。[31]提出了一种基于全向超声传感的定位方法。该方法利用反射锥体产生 360° 传播超声波束，以解决传统超声波传感器的检测角度限制。超声波直接发射到圆锥体上，圆锥体将超声波向各个方向反射。超声波信号由环境中安装的接收器接收。移动机器人的位置通过ToF（飞行时间）方法和三边定位方法确定。

超声波传感器容易受到环境噪声和远距离强度衰减的影响[32]，并且不够准确，无法测量环境的详细特征。高精度室内导航应用中采用超声波传感器进行定位的案例相对较少。然而，超声波传感器在障碍物检测方面具有明显的优势。它们在检测玻璃和镜子等高透射或反射物体时非常准确，这对许多光学传感器来说可能是一个挑战。H.高井等人。[33]使用超声波立体声纳和单个图像传感器来检测工作空间中的障碍物。J.-H. 让等人。[34]主要利用超声波传感器收集的环境距离信息，使用基于势场法的转向控制器沿着墙壁基线引导机器人。格拉米·T.等人。[35]设计了一个配备七个超声波传感器的移动机器人，并使用粒子过滤方法来实现其定位功能。移动机器人可以自由移动，同时避开障碍物。玛丽娜·德卡赫等人。[36]利用四个超声波传感器在移动机器人中避障。他们提出的算法使用线性递归卡尔曼滤波器来处理传感器数据，使机器人能够避开额外的障碍物。

2.3. Infrared Sensor

红外线是一种波长介于可见光和微波之间的电磁波。红外传感器使用发射器发射红外光束，并使用接收器测量光束的强度。红外光束的强度随着行进距离的增加而减弱。除了ToF（飞行时间）方法之外，红外传感器还可通过评估光束强度来测量距离。与超声波传感器类似，红外传感器采用反射式、单向测距方法。

有许多早期研究将红外传感器应用于移动机器人定位。埃里克·布拉萨尔特等人。[37]设计了一种基于红外信标的移动机器人定位系统。在该系统中，安装在天花板上的红外信标发送编码的红外信号，然后由安装在移动机器人上的 CCD 摄像机接收。机器人的位置是使用三角测量和三边测量方法确定的。J.克雷萨等人。[38]提出了一种使用红外信标的移动机器人的内部定位方法。信标放置在环境中的已知位置，并使用具有 16 个接收器的环形信标扫描设备来接收红外信号。扫描仪安装在机器人上，用于测量机器人与信标之间的相对角度。通过扩展卡尔曼滤波器将方向信息与运动控制器命令或里程表融合来估计移动机器人的位置。红外传感器可以是

used to track targets when the transmitter and receiver are mounted on different robots. Tzue-Hseng S. Li et al. [39] have designed a tracking control scheme for an autonomous mobile robot by using infrared sensors. The scheme involves a target mobile robot with an infrared reflector and a tracking mobile robot with an infrared receiver and a reflective sensor. The latter was designed to track the target robot. Juang J. G. et al. [40] designed wheeled mobile robots with integrated ultrasonic and infrared sensors that can move along walls and avoid obstacles. Ultrasonic and infrared sensors were used to detect obstacles and identify unknown environments. Infrared-based localization technology is currently mature. However, it is difficult for it to penetrate objects and is susceptible to its surroundings. Additionally, obstacles can directly block infrared rays in the infrared beacon-based localization method. As a result, the application of infrared sensors in mobile robot localization is limited.

In obstacle detection, infrared sensors are frequently used because of their rapid response and compact size [41–43]. Jianwei Zhao et al. [44] designed a multi-sensor mobile robot that uses infrared sensors mounted in front of the two front wheels at the bottom of the robot to detect ground pits.

2.4. LiDAR

LiDAR (Light Detection and Ranging) uses laser beams to measure the position and distance of target objects. Multiple laser beams are sent out from transmitters, and when they hit objects, they reflect back. The reflected beams are received by the LiDAR receivers. LiDAR measures the distance traveled by each beam using time-of-flight (ToF) and combines the distance information from all the laser beams to obtain information about the surroundings. LiDAR operates similarly to radar, but with a higher accuracy, measurement speed, and measurement distance compared to conventional radar.

In indoor mobile robot navigation, mobile robots mainly use 2D LiDAR to sense environmental information, while 3D LiDAR is more often used in the field of intelligent driving because it needs more information to process [45]. LiDAR is widely used for mobile robot localization due to its excellent distance measurement accuracy. Localization with LiDAR is based on an environment map [46]. Dirk Hähnel et al. [47] presented an algorithm for acquiring 3D models with mobile robots. Xipeng Wang et al. [48] devised a method for global localization using only schematic floor plans as prior maps. The method achieves global localization by matching features observed from LiDAR with features in the floor plan. The method performs comparably to a baseline system using a conventional LiDAR-based prior map. Haofei Kuang et al. [49] proposed a global localization method for mobile robots using 2D LiDAR. In this method, the neural occupancy field uses a neural network to implicitly represent the scene, and 2D LiDAR scans of arbitrary robot poses are synthesized by using a pre-trained network. The MCL system integrates the similarity between the synthesized scans and the actual scans as an observation model to perform accurate localization based on the implicit representation.

As LiDAR distance measurement is based on the principle of light reflection, the measurement has a large error in environments with glass obstacles or mirror obstacles. To overcome the limitations, J. Kim et al. [50] proposed a method for LiDAR-based localization schemes in glass wall environments. In this method, all candidate distances from the glass wall are pre-calculated based on the occupancy grid map and the reflection characteristics of the laser beam. The type of reflection phenomenon on the glass surface is then estimated on the basis of actual measurements. The robot's local position tracking is performed by using a scan-matching method that considers the estimated results. Additionally, the fusion of ultrasonic sensors and LiDAR during obstacle detection can achieve a higher detection accuracy as ultrasonic sensors are not affected by glass or mirror obstacles.

In the absence of a prior environment map, LiDAR SLAM uses sensors to measure the surrounding environment and obtain a LiDAR point cloud, which is then used to estimate its position and construct a map based on the position information.

当发射器和接收器安装在不同的机器人上时，用于跟踪目标。Tzuu-Hseng S. Li 等人。[39]利用红外传感器设计了一种自主移动机器人的跟踪控制方案。该方案涉及一个带有红外反射器的目标移动机器人和一个带有红外接收器和反射传感器的跟踪移动机器人。后者旨在跟踪目标机器人。Juang J.G.等。[40]设计了带有集成超声波和红外传感器的轮式移动机器人，可以沿着墙壁移动并避开障碍物。超声波和红外传感器用于检测障碍物并识别未知环境。基于红外的定位技术目前已经成熟。但它很难穿透物体并且容易受到周围环境的影响。另外，在基于红外信标的定位方法中，障碍物可以直接阻挡红外线。因此，红外传感器在移动机器人定位中的应用受到限制。

在障碍物检测中，红外传感器因其快速响应和紧凑的尺寸而被频繁使用[41-43]。赵建伟等人。[44]设计了一种多传感器移动机器人，利用安装在机器人底部两个前轮前面的红外传感器来检测地坑。

2.4. LiDAR

LiDAR（光检测和测距）使用激光束来测量目标物体的位置和距离。发射器发出多束激光束，当它们击中物体时，它们会反射回来。反射光束由激光雷达接收器接收。LiDAR使用飞行时间 (ToF) 测量每个光束行进的距离，并结合所有激光束的距离信息来获取周围环境的信息。激光雷达的工作原理与雷达类似，但与传统雷达相比，其精度、测量速度和测量距离更高。

在室内移动机器人导航中，移动机器人主要使用2D LIDAR来感知环境信息，而3D LIDAR更常用于智能驾驶领域，因为它需要更多的信息来处理[45]。激光雷达因其出色的测距精度而被广泛应用于移动机器人定位。LiDAR定位基于环境地图[46]。德克·哈内尔等人。[47]提出了一种使用移动机器人获取3D模型的算法。王西鹏等人。[48]设计了一种仅使用示意性平面图作为先前地图的全局定位方法。该方法通过将激光雷达观测到的特征与平面图中的特征进行匹配来实现全局定位。该方法的性能与使用传统的基于LiDAR的先验地图的基线系统相当。匡浩飞等。[49]提出了一种使用2D LiDAR的移动机器人的全局定位方法。在该方法中，神经占用场使用神经网络隐式表示场景，并使用预训练的网络合成任意机器人姿势的2D LiDAR扫描。MCL系统将合成扫描与实际扫描之间的相似性整合为观察模型，以基于隐式表示进行精确定位。

由于激光雷达测距是基于光反射原理，因此在有玻璃障碍物或镜面障碍物的环境中测量误差较大。为了克服这些限制，J. Kim 等人。[50]提出了一种在玻璃墙环境中基于激光雷达的定位方案的方法。在该方法中，所有距玻璃墙的候选距离都是根据占用网格图和激光束的反射特性预先计算的。然后根据实际测量来估计玻璃表面上的反射现象的类型。机器人的局部位置跟踪是通过使用考虑估计结果的扫描匹配方法来执行的。此外，超声波传感器与激光雷达的融合在障碍物检测中可以实现更高的检测精度，因为超声波传感器不受玻璃或镜子障碍物的影响。

在没有先验环境地图的情况下，LiDAR SLAM使用传感器测量周围环境并获得LiDAR点云，然后使用该点云来估计其位置并根据位置信息构建地图。

Typical 2D LiDAR SLAM methods include Gmapping [51], Hector SLAM [52], Karto SLAM [53], and Cartographer [54]. The Gmapping algorithm framework is based on the RBPF (Rao-Blackwellized Particle Filters) algorithm, and Gmapping makes two main improvements, including improving proposal distribution and selectively resampling. The Hector SLAM algorithm framework is based on the EKF (extended Kalman filter) algorithm. Hector SLAM can be used without an odometer and is suitable for airborne or uneven road environments, but it is prone to drift errors when the LiDAR is rotating too fast. Karto SLAM is the first open-source algorithm based on the PGO (pose graph optimization) algorithm. It relies on efficient linear matrix construction and sparse non-iterative Cholesky decomposition to efficiently represent and solve large sparse pose graphs. The most important point of Karto SLAM is the introduction of back-end optimization and loop closure detection. Compared with Gmapping and Hector SLAM, Karto SLAM is more advantageous for mapping in large environments. Cartographer is based on the PGO (pose graph optimization) algorithm. It eliminates the cumulative errors generated during the map building process mainly through loop closure detection, in which the submap is the basic unit. It accelerates loop closure detection by combining local and global data and has high real-time performance, which makes it a reliable 2D LiDAR SLAM method.

In recent years, many improvements have been made based on these SLAM methods. Xiang Y. et al. [55] designed a LiDAR-based localization and mapping method for indoor mobile robots. The method makes improvements in distribution and limiting the number of resamplings based on the original RBPF-SLAM algorithm. Han Wang et al. [56] proposed an intensity-assisted SLAM framework for LiDAR-based localization systems. The proposed SLAM includes intensity-based front-end odometry estimation and intensity-based back-end optimization. Yingzhong Tian et al. [57] proposed a method to improve the traditional ICP method by incorporating intensity into the calculation. This method reduces the number of iterations and arithmetic required. The method uses an objective function that considers the distance and intensity residuals of the LiDAR point cloud to determine the optimal initial transformation estimation. Weiwei Hu et al. [58] proposed keyframe extraction by clustering 2D LiDAR point clouds in indoor environments. In this method, firstly, the dimension of the scan is reduced to a histogram, and the key frames are extracted using the histogram. Then, the laser points in the key frames are divided into different regions using a region-segmentation method. Next, the points are clustered in separate regions, and the point sets from neighboring regions are merged. Finally, the sets with laser points below a threshold are discarded as anomalous clusters. Saike Jiang et al. [59] designed an autonomous navigation method for indoor mobile robots by fusing LiDAR, IMU, and encoder data. The method optimizes and combines the 3D point cloud information from multi-line LiDAR into a 2D LiDAR point cloud. This reduces computational power consumption in the SLAM process and improves map building efficiency. Additionally, this method can recognize obstacles at varying heights, overcoming the limitations of 2D LiDAR in acquiring height-related information and enhancing the safety of mobile robot movement. Weipeng Guan et al. [60] proposed a loosely coupled multi-sensor fusion method for VLP and LiDAR SLAM. In this method, the LiDAR sensor detects the surroundings to avoid obstacles during navigation and compensates for the cumulative error of the odometer. And VLP is used to provide high-precision pose initialization and correction for the LiDAR SLAM and odometer.

For obstacle detection, compared to ultrasonic sensors, LiDAR has significant advantages in distance measurement accuracy and efficiency for obstacles. J. H. Lee et al. [61] proposed a method for tracking multiple walking humans based on 2D LiDAR. The algorithm considers the fact that 2D LiDAR is often positioned at the height of human legs and uses the geometric characteristics of the legs to detect their position. The method utilizes a pendulum model of the angle between the two legs as well as an extended Kalman filter to extract the frequency and phase of the walking motion. Mozos O. M. et al. [62] used a multilayer 2D LiDAR for person detection. The multilayer LiDAR method scans different parts of the human body using different layers. A supervised learning approach

典型的2D LiDAR SLAM方法包括Gmapping [51]、Hector SLAM [52]、Karto SLAM [53]和Cartographer [54]。Gmapping算法框架基于RBPF (Rao-Blackwellized Particle Filters) 算法, Gmapping主要做了两个改进, 包括改进proposal分布和选择性重采样。Hector SLAM算法框架基于EKF (扩展卡尔曼滤波器) 算法。Hector SLAM可以在没有里程表的情况下使用, 适用于机载或不平坦的道路环境, 但当激光雷达旋转过快时容易出现漂移误差。Karto SLAM是第一个基于PGO (位姿图优化) 算法的开源算法。它依靠高效的线性矩阵构造和稀疏非迭代 Cholesky 分解来有效地表示和求解大型稀疏位姿图。Karto SLAM最重要的一点是后端优化和闭环检测的引入。与Gmapping和Hector SLAM相比, Karto SLAM对于大型环境下的建图更有优势。Cartographer 基于 PGO (位姿图优化) 算法。它主要通过闭环检测来消除地图构建过程中产生的累积误差, 其中子图是基本单元。它通过结合局部和全局数据来加速闭环检测, 并且具有高实时性, 这使其成为可靠的2D LiDAR SLAM方法。

近年来, 基于这些SLAM方法已经做出了许多改进。向Y.等人。[55]设计了一种基于激光雷达的室内移动机器人定位和建图方法。该方法在原有RBPF-SLAM算法的基础上对分布和重采样次数限制进行了改进。王汉等人。[56]提出了一种用于基于 LiDAR 的定位系统的强度辅助 SLAM 框架。所提出的 SLAM 包括基于强度的前端里程计估计和基于强度的后端优化。田英中等人。[57]提出了一种通过将强度纳入计算来改进传统ICP方法的方法。该方法减少了所需的迭代次数和算术次数。该方法使用考虑LiDAR点云的距离和强度残差的目标函数来确定最佳初始变换估计。胡伟伟等人。[58]提出了通过在室内环境中聚类 2D LiDAR 点云来提取关键帧。在该方法中, 首先将扫描的维数降维为直方图, 并利用直方图提取关键帧。然后, 使用区域分割方法将关键帧中的激光点划分为不同的区域。接下来, 将点聚集在不同的区域中, 并合并来自相邻区域的点集。最后, 激光点低于阈值的集合将作为异常簇被丢弃。江赛克等。[59]通过融合LiDAR、IMU和编码器数据, 设计了一种室内移动机器人的自主导航方法。该方法将多线激光雷达的 3D 点云信息优化并组合成 2D 激光雷达点云。这减少了SLAM过程中的计算功耗并提高了地图构建效率。此外, 该方法可以识别不同高度的障碍物, 克服了2D LiDAR在获取高度相关信息方面的局限性, 增强了移动机器人运动的安全性。关伟鹏等。[60]提出了一种用于VLP和LiDAR SLAM的松耦合多传感器融合方法。在这种方法中, 激光雷达传感器在导航过程中检测周围环境以避免障碍物, 并补偿里程表的累积误差。VLP用于为LiDAR SLAM和里程计提供高精度位姿初始化和校正。

对于障碍物检测, 与超声波传感器相比, 激光雷达在障碍物测距精度和效率方面具有显著优势。J. H. Lee 等人[61]提出了一种基于2D LiDAR跟踪多个行走人体的方法。该算法考虑了 2D LiDAR 通常位于人体腿部高度的事实, 并利用腿部的几何特征来检测其位置。该方法利用两腿之间角度的摆模型以及扩展卡尔曼滤波器来提取步行运动的频率和相位。Mozos O. M. 等人。[62]使用多层 2D LiDAR 进行人员检测。多层激光雷达方法使用不同的层扫描人体的不同部位。监督学习方法

is used to obtain a classifier for each layer of LiDAR data, which is then used to detect and recognize specific body parts. The results of each classifier are ultimately combined to achieve human body detection. Ángel Manuel Guerrero-Higueras et al. [63] devised a 2D LiDAR-based method for tracking people. The 2D LiDAR is mounted at knee height in a mobile robot, and the method is based on an offline trained full Convolutional Neural Network that can track pairs of legs in the environment. Yan, Z. et al. [64] proposed an online learning framework based on 3D LiDAR scanning for human detection. Ailing Zou et al. [65] proposed a nearest neighbor clustering algorithm based on the adaptive threshold to detect obstacles in the path of indoor robots by using 2D LiDAR.

2.5. Vision-Based Sensor

Visual sensors are devices that mimic the function of the human visual system. They capture, perceive, and interpret visible light information in the environment and convert it into digital signals or other forms of output for analysis, processing, and decision-making by robots or other systems. Compared to other sensors, visual sensors have the ability to represent the surrounding environment in the form of an image, making them capable of describing complex texture features in the environment. Moreover, vision sensors are widely utilized in the field of mobile robotics due to their relatively low cost. The main types of vision sensors are monocular cameras, binocular cameras, RGBD cameras [66], and event cameras [13].

The imaging principle of a monocular camera is similar to that of a normal camera and is usually represented by the pinhole camera model. Light enters through the camera's aperture and is projected onto an imaging plane, typically a CMOS or CCD sensor [67]. The projected image is de-distorted and aligned with pixels to present a two-dimensional image similar to what is observed by the human eye. Despite its low cost, there are still significant technical challenges in research and development due to the difficulty of directly computing depth information. A stereo camera is a camera sensing system consisting of two monocular cameras, similar to the structure of the human eye. The depth information can be calculated from the baseline distance between the optical centers of the two cameras [68]. Depth estimation of a pixel point can be achieved by the underlying geometric model, as shown in Equation (1).

$$z = \frac{fb}{d} \quad (1)$$

where z is the actual depth estimation, f is the camera focal length, b is the actual baseline distance, and d is the parallax between the pixel points in the left and right screen projections. From this equation, it can be seen that, due to the influence of the baseline distance b , a truncation error will occur if the depth z is too large, resulting in inaccurate depth estimation.

RGBD cameras are sensors obtained by integrating a depth sensor with a monocular sensor. The depth camera actively provides depth measurements at each pixel. Currently, the main depth sensors for RGBD cameras are infrared structured-light sensors and ToF (time-of-flight) sensors [69]. The main difference between the two cameras is the depth estimation principle. The infrared structured-light sensor calculates the depth information by recognizing the structured light pattern. The Time of Flight (ToF) sensor calculates the depth information of each pixel point in the camera by sending pulses of light and measuring the time it takes for the light to return. The depth map is then mapped to the RGB image using the relative position relationship between the sensors, resulting in a composite image of color and depth. However, due to the limitations of the depth sensor, searching for depth information is not feasible in large-scale and large-range spatial environments.

Event cameras are biologically inspired to work, so their imaging principles are not the same as those cameras described above. In traditional cameras, the main camera unit is a monocular camera, where the sensor captures the surrounding image at a constant frame rate to obtain environmental information. However, considering the frame rate and aperture factors, it will appear shadowy or blurred when facing environmental factors such

用于获得每一层激光雷达数据的分类器，然后用于检测和识别特定的身体部位。最终将各个分类器的结果结合起来，实现人体检测。安赫尔·曼努埃尔·格雷罗·伊格拉斯等人。[63]设计了一种基于 2D LiDAR 的人体跟踪方法。2D LiDAR 安装在移动机器人的膝盖高度，该方法基于离线训练的全卷积神经网络，可以跟踪环境中的双腿。严，Z.等人。[64]提出了一种基于 3D LiDAR 扫描的人体检测在线学习框架。邹爱玲等。[65]提出了一种基于自适应阈值的最近邻聚类算法，利用 2D LiDAR 检测室内机器人路径中的障碍物。

2.5. Vision-Based Sensor

视觉传感器是模仿人类视觉系统功能的设备。它们捕获、感知和解释环境中的可见光信息，并将其转换为数字信号或其他形式的输出，供机器人或其他系统分析、处理和决策。与其他传感器相比，视觉传感器能够以图像的形式表示周围环境，从而能够描述环境中复杂的纹理特征。此外，视觉传感器由于其相对较低的成本而广泛应用于移动机器人领域。视觉传感器的主要类型有单目相机、双目相机、RGBD相机[66]和事件相机[13]。

单目相机的成像原理与普通相机类似，通常用针孔相机模型来表示。光线通过相机的光圈进入并投射到成像平面上，通常是 CMOS 或 CCD 传感器 [67]。投影图像经过畸变并与像素对齐，以呈现类似于人眼观察到的二维图像。尽管其成本较低，但由于直接计算深度信息的困难，在研发方面仍然存在重大技术挑战。立体相机是由两个单目相机组成的相机传感系统，类似于人眼的结构。深度信息可以根据两个相机光学中心之间的基线距离计算出来[68]。像素点的深度估计可以通过底层几何模型来实现，如方程（1）所示。

$$z = \frac{fb}{d} \quad (1)$$

其中 z 是实际深度估计， f 是相机焦距， b 是实际基线距离， d 是左右屏幕投影中像素点之间的视差。从该方程可以看出，由于基线距离 b 的影响，如果深度 z 过大，就会出现截断误差，导致深度估计不准确。

RGBD相机是深度传感器与单目传感器集成而成的传感器。深度相机主动提供每个像素的深度测量。目前，RGBD相机的主要深度传感器是红外结构光传感器和ToF（飞行时间）传感器[69]。两种相机的主要区别在于深度估计原理。红外结构光传感器通过识别结构光图案来计算深度信息。飞行时间（ToF）传感器通过发送光脉冲并测量光返回所需的时间来计算相机中每个像素点的深度信息。然后利用传感器之间的相对位置关系将深度图映射到RGB图像，从而得到颜色和深度的合成图像。然而，由于深度传感器的限制，在大范围、大范围的空间环境中搜索深度信息并不可行。

事件相机的工作原理是受生物学启发，因此它们的成像原理与上述相机不同。在传统相机中，主摄像头单元是单目相机，传感器以恒定的帧速率捕获周围图像以获取环境信息。但考虑到帧率和光圈因素，面对环境因素时会出现阴影或模糊的情况

as high dynamic and strong exposure. The event camera changes the recording method of the captured image to the recording of pixel changes. This means that the change in the pixels on the screen is recorded as an “event”, and the environmental information is read by the change in pixel brightness in the image on the output screen. As a result, the event camera can calculate quickly, with a frame rate of up to 1 KHz, and will not overexpose in high light intensity environments.

The light field camera is based on a monocular camera. A microlens array is added to the monocular camera to achieve a composite collection of environmental information. Due to the addition of the microlens array, the light sensor is able to sense light information in different directions at the same pixel point. The light field can be solved by the light information in different directions to achieve depth estimation in the surrounding environment. Similar to the binocular camera, it is able to estimate the depth of the surroundings from the line of sight in one frame. However, in contrast to the binocular camera’s simple two-frame camera, the light field camera can provide richer depth information for depth estimation or scene reconstruction at a later stage. The table below (Table 1) shows the pros and cons of each visual sensor.

Table 1. The advantages and disadvantages of each sensor and the related representative SLAM work applied for the first time to autonomous mobile robot perception.

Camera Type	Application Time	Characteristic	Advantages and Disadvantages
Monocular Camera	2007 [70]	Contains only one lens, the basic unit of the visual sensor	Pros: easy to integrate, low cost Cons: cannot directly provide in-depth information
Stereo Camera	2008 [71]	Simultaneous generation of left and right images for stereoscopic visual	Pros: produces two images (left and right), and can be used for stereo visual Cons: calibrates camera, computationally complex, limited depth estimation range
RGBD Camera	2011 [72]	RGB images and depth images can be provided directly	Pros: Provides accurate depth information Cons: Depth information is susceptible to environmental influences
Event Camera	2017 [73]	Uses pixel changes as events. Captures luminance changes with microsecond time resolution	Pros: able to cope with high dynamics and scenes with large lighting variations, low power consumption Cons: requires image pre-processing to obtain traditional image information
Light Field Camera	2017 [74]	Adding microlens arrays to camera lenses to add scale information to monocular cameras	Pros: provides scale information for monocular cameras Cons: high computational volume, difficult to achieve real-time computing

In indoor environments, the effects of lighting, dynamic objects, and unstructured scenes can cause images that do not accurately represent the indoor environment. Additionally, before performing localization, the scene must be depicted as a scene that can be modified by the mobile robot. Therefore, it is necessary to pre-process the images to facilitate the extraction of environmental information by the camera.

Image pre-processing is mainly divided into the following aspects:

- 1: Eliminating image distortion.
- 2: Assigning semantic labels to specific objects. The extraction of specific objects or features in the picture can be achieved by neural networks or traditional feature extraction algorithms. Assigning semantic labels to them is more suitable for human understanding of environmental features.
- 3: Image Partial Reconstruction. In indoor environments, deep learning techniques like generative adversarial networks (GAN) and diffusion should be considered for

高动态和强曝光。事件相机将拍摄图像的记录方式改为记录像素变化。这意味着屏幕上像素的变化被记录为“事件”，并且通过输出屏幕上图像中像素亮度的变化来读取环境信息。因此，事件相机可以快速计算，帧率高达1KHz，并且在高光强环境下也不会曝光过度。

光场相机是基于单目相机的。在单目相机中加入微透镜阵列，实现环境信息的复合采集。由于微透镜阵列的加入，光传感器能够在同一像素点感知不同方向的光信息。通过不同方向的光信息可以求解光场，从而实现周围环境的深度估计。与双目相机类似，它能够在一帧内从视线估计周围环境的深度。然而，与双目相机简单的两帧相机相比，光场相机可以为后期的深度估计或场景重建提供更丰富的深度信息。下表（表 1）显示了每种视觉传感器的优缺点。

表1. 每种传感器的优缺点以及首次应用于自主移动机器人感知的相关代表性SLAM工作。

Camera Type	Application Time	Characteristic	Advantages and Disadvantages
Monocular Camera	2007 [70]	Contains only one lens, the basic unit of the visual sensor	Pros: easy to integrate, low cost Cons: cannot directly provide in-depth information
Stereo Camera	2008 [71]	Simultaneous generation of left and right images for stereoscopic visual	Pros: produces two images (left and right), and can be used for stereo visual Cons: calibrates camera, computationally complex, limited depth estimation range
RGBD Camera	2011 [72]	RGB images and depth images can be provided directly	Pros: Provides accurate depth information Cons: Depth information is susceptible to environmental influences
Event Camera	2017 [73]	Uses pixel changes as events. Captures luminance changes with microsecond time resolution	Pros: able to cope with high dynamics and scenes with large lighting variations, low power consumption Cons: requires image pre-processing to obtain traditional image information
Light Field Camera	2017 [74]	Adding microlens arrays to camera lenses to add scale information to monocular cameras	Pros: provides scale information for monocular cameras Cons: high computational volume, difficult to achieve real-time computing

在室内环境中，光照、动态物体和非结构化场景的影响可能会导致图像不能准确地表示室内环境。此外，在执行定位之前，必须将屏幕描绘为移动机器人可以修改的场景。因此，需要对图像进行预处理，以方便相机提取环境信息。图像预处理主要分为以下几个方面：1：消除图像畸变。2：为特定对象分配语义标签。对图片中特定物体或特征的提取可以通过神经网络或传统的特征提取算法来实现。给它们分配语义标签更适合人类对环境特征的理解。3：图像部分重建。在室内环境中，应考虑生成对抗网络（GAN）和扩散等深度学习技术

reconstructing specific images due to the significant impact of light and dynamic objects on environmental representation. This reduces or reconstructs interfering factors in the image for the map construction process, such as dynamic objects, visible light, and shadows from occlusions. The goal is to acquire suitable images for scene localization.

Liu et al. [75] solved the issue of depth estimation between day and night by using the DCNN neural network. Considering the similarities and differences of the images between the environment with and without lighting, the images are divided into a private domain (lighting conditions) and a public domain (texture features). The reconstruction of a night image from a day image is achieved by the GAN neural network. This reduces the effect of the day/night boundary on the position and image construction errors. Bescos et al. [76] utilized CNN neural networks to implement multi-category semantic segmentation for the removal of dynamic objects and their accompanying shadows. GANs were employed along with a loss function based on image steganalysis techniques to achieve an absolutely static environment in the scene.

The conventional method for localizing mobile robots using visual sensors involves capturing image information with a camera and then comparing the images to determine the robot's current position [77]. It is mainly divided into two major steps: feature extraction and data association. The decision of whether to extract features is divided between the feature point method and the direct method. For the data association part, the main goal is to achieve a real-time estimation of the camera's own position change through the image comparison between frames. As the monocular camera cannot detect depth information directly, the relative positional relationship between the front and back frames of the camera corresponds to the rotation and translation in 3D space, which is achieved by establishing the rotational position between the front and back frames. That is, the pair of pole geometry problem, as shown in the following equation:

$$E = t^{\wedge} R \quad x_2^T E x_1 = 0 \quad E = \begin{bmatrix} e_1 & e_2 & e_3 \\ e_4 & e_5 & e_6 \\ e_7 & e_8 & e_9 \end{bmatrix} \quad (2)$$

where E is the essential matrix, which represents the positional transformation of the robot's translation and rotation, and x_1 and x_2 denote the normalized coordinates of the corresponding pixel points of both on the projection plane. Since e_9 can be directly designated as one, it is possible to solve the polar geometry problem for eight pairs of pixel points corresponding to the eight pairs in the frame, thus realizing the position estimation of the camera. The depth information of the environment in the camera positioning scene is solved by triangulation, that is, the positional transformation relationship between the two frames and the corresponding pixel projection coordinates in the two frames are known, so as to calculate the corresponding depth information of the pixel points. This can be shown in the following equation:

$$s_1 x_1^{\wedge} x_1 = 0 = s_2 x_1^{\wedge} R x_2 + x_1^{\wedge} t \quad (3)$$

Although the monocular camera can realize the position estimation of the robot from eight mutually matching pixel points, it still has problems such as the large initialization error and degradation of position estimation due to pure rotation. For the stereo camera and depth camera, they can directly glean the pixel information from the environment and its corresponding depth information. Therefore, the position calculation is relatively simple, i.e., through the PNP algorithm [78]. As shown in the following equation:

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} t_1 & t_2 & t_3 & t_4 \\ t_5 & t_6 & t_7 & t_8 \\ t_9 & t_{10} & t_{11} & t_{12} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \quad (4)$$

由于光和动态物体对环境表征的显着影响而重建特定图像。这减少或重建了地图构建过程中图像中的干扰因素，例如动态对象、可见光和遮挡造成的阴影。目标是获取适合场景定位的图像。

刘等人。[75]利用DCNN神经网络解决了白天和夜晚的深度估计问题。考虑到有光照和无光照环境下图像的异同，将图像分为私有域（光照条件）和公共域（纹理特征）。从白天图像重建夜间图像是通过 GAN 神经网络实现的。这减少了昼/夜边界对位置和图像构建误差的影响。贝斯科斯等人。[76]利用CNN神经网络实现多类别语义分割，以去除动态对象及其伴随的阴影。GAN 与基于图像隐写分析技术的损失函数一起使用，以在场景中实现绝对静态的环境。

使用视觉传感器定位移动机器人的传统方法是用相机捕获图像信息，然后比较图像以确定机器人的当前位置[77]。主要分为两大步骤：特征提取和数据关联。是否提取特征的决策分为特征点法和直接法。对于数据关联部分，主要目标是通过帧之间的图像比较来实现相机自身位置变化的实时估计。由于单目相机无法直接检测深度信息，因此相机前后帧之间的相对位置关系对应于3D空间中的旋转和平移，这是通过建立前后帧之间的旋转位置来实现的。即双极几何问题，如下式所示：

$$E = t^{\wedge} R \quad x_2^T E x_1 = 0 \quad E = \begin{bmatrix} e_1 & e_2 & e_3 \\ e_4 & e_5 & e_6 \\ e_7 & e_8 & e_9 \end{bmatrix} \quad (2)$$

其中 E 是基本矩阵，表示机器人平移和旋转的位置变换， x_1 和 x_2 表示两者在投影平面上对应像素点的归一化坐标。由于 e_9 可以直接指定为1，因此可以解决帧中8对对应的8对像素点的极坐标几何问题，从而实现相机的位置估计。通过三角测量来求解相机定位场景中环境的深度信息，即已知两帧之间的位置变换关系以及两帧中对应的像素投影坐标，从而计算出像素点对应的深度信息。这可以用以下等式表示：

$$s_1 x_1^{\wedge} x_1 = 0 = s_2 x_1^{\wedge} R x_2 + x_1^{\wedge} t \quad (3)$$

虽然单目相机可以从8个相互匹配的像素点实现机器人的位置估计，但仍然存在初始误差大、纯旋转导致的位置估计退化等问题。对于立体相机和深度相机来说，它们可以直接从环境中收集像素信息及其相应的深度信息。因此，位置计算比较简单，即通过PNP算法[78]。如下式所示：

$$s \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} t_1 & t_2 & t_3 & t_4 \\ t_5 & t_6 & t_7 & t_8 \\ t_9 & t_{10} & t_{11} & t_{12} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} \quad (4)$$

where u, v are the pixel points in normalized coordinates and X, Y, Z are the 3D coordinates corresponding to the pixel point. The T matrix represents the camera's positional transformation. From the above, it can be concluded that the positional estimation of the camera can be achieved by six pairs of well-matched pixel points. Due to the addition of depth information, other constraints can be added to the PNP algorithm to achieve fewer points for the pose estimation, such as the P3P algorithm, the EPNP algorithm, etc., [68]. The event camera serves as the basic unit of the camera. Unlike monocular camera processing, the event camera performs feature matching by matching time, pixel coordinates, and light intensity changes in the event stream to generate environmental features in the frame that can be correlated with the data [79].

The position estimation of mobile robots can be implemented using a neural implicit approach due to the excellent generalization and iterative capabilities of neural networks. The traditional explicit representation of position estimation is to represent the relevant objects in the frame and the relevant features as the exact six-degree-of-freedom relevant point, line, and surface features. Neural implicit methods favor the representation of the features in the frame over functions or neural networks, such as voxels, meshes, distance fields, etc. The advantage of implicit representation is that it can effectively represent complex texture features and more effectively represent complex surfaces such as voids and surfaces as opposed to explicit point clouds or grids. Furthermore, neural implicit methods have the advantage of compact spatial representation and the ability to store a large amount of information, allowing them to represent a larger map space with less memory compared to explicit networks. By using neural networks to predict and update maps, neural implicit methods have end-to-end learning capabilities that allow for greater flexibility in updating maps and estimating poses in real-time applications. Compared to traditional feature extraction and matching-based methods, neural implicit methods have better adaptability and robustness when dealing with dynamic scenes or fast movements. Zhu et al. [80] employs a neural implicit representation to capture the geometric and appearance attributes of a scene in terms of hierarchical network features. They utilize a pre-trained neural implicit decoder encoded at different spatial resolutions. By rendering the generated depth and colour images, they optimise the feature mesh within the field of view cone by minimising the re-rendering loss function.

There are many types of visual map building, but map functions generally have the following characteristics:

- 1: The ability to effectively represent the characteristics of the surrounding environment.
- 2: The amount of computation and storage required can match the hardware carried by the robot.
- 3: The ability to achieve the functionality required by the mobile robot.

As for map types, the main ones are point cloud maps [81,82], 3D occupancy grid maps [83], TSDF maps [84], semantic maps [85], etc.

A point cloud map generally consists of a camera frame that represents the camera's position, the camera's historical trajectory, and a pixel point cloud. It accurately depicts the surrounding environment and the camera's position transformation. However, the redundant point cloud features result in large storage space and construction costs. Vijayanarasimhan et al. [86] introduced the SfM-Net network, which retrieves depth maps and performs camera pose estimation using a single video stream. The network generates a corresponding depth image from the input single image and fuses the depth point cloud to create a comprehensive representation. Finally, the positional relationship between the resulting images is computed based on the input pair of consecutive frames. Furthermore, the method identifies and segments any moving objects within the scene, presenting them in the form of a mask.

The 3D occupancy grid map divides space into occupied subspaces based on a certain volume. If an object is present in the occupied subspace, it is considered occupied. While it sacrifices clarity in environmental information representation compared to the point cloud map, it significantly reduces computation and map information storage requirements. This

其中 u, v 是归一化坐标中的像素点, X, Y, Z 是像素点对应的3D坐标。 T 矩阵表示相机的位置变换。由上可知, 相机的位置估计可以通过六对匹配良好的像素点来实现。由于增加了深度信息, 可以在PNP算法中添加其他约束以实现更少的点位姿估计, 例如P3P算法、EPNP算法等[68]。事件相机作为相机的基本单元。与单目相机处理不同, 事件相机通过匹配事件流中的时间、像素坐标和光强度变化来执行特征匹配, 以在帧中生成可以与数据相关的环境特征[79]。

由于神经网络出色的泛化和迭代能力, 移动机器人的位置估计可以使用神经隐式方法来实现。位置估计的传统显式表示是将框架中的相关物体和相关特征表示为精确的六自由度相关点、线、面特征。与函数或神经网络相比, 神经隐式方法有利于表示帧中的特征, 例如体素、网格、距离场等。隐式表示的优点在于, 与显式点云或网格相比, 它可以有效地表示复杂的纹理特征, 并且更有效地表示复杂的表面, 例如空隙和表面。此外, 神经隐式方法具有紧凑的空间表示和存储大量信息的能力的优点, 与显式网络相比, 允许它们用更少的内存表示更大的地图空间。通过使用神经网络来预测和更新地图, 神经隐式方法具有端到端学习功能, 可以在实时应用中更灵活地更新地图和估计姿势。与传统的基于特征提取和匹配的方法相比, 神经隐式方法在处理动态场景或快速运动时具有更好的适应性和鲁棒性。朱等人。[80]采用神经隐式表示来根据分层网络特征捕获场景的几何和外观属性。他们利用以不同空间分辨率编码的预训练神经隐式解码器。通过渲染生成的深度和彩色图像, 他们通过最小化重新渲染损失函数来优化视锥内的特征网格。

可视化地图构建的类型有很多种, 但地图功能一般具有以下特点:

1: 有效表现周围环境特征的能力。 2: 所需的计算量和存储量可以与机器人携带的硬件相匹配。 3: 能够实现移动机器人所需的功能。

至于地图类型, 主要有点云地图[81,82]、3D占用网格地图[83]、TSDF地图[84]、语义地图[85]等。

点云图通常由代表相机位置的相机框架、相机的历史轨迹和像素点云组成。它准确地描绘了周围环境和相机的位置变换。然而, 冗余的点云特征导致存储空间和建设成本都较大。维贾亚纳拉西姆汉等人。[86]介绍了 SfM-Net 网络, 该网络检索深度图并使用单个视频流执行相机姿态估计。网络从输入的单图像生成相应的深度图像, 并融合深度点云以创建综合表示。最后, 根据输入的连续帧对计算结果图像之间的位置关系。此外, 该方法还可以识别并分割场景中的任何移动对象, 并以掩模的形式呈现它们。

3D占用网格图根据一定的体积将空间划分为占用的子空间。如果一个对象存在于占用的子空间中, 则认为它被占用。虽然与点云地图相比, 它牺牲了环境信息表示的清晰度, 但它显着降低了计算和地图信息存储要求。这

is advantageous for constructing large-scale maps. Cao et al. [87] proposed a 3D semantic occupancy raster map completion method for inferring indoor and outdoor scenes in a single RGB image. The method is applicable to a wide range of scenes and environments and does not require additional depth sensors or specific scene types. The article introduces the FLoSP mechanism to correlate 2D and 3D networks. Additionally, a new loss function is proposed to optimize inference results by taking into account the semantic relationships and local structure of voxel groups. This improves the quality and consistency of scene completion.

TSDF maps are maps represented by the TSDF notation (Truncated Signed Distance Function). Similar to three-dimensional occupancy grid maps, TSDF divides the space equally into an infinite number of voxel representations denoted by (x, y, z, v) . The map is iteratively updated by a mobile robot in a continuous positioning process. Since updating the voxels directly is required, RGBD cameras are generally used to implement TSDF map construction. Kim et al. [88] predicted 3D Truncated Signed Distance Function (TSDF) voxels from a single RGB image by integrating depth prediction and TSDF generation processes seamlessly. This approach enables the generation of highly accurate TSDF maps with greater efficiency compared to TSDF conversion from depth prediction alone. The comprehensive TSDF model is utilized to improve the precision and robustness of camera pose estimation, resulting in more effective scene facilitation.

Semantic maps differ from the above three maps. They are created by using image processing or deep learning network methods to extract the physical properties of points, lines, surfaces, or object features from an image and assign corresponding semantic information during the map construction process. Thus, we can create maps that align with human thinking and are more conducive to human-machine interactions based on the aforementioned maps. Yu et al. [89] proposed a methodology for extracting semantic segmentation information from the conventional SLAM framework. This process is executed as an independent thread within the image preprocessing module. The approach effectively identifies and eliminates dynamic objects present in each frame by combining semantic data with the concurrent detection of motion feature points. A dense semantic octree map is generated to meet the specific task requirements associated with advanced robot decision-making and path planning.

The development of mapless navigation can improve the computational speed of indoor mobile robots, which is conducive to autonomous decision making and path planning in dynamic environments. However, how to set up an effective reward mechanism and network structure is a major research challenge in mapless navigation. Kulhanek et al. [90] used auxiliary tasks consisting of reward prediction, pixel control, and semantic segmentation prediction to achieve effective training of neural net structure, and introduced LSTM as a memory module in the process of the network structure to cope with the before and after state correlation problem in the POMDP problem. Xiao et al. [91] addressed the problem of how to make a mobile robot learn to find the target point by passing through corners and such places when the initial target is not visible, creating arrival rewards, collision penalties, and proximity rewards. An “action consistency” mechanism was designed to achieve consistency between the learned behaviour of the mobile robot and its accumulated experience. Xie et al. [92] proposed a framework for a mapless navigation method for mobile robots based on a hierarchical approach with DL and DRL. It is assumed that the mobile robot can be provided with navigation instructions and images of key locations before task execution. The upper layer “watches” the navigation instructions and the current state to determine whether it is time to reach the place where it needs to act according to the navigation instructions. The lower layer is responsible for completing the navigation function of the mobile robot in between the upper layer’s updated decisions.

Vision cameras have a significant advantage in obstacle recognition compared to other sensors, making them a popular choice for obstacle avoidance in mobile robots. The obstacle avoidance system designed by Tai et al. [93] does not provide real-time estimations of the robot’s position during obstacle avoidance but builds a convolutional

有利于构建大比例尺地图。曹等人。[87]提出了一种 3D 语义占用栅格地图补全方法，用于推断单个 RGB 图像中的室内和室外场景。该方法适用于广泛的场景和环境，不需要额外的深度传感器或特定的场景类型。本文介绍了 FLoSP 机制来关联 2D 和 3D 网络。此外，还提出了一种新的损失函数，通过考虑体素组的语义关系和局部结构来优化推理结果。这提高了场景完成的质量和一致性。

TSDF 地图是由 TSDF 符号（截断符号距离函数）表示的地图。与三维占用网格图类似，TSDF 将空间平均划分为无限数量的体素表示，用 (x, y, z, v) 表示。移动机器人在连续定位过程中迭代更新地图。由于需要直接更新体素，因此通常使用 RGBD 相机来实现 TSDF 地图构建。金等人。[88]通过无缝集成深度预测和 TSDF 生成过程，从单个 RGB 图像预测 3D 截断符号距离函数 (TSDF) 体素。与仅从深度预测进行 TSDF 转换相比，这种方法能够以更高的效率生成高精度 TSDF 地图。利用综合的 TSDF 模型来提高相机位姿估计的精度和鲁棒性，从而实现更有效的场景引导。

语义图与上述三个图不同。它们是通过使用图像处理或深度学习网络方法从图像中提取点、线、面或物体特征的物理属性并在地图构建过程中分配相应的语义信息来创建的。这样，我们就可以在上述地图的基础上创建出符合人类思维、更有利于人机交互的地图。于等人。[89]提出了一种从传统 SLAM 框架中提取语义分割信息的方法。该过程作为图像预处理模块内的独立线程执行。该方法通过将语义数据与运动特征点的并发检测相结合，有效地识别和消除每帧中存在的动态对象。生成密集的语义八叉树图以满足与高级机器人决策和路径规划相关的特定任务要求。

无地图导航的发展可以提高室内移动机器人的计算速度，有利于动态环境下的自主决策和路径规划。然而，如何建立有效的奖励机制和网络结构是无地图导航的一大研究挑战。库尔哈内克等人。[90]利用奖励预测、像素控制和语义分割预测组成的辅助任务来实现神经网络结构的有效训练，并在网络结构过程中引入 LSTM 作为记忆模块来应对 POMDP 问题中的前后状态相关性问题。肖等人。[91]解决了当初始目标不可见时如何让移动机器人学会通过拐角等地方找到目标点，创建到达奖励、碰撞惩罚和接近奖励。设计了“动作一致性”机制，以实现移动机器人学习的行为与其积累的经验之间的一致性。谢等人。[92]提出了一种基于深度学习和深度强化学习分层方法的移动机器人无地图导航方法框架。假设在任务执行之前可以向移动机器人提供导航指令和关键位置的图像。上层“观察”导航指令和当前状态，以根据导航指令确定是否到达需要采取行动的地点。下层负责在上层更新决策之间完成移动机器人的导航功能。

与其他传感器相比，视觉相机在障碍物识别方面具有显著优势，使其成为移动机器人避障的热门选择。Tai 等人设计的避障系统。[93]没有提供避障过程中机器人位置的实时估计，而是构建了一个卷积

neural network for the original depth image. Subsequently, it establishes a fully connected network with five decisions on obstacle avoidance for the mobile robot, which enables the mobile robot to autonomously realize obstacle avoidance planning during indoor movement. Cristóforis et al. [94] achieved autonomous obstacle avoidance by building a “teach-learning” system that only requires texture information about the path of the screen rather than the position estimated since localization.

The perception of indoor mobile robots covers a variety of map representations. Point cloud maps are discrete point cloud data acquired by sensors such as LIDAR or camera sensors that provide highly accurate information about the structure of the environment. Three-dimensional occupancy raster maps divide the environment into a three-dimensional grid that is used to represent the location and occupancy of objects. TSDF maps provide highly accurate geometric information by modelling the surface of an object. Semantic maps further incorporate semantic information to enable the robot to understand the semantic meaning of different areas in the environment. In contrast to traditional navigation techniques, the mapless navigation function based on image information is a lightweight and efficient navigation method. It acquires environmental images through sensors such as cameras, and uses deep reinforcement learning and other methods to achieve end-to-end perception and decision making by learning the environmental features in the images, so as to realize autonomous navigation for robots. As the method does not need to pre-construct maps, it alleviates the dependence on expensive sensors, while showing a strong adaptability in dynamic environments. Therefore, mapless navigation based on image information is robust, with a small computation and memory footprint, making it perform well on resource-constrained mobile robot systems.

2.6. Radio Frequency Technologies

Radio frequency (RF)-based sensing technology is mostly used for indoor mobile robot localization. The RF-based localization system comprises beacon stations (BS) that transmit radio signals and mobile robots that receive them. Unlike infrared and ultrasound-based beacon location systems, RF waves can penetrate doors and walls, providing ubiquitous coverage of buildings. RF-based localization systems applied to indoor mobile robots typically include: WiFi [95], Bluetooth [96], Zigbee [97], Ultra-Wideband (UWB) [98], and Radio Frequency Identification (RFID). The advantages and disadvantages of the RF techniques [99] are shown in Table 2.

Table 2. The characteristics of different RF technologies used for indoor localization.

RF	Range (m)	Cost	Advantages	Disadvantages
WiFi	250	high	Implementation simplicity, large coverage, high transmission rate	High power consumption, meterlevel accuracy, vulnerable to NLOS path
Zigbee	100	medium	Extremely low energy consumption, low system cost	Low transmission rate, high latency, vulnerable to NLOS path
Bluetooth	100	low	Implementation simplicity, low energy consumption	Small coverage, low transmission rate, vulnerable to NLOS path
UWB	100	high	High accuracy, extremely high transmission rate, low latency, immune to interference, robustness to NLOS path	High energy consumption and system cost
RFID	5	low	Extremely low energy consumption, implementation simplicity, low system cost, high accuracy with specific approach	Small coverage, vulnerable to NLOS path

2.6.1. Radio Frequency Technology Localization Algorithms

RF-based indoor localization technologies use two main methods: geometry and fingerprinting. The geometry-based localization method determines the location of the mobile robot using geometric knowledge, either by measuring the distance to fixed beacon stations (BS) or by calculating the received signal angle. Several typical RF-based position-

原始深度图像的神经网络。随后，为移动机器人建立了具有五种避障决策的全连接网络，使移动机器人能够在室内运动过程中自主实现避障规划。克里斯托福里斯等人。[94]通过构建一个“教学”系统实现了自主避障，该系统只需要有关屏幕路径的纹理信息，而不是定位以来估计的位置。

室内移动机器人的感知涵盖了多种地图表征。点云地图是由激光雷达或相机传感器等传感器获取的离散点云数据，可提供有关环境结构的高度准确的信息。三维占用栅格地图将环境划分为三维网格，用于表示物体的位置和占用情况。TSDF 地图通过对物体表面进行建模来提供高精度的几何信息。语义地图进一步融合语义信息，使机器人能够理解环境中不同区域的语义。与传统导航技术相比，基于图像信息的无地图导航功能是一种轻量级、高效的导航方法。它通过摄像头等传感器获取环境图像，利用深度强化学习等方法，通过学习图像中的环境特征来实现端到端的感知和决策，从而实现机器人的自主导航。由于该方法不需要预先构建地图，减轻了对昂贵传感器的依赖，同时在动态环境中表现出很强的适应性。因此，基于图像信息的无地图导航具有鲁棒性，计算量和内存占用小，使其在资源受限的移动机器人系统上表现良好。

2.6. Radio Frequency Technologies

基于射频（RF）的传感技术主要用于室内移动机器人定位。基于射频的定位系统包括发射无线电信号的信标站 (BS) 和接收无线电信号的移动机器人。与基于红外和超声波的信标定位系统不同，射频波可以穿透门和墙壁，实现建筑物的无处不在的覆盖。应用于室内移动机器人的基于射频的定位系统通常包括：WiFi [95]、蓝牙[96]、Zigbee [97]、超宽带（UWB）[98]和射频识别（RFID）。RF技术[99]的优点和缺点如表2所示。

表 2. 用于室内定位的不同射频技术的特性。

RF	Range (m)	Cost	Advantages	Disadvantages
WiFi	250	high	Implementation simplicity, large coverage, high transmission rate	High power consumption, meterlevel accuracy, vulnerable to NLOS path
Zigbee	100	medium	Extremely low energy consumption, low system cost	Low transmission rate, high latency, vulnerable to NLOS path
Bluetooth	100	low	Implementation simplicity, low energy consumption	Small coverage, low transmission rate, vulnerable to NLOS path
UWB	100	high	High accuracy, extremely high transmission rate, low latency, immune to interference, robustness to NLOS path	High energy consumption and system cost
RFID	5	low	Extremely low energy consumption, implementation simplicity, low system cost, high accuracy with specific approach	Small coverage, vulnerable to NLOS path

2.6.1. 射频技术定位算法

基于射频的室内定位技术使用两种主要方法：几何和指纹。基于几何的定位方法使用几何知识确定移动机器人的位置，通过测量到固定信标站（BS）的距离或通过计算接收信号角度。几种典型的基于射频的定位

ing algorithms are Time of Arrival (TOA), Time Difference of Arrival (TDOA), Angle of Arrival (AOA), and Received Signal Strength Indication (RSSI), as shown in Figure 6.

TOA is a localization method that measures signal propagation time. A signal sent from a transmitter on a mobile robot travels through the air at approximately the speed of light. When the signal reaches receivers mounted at multiple beacon stations (BSs), the receivers record their respective arrival times [100]. The distance of signal propagation is calculated by measuring the propagation time. This allows for the localization of the mobile robot using the triangulation method, as shown in Figure 6a. The signals emitted by the transmitters are time-stamped to calculate the propagation time. It is crucial to synchronize the time of the beacon stations with the time of the mobile robot [101].

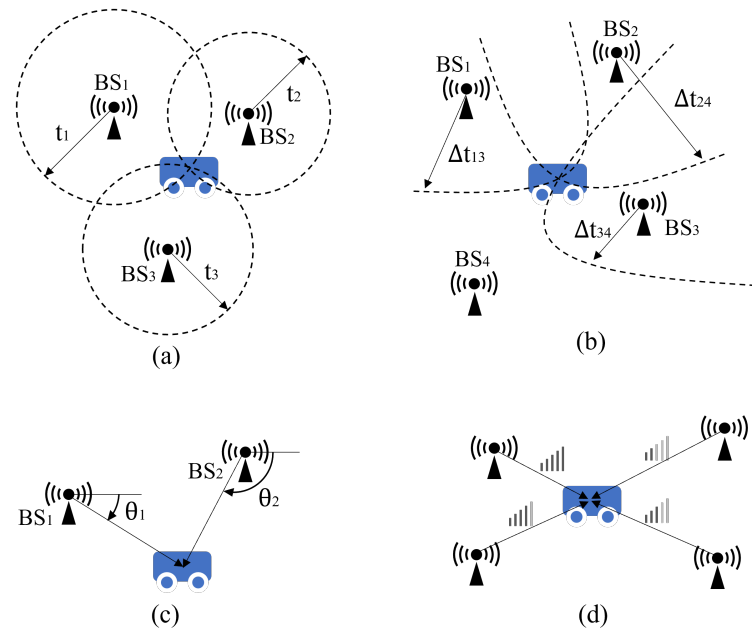


Figure 6. RF localization technologies: (a) Time of Arrival (TOA). (b) Time Difference of Arrival (TDOA). (c) Angle of Arrival (AOA). (d) Received Signal Strength Indication (RSSI).

TDOA is a method used to determine the location of a signal transmitter by measuring the time difference between the signal's arrival at different receivers [102]. The arrival time of the same signal is recorded by multiple signal receivers, and the time differences between the different receivers are calculated. These time differences are then used to construct a system of equations. Each equation represents the positional constraint of the corresponding receiver on the transmitter. These constraints are reflected in the coordinate graph as hyperbolas. The position of the signal transmitter, i.e., the mobile robot, is determined by solving these equations, as shown in Figure 6b. Unlike the TOA method, the TDOA method only requires time synchronization between the beacon sites, not between them and the mobile robot [103].

AOA is a method used to determine the position of a signal transmitter by measuring the angle at which the signal arrives at different receivers [104]. The signal receiver consists of multiple signal-receiving elements, and the angle of arrival is calculated by comparing the arrival times or phase differences of the signals between the different receiving elements. The angle of arrival of a signal is recorded by a plurality of signal receivers. Geometric relationships are then used to calculate the position of the signal transmitter, which is the mobile robot, as shown in Figure 6c.

The received signal strength indicator (RSSI) is a measurement of the power of a radio signal that has been received. Signal strength decays during propagation, which is characterized by the propagation power loss model. Therefore, the strength of the received

这些算法包括到达时间 (TOA)、到达时间差 (TDOA)、到达角度 (AOA) 和接收信号强度指示 (RSSI)，如图 6 所示。

TOA 是一种测量信号传播时间的定位方法。从移动机器人上的发射器发送的信号以大约光速在空气中传播。当信号到达安装在多个信标站 (BS) 的接收器时，接收器记录它们各自的到达时间[100]。通过测量传播时间来计算信号传播的距离。这允许使用三角测量方法来定位移动机器人，如图 6a 所示。发射机发出的信号带有时间戳以计算传播时间。将信标站的时间与移动机器人的时间同步至关重要[101]。

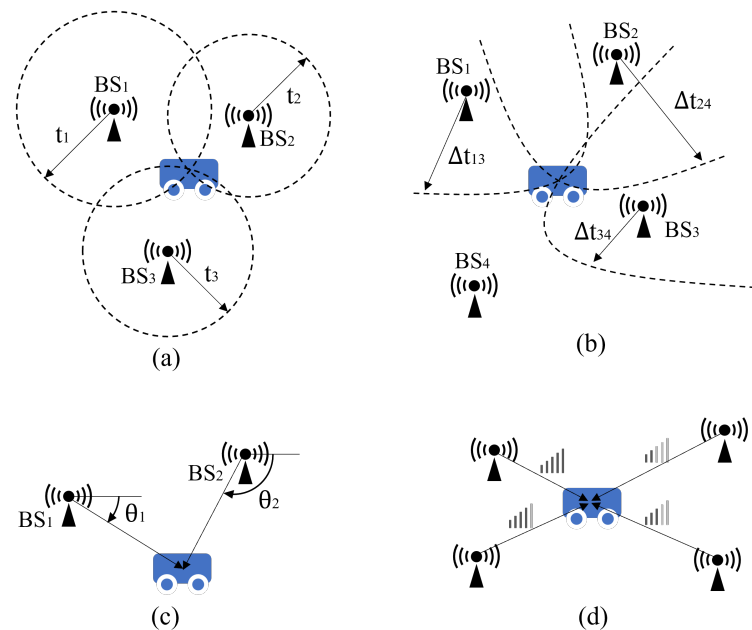


图 6. RF 定位技术: (a) 到达时间 (TOA)。 (b) 到达时间差 (TDOA)。 (c) 到达角 (AOA)。 (d) 接收信号强度指示 (RSSI)。

TDOA 是一种通过测量信号到达不同接收器之间的时间差来确定信号发射器位置的方法[102]。多个信号接收器记录同一信号的到达时间，并计算不同接收器之间的时间差。然后使用这些时间差构建方程组。每个方程代表相应接收器在发射器上的位置约束。这些约束在坐标图中反映为双曲线。信号发射器（即移动机器人）的位置通过求解这些方程来确定，如图 6b 所示。与 TOA 方法不同，TDOA 方法只需要信标站点之间的时间同步，而不需要信标站点和移动机器人之间的时间同步[103]。

AOA 是一种通过测量信号到达不同接收器的角度来确定信号发射器位置的方法[104]。信号接收器由多个信号接收元件组成，通过比较不同接收元件之间信号的到达时间或相位差来计算到达角。信号的到达角度由多个信号接收器记录。然后使用几何关系来计算信号发射器（即移动机器人）的位置，如图 6c 所示。

接收信号强度指示器 (RSSI) 是对已接收无线电信号功率的测量。信号强度在传播过程中衰减，其特征在于传播功率损耗模型。因此，接收到的强度

signal can also be used to calculate the signal propagation distance. Triangulation is then used to measure the position of the mobile robot [105].

Fingerprint localization is a technique used to determine location by comparing received signals with a pre-established fingerprint database [106]. It consists of fingerprint database construction and location estimation. The process involves deploying a set of RF sensor receiver stations throughout the area to collect RF signals at each sampling point and record their characteristic parameters. The sampling point locations are correlated with these characterization parameters to construct a fingerprint database. This database contains pre-measured and recorded correspondences between signal features and locations. The most commonly used signal characteristic parameter is RSSI. Once the fingerprint database is established, the RF sensor stations collect real-time wireless signal parameters, which are then matched and compared with the fingerprint information in the database [107]. Accurate target location is estimated through the use of comparison and inference algorithms. Compared to the geometry-based method, the fingerprint localization method is more accurate. However, constructing the fingerprint database requires significant effort, and it must be re-collected and constructed if the signal receiver station's location changes [103].

2.6.2. WiFi

WiFi is a widely used wireless technology that follows the IEEE 802.11 standards. WiFi utilizes wireless communication frequencies of 2.4 GHz and 5 GHz. It can support multiple device connections simultaneously and is widely used in indoor settings such as factories and offices. While WiFi technology offers advantages such as convenience, flexibility, and high speed, it has limitations when used for indoor localization. Factors such as the volatility of the WiFi signal and the non-line-of-sight (NLOS) path may affect the accuracy of localization.

For WiFi-based positioning methods, RSSI is most commonly used because of its high accuracy. Hemin Ye et al. [108] optimized the traditional WiFi fingerprint positioning method. Compared to traditional methods for fingerprint localization, this approach reduces the distance between sampling points and improves fingerprint matching accuracy by collecting and normalizing WiFi signals from different time periods. Additionally, the method uses Mahalanobis distance as a similarity reference and filters out noise using an improved adaptive K-value WKNN algorithm to enhance location estimation accuracy. Zhang L. et al. [109] optimized WiFi-based RSSI localization with Deep Fuzzy Forest. In their work, a deep fuzzy random forest is used as a mapping function, which estimates the corresponding coordinates of fingerprint vectors from an offline database. The deep fuzzy forest can inherit the merits of decision trees and deep neural networks within an end-to-end trainable architecture. Guangbing Zhou et al. [110] combine WiFi, vision, and LIDAR for the indoor localization of mobile robots. The WiFi-based RSSI fingerprinting localization method was used for coarse area estimation.

2.6.3. Zigbee

Zigbee is a wireless communication technology based on the IEEE 802.15.4 standard. It uses a distributed network topology that supports multiple devices communicating with each other, with all terminators communicating directly with the coordinator. ZigBee has a low cost, low power consumption, and low data transfer rate compared to WiFi standards, as well as a longer latency time.

Zhang Lei et al. [111] conducted an experimental analysis of Zigbee location using RSSI technology for the influence of reference node layout and number on the location accuracy. The experimental results indicate that Zigbee reference nodes with triangular layouts achieve higher localization accuracy than those with rectangular and prototypical layouts. Anbalagan Loganathan et al. [97] combined Zigbee-based RSSI and odometry for the indoor localization of mobile robots. Their method fuses the results of Zigbee-based RSSI triangulation with odometer data through an adaptive filtering method to find the

信号还可用于计算信号传播距离。然后使用三角测量来测量移动机器人的位置[105]。

指纹定位是一种通过将接收到的信号与预先建立的指纹数据库进行比较来确定位置的技术[106]。它包括指纹数据库构建和位置估计。该过程涉及在整个区域部署一组射频传感器接收站，以收集每个采样点的射频信号并记录其特征参数。采样点位置与这些特征参数相关联以构建指纹数据库。该数据库包含信号特征和位置之间预先测量和记录的对应关系。最常用的信号特征参数是RSSI。一旦指纹数据库建立，射频传感器站就会收集实时无线信号参数，然后将其与数据库中的指纹信息进行匹配和比较[107]。通过使用比较和推理算法来估计准确的目标位置。与基于几何的方法相比，指纹定位方法更加准确。然而，构建指纹数据库需要付出巨大的努力，如果信号接收站的位置发生变化，则必须重新收集和构建[103]。

2.6.2. 无线上网

WiFi 是一种广泛使用的无线技术，遵循 IEEE 802.11 标准。WiFi 使用 2.4 GHz 和 5 GHz 的无线通信频率。可支持多设备同时连接，广泛应用于工厂、办公室等室内环境。虽然 WiFi 技术具有便利、灵活和高速等优点，但用于室内定位时却存在局限性。WiFi 信号的波动性和非视距（NLOS）路径等因素可能会影响定位的准确性。

对于基于 WiFi 的定位方法，RSSI 由于其高精度而最常用。叶和民等人。[108]优化了传统WiFi指纹定位方法。与传统的指纹定位方法相比，该方法通过收集不同时间段的 WiFi 信号并对其进行归一化，缩短了采样点之间的距离，提高了指纹匹配精度。此外，该方法使用马哈拉诺比斯距离作为相似性参考，并使用改进的自适应K值WKNN算法滤除噪声，以提高位置估计的准确性。张L.等人。[109]使用深度模糊森林优化基于 WiFi 的 RSSI 定位。在他们的工作中，使用深度模糊随机森林作为映射函数，从离线数据库中估计指纹向量的相应坐标。深度模糊森林可以在端到端可训练架构中继承决策树和深度神经网络的优点。周光兵等人。[110]结合WiFi、视觉和LIDAR进行移动机器人的室内定位。基于WiFi的RSSI指纹定位方法用于粗略区域估计。

2.6.3. 紫蜂

Zigbee 是一种基于 IEEE 802.15.4 标准的无线通信技术。它采用分布式网络拓扑，支持多个设备相互通信，所有终结器直接与协调器通信。与 WiFi 标准相比，ZigBee 具有低成本、低功耗、数据传输速率低以及延迟时间较长的特点。

张雷等.文献[111]利用 RSSI 技术对 Zigbee 定位进行了实验分析，研究参考节点布局 and 数量对定位精度的影响。实验结果表明，三角形布局的Zigbee参考节点比矩形和原型布局的Zigbee参考节点具有更高的定位精度。安巴拉甘·洛加纳坦等人。[97]将基于 Zigbee 的 RSSI 和里程计结合起来用于移动机器人的室内定位。他们的方法通过自适应滤波方法将基于 Zigbee 的 RSSI 三角测量结果与里程表数据融合，以找到

robot's position. Zhu Wang et al. [112] designed a Zigbee-based localization algorithm for mobile robots. The method corrects the RSSI ranging results by least squares and uses triangulation to calculate the robot's position. The coordination nodes and mobile nodes of the ZigBee network are embodied in mobile robots, thus enhancing the mobility and flexibility of the network.

2.6.4. Bluetooth

Bluetooth is a standardized wireless communication technology based on the IEEE 802.15 standard. It is commonly used for short-range data transmission due to its low power consumption, low cost, high security, and ease of use. Bluetooth technology is widely used in various electronic products, automobiles, and medical devices. However, the accuracy of indoor positioning is affected by the non-line-of-sight (NLOS) path due to obstacles affecting Bluetooth signal propagation.

Aswin N. Raghavan et al. [113] proposed a method for the localization of mobile robots using Bluetooth. This method is based on RSSI for distance estimation and uses trilateration measurements for localization. The localization accuracy is within 1 m. Y. Yamami et al. [114] designed a Bluetooth-based AOA method for indoor mobile robot localization. The method makes use of Bluetooth 5.1 with the Constant Tone Extension (CTE) at the end of a packet, enabling the Bluetooth module to receive signals from multiple antennas simultaneously. In the study conducted by Katrina Weinmann et al. [115], a mobile robot controller employs Bluetooth as a sensor to measure the localization of the target object. The Bluetooth beacon is mounted on the target object, while the mobile robot is equipped with a Bluetooth antenna array. The mobile robot achieves target tracking by estimating the direction and distance of the target object. The direction is estimated using the AoA method, while the distance is estimated using RSSI. Astafiev Alexandr et al. [116] experimentally analyzed indoor localization based on a sensor network using Bluetooth Low Energy (BLE) beacons. The experiments demonstrate that the transmission of BLE signals can be affected by the presence of other radio signals in the localization area. The level of signal interference directly impacts BLE-based localization accuracy, resulting in greater errors and a lower accuracy. Additionally, obstacles such as people interfere with the RSSI measurement of BLE, further affecting positioning accuracy. To enhance the anti-interference capability of the BLE positioning system, it can be combined with other sensors, such as the IMU.

2.6.5. Ultra-Wideband (UWB)

UWB (Ultra-Wideband) technology is a wireless communication technology that uses short, non-continuous pulse signals to transmit data. It has a wide frequency bandwidth and offers advantages such as low power consumption, fast transmission rate, and strong anti-interference capability. Additionally, UWB is theoretically less affected by non-line-of-sight (NLOS) paths, making it highly accurate and stable for indoor localization.

Dongqing Shi et al. [117] designed a positioning system based on UWB sensors. To optimize localization results in the presence of nonlinearity and noise in UWB ranging, the system employs the gradient descent method and the least squares method. Jiajun Leng et al. [118] designed a two-stage UWB positioning algorithm for indoor mobile robots. First, the distance measured by the TDOA method was optimized by Gaussian filtering. Then, the final location coordinates were obtained by weighting the corrected range values with coordinates.

To reduce the influence of the LOS/NLOS environment on positioning accuracy, Peisen LI et al. [119] designed a positioning method combining the inertial navigation system (INS) with Ultra-Wideband (UWB). In this method, an interactive multiple model (IMM) algorithm is proposed to integrate the different distance error characteristics of LOS and NLOS states. The algorithm uses two Kalman filter models for different states and transforms them using the Markov chain. The filter results are then fused using weighting for position estimation. The localization method corrects the UWB's estimated position

机器人的位置。朱旺等人。[112]设计了一种基于 Zigbee 的移动机器人定位算法。该方法通过最小二乘法修正RSSI测距结果，并利用三角测量来计算机器人的位置。ZigBee网络的协调节点和移动节点体现为移动机器人，从而增强了网络的移动性和灵活性。

2.6.4. 蓝牙

蓝牙是一种基于 IEEE 802.15 标准的标准化无线通信技术。由于其低功耗、低成本、高安全性和易用性，通常用于短距离数据传输。蓝牙技术广泛应用于各种电子产品、汽车、医疗器械等领域。然而，由于障碍物影响蓝牙信号传播，室内定位的精度受到非视距（NLOS）路径的影响。

阿斯温·拉加万 (Aswin N. Raghavan) 等人[113]提出了一种使用蓝牙定位移动机器人的方法。该方法基于 RSSI 进行距离估计，并使用三边测量进行定位。定位精度在1m以内。Y.山见等人。[114]设计了一种基于蓝牙的AOA方法用于室内移动机器人定位。该方法利用蓝牙 5.1，并在数据包末尾添加恒定音调扩展 (CTE)，使蓝牙模块能够同时接收来自多个天线的信号。在 Katrina Weinmann 等人进行的研究中。[115]，移动机器人控制器采用蓝牙作为传感器来测量目标物体的定位。蓝牙信标安装在目标物体上，而移动机器人则配备蓝牙天线阵列。移动机器人通过估计目标物体的方向和距离来实现目标跟踪。使用 AoA 方法估计方向，而使用 RSSI 估计距离。阿斯塔菲耶夫·亚历山大等人。[116]使用蓝牙低功耗（BLE）信标对基于传感器网络的室内定位进行了实验分析。实验表明，BLE 信号的传输可能会受到定位区域中其他无线电信号存在的影响。信号干扰的程度直接影响BLE的定位精度，导致误差增大、精度降低。此外，人等障碍物会干扰BLE的RSSI测量，进一步影响定位精度。为了增强BLE定位系统的抗干扰能力，可以与其他传感器结合使用，例如IMU。

2.6.5. 超宽带 (UWB)

UWB（超宽带）技术是一种利用短的、非连续的脉冲信号来传输数据的无线通信技术。它具有频率带宽宽、功耗低、传输速率快、抗干扰能力强等优点。此外，UWB 理论上受非视距 (NLOS) 路径的影响较小，使其室内定位高度准确且稳定。

史冬青等人。[117]设计了一种基于UWB传感器的定位系统。为了在UWB测距中存在非线性和噪声的情况下优化定位结果，系统采用了梯度下降法和最小二乘法。冷家俊等人。[118]设计了一种用于室内移动机器人的两阶段UWB定位算法。首先，通过高斯滤波对TDOA方法测量的距离进行优化。然后，通过坐标对校正后的范围值进行加权，得到最终的位置坐标。

为了减少LOS/NLOS环境对定位精度的影响，李培森等人提出了一种方法。[119]设计了一种将惯性导航系统（INS）与超宽带（UWB）相结合的定位方法。在该方法中，提出了一种交互式多模型（IMM）算法来集成LOS和NLOS状态的不同距离误差特征。该算法针对不同状态使用两个卡尔曼滤波器模型，并使用马尔可夫链对其进行转换。然后使用加权来融合滤波器结果以进行位置估计。定位方法修正了UWB的估计位置

with the INS position estimation results, resulting in the final localization estimation. Tao Xu et al. [120] proposed a Weighted Adaptive Kalman Filter (WAKF) positioning method based on UWB and odometry. To minimize the effect of LOS/NLOS scenarios on UWB positioning accuracy, a Kalman filter is used to integrate the odometry data. Additionally, the method uses power differences to distinguish between LOS and NLOS environments, and it adjusts the Kalman filter weights accordingly. Haoyu Zhou et al. [121] designed an online multi-robot SLAM system based on Lidar/UWB fusion. The system fuses distance measurements provided by UWB sensors with Lidar data provided by different mobile robots and constructs a globally consistent map based on the UWB coordinate system.

2.6.6. Radio Frequency Identification (RFID)

RFID (Radio Frequency Identification) technology uses radio waves for data transmission and identification. It typically consists of three components: a tag, a reader, and a data processing system. RFID tags are small devices that contain a chip and an antenna and are used for data storage and transmission. The reader communicates with the tag via radio and reads or writes the tag's information. The RFID data are processed and managed by the data processing system. RFID technology offers the benefits of low energy consumption and low system costs. However, it has a short signal transmission distance of about 1–5 m, which requires the presetting of numerous beacon sites for large-area localization. RFID is advantageous for short-range target tracking and identification.

Haibing Wu et al. [122] proposed a method for mobile robot navigation based on RFID technology to track targets. The method uses a particle filter to measure the real-time relative position between the mobile robot and the RFID-tagged object based on the RFID phase difference observation model. Jun Wang et al. [123] proposed a SLAM method for mobile robots based on high-frequency band RFID by using the particle smoother for landmark mapping and the particle filter for the self-localization of the mobile robot. F. Shamsfakhr et al. [124] proposed a mobile robot localization method based on passive UHF radio frequency identification technology. The method combines odometry and RSSI-based position estimation results to determine the robot pose by the linear least squares method.

2.7. Summary and Discussion

IMU sensors have a high output frequency and are extensively adopted in mobile robot localization and SLAM fields. As a proprioceptive sensor, it is not influenced by the environment, but it does not have a role in obstacle detection. To mitigate error accumulation, an IMU is often used in conjunction with encoders as a short-term odometer and is part of the sensor fusion method in mobile robot positioning and SLAM. Ultrasonic sensors are less accurate in distance detection than infrared sensors and LiDAR. Therefore, they are rarely used as primary sensors in mobile robot localization and SLAM in recent studies. However, they have significant advantages over other sensors in obstacle detection and are widely used in mobile robot obstacle avoidance. Infrared sensors, as optical sensors, are susceptible to other light sources or reflections in the environment. However, infrared sensors are very low cost and are often used as auxiliary sensors in mobile robots, responsible for obstacle or target distance detection. LiDAR and vision sensors are the most commonly used sensors in SLAM for mobile robots, and they can both be used for navigation in unknown environments. In contrast, LiDAR has a higher distance measurement accuracy, and vision sensors can collect richer information about the working scenes. However, the accuracy of LiDAR measurements can be affected by transparent or reflective objects in the environment. Vision sensors are advantageous for detecting obstacles or targets due to their excellent object recognition technology. They can efficiently filter out obstacle data, improving localization precision during robot navigation. Radio frequency (RF) sensors are widely adopted in mobile robot localization, and the localization accuracy will not be influenced by environmental factors such as low-visibility conditions. However, RF sensor-based localization requires additional anchor nodes with prior position information

与INS位置估计结果相结合, 得到最终的定位估计。徐涛等人。[120]提出了一种基于UWB测距的加权自适应卡尔曼滤波器(WAKF)定位方法。为了最大限度地减少视距/非视距场景对UWB定位精度的影响, 使用卡尔曼滤波器对里程计数据进行积分。此外, 该方法使用功率差异来区分LOS和NLOS环境, 并相应地调整卡尔曼滤波器权重。周浩宇等人。[121]设计了一种基于激光雷达/UWB融合的在线多机器人SLAM系统。该系统将UWB传感器提供的距离测量与不同移动机器人提供的激光雷达数据融合, 并基于UWB坐标系构建全局一致的地图。

2.6.6. 射频识别 (RFID)

RFID(射频识别)技术利用无线电波进行数据传输和识别。它通常由三个组件组成: 标签、阅读器和数据处理系统。RFID标签是包含芯片和天线的小型设备, 用于数据存储和传输。阅读器通过无线电与标签通信并读取或写入标签的信息。RFID数据由数据处理系统进行处理和管理。RFID技术具有低能耗和低系统成本的优点。但其信号传输距离较短, 约为1~5 m, 需要预先设置大量信标站点进行大面积定位。RFID有利于短距离目标跟踪和识别。

吴海兵等人。[122]提出了一种基于RFID技术跟踪目标的移动机器人导航方法。该方法基于RFID相位差观测模型, 利用粒子滤波器测量移动机器人与RFID标签物体之间的实时相对位置。王军等人。[123]提出了一种基于高频RFID的移动机器人SLAM方法, 利用粒子平滑器进行地标映射, 并使用粒子滤波器进行移动机器人的自定位。F.Shamsfakhr等人。[124]提出了一种基于无源UHF射频识别技术的移动机器人定位方法。该方法结合里程计和基于RSSI的位置估计结果, 通过线性最小二乘法确定机器人位姿。

2.7. Summary and Discussion

IMU传感器具有高输出频率, 广泛应用于移动机器人定位和SLAM领域。作为本体感受传感器, 它不受环境影响, 但不起到障碍物检测的作用。为了减轻误差累积, IMU通常与编码器结合使用作为短期里程计, 并且是移动机器人定位和SLAM中传感器融合方法的一部分。超声波传感器的距离检测精度不如红外传感器和激光雷达。因此, 在最近的研究中, 它们很少被用作移动机器人定位和SLAM中的主要传感器。然而, 它们在障碍物检测方面比其他传感器具有显着优势, 并广泛应用于移动机器人避障。红外传感器与光学传感器一样, 容易受到环境中其他光源或反射的影响。然而, 红外传感器成本非常低, 通常用作移动机器人中的辅助传感器, 负责障碍物或目标距离检测。激光雷达和视觉传感器是移动机器人SLAM中最常用的传感器, 它们都可以用于未知环境中的导航。相比之下, 激光雷达具有更高的测距精度, 视觉传感器可以收集更丰富的工作场景信息。然而, LiDAR测量的准确性可能会受到环境中透明或反射物体的影响。视觉传感器因其出色的物体识别技术而有利于检测障碍物或目标。它们可以有效过滤掉障碍物数据, 提高机器人导航过程中的定位精度。射频(RF)传感器广泛应用于移动机器人定位, 定位精度不会受到低能见度条件等环境因素的影响。然而, 基于射频传感器的定位需要具有先前位置信息的额外锚节点

and cannot provide orientation information for mobile robots. A comparison of results of the relating sensors is demonstrated in Table 3.

Table 3. The characteristics of the sensors.

Sensors	Advantages	Disadvantages
IMU	High output frequency, free from environmental interference.	Initial conditions required, error accumulation, accelerometer is susceptible to vibration.
Ultrasonic sensor	Low cost, not influenced by transparent or reflective objects	long-range intensity attenuation, vulnerable to ambient noise.
Infrared sensor	Low cost.	Susceptible to environmental interference (light source, reflection).
LiDAR	High distance measurement accuracy.	Restricted in environments surrounded by transparent or reflective objects.
Vision-based sensor	Rich collection of information	Large computational volume, vulnerable to light.
Radio Frequency Sensor	Environment and obstacles have less effect on positioning.	Not suitable for unknown or unpredictable environments, unable to measure orientation.

3. Overview of Multi-Sensor Fusion Sensing Technologies

Various sensors can gather diverse forms of measurement data. Sensor fusion allows for a more comprehensive acquisition of sensory information, thereby enhancing localization efficiency in task completion. It is important to note that each sensor has its own superiorities, limitations, and potential scenarios. It is noted that sensor information fusion is able to enhance mobile robot positioning accuracy, obstacle recognition efficiency, and other characteristics by leveraging strengths and compensating for weaknesses. This approach can meet the requirements of diverse working environments. For instance, visual sensors provide a vast amount of image data that is valuable for recurrent closed-loop detection and optimization of the same scene in large-scale environments. This method enables the elimination of cumulative sensor errors. However, visual sensors in indoor environments are also limited by their image information. They suffer from long image processing times (generally 30 fps is required for the vision SLAM image in real-time), and environmental information is susceptible to occlusion. Therefore, improving visual sensor performance through other sensors is also a hot topic for mobile robot perception. As for multi-sensors, sensors that are often mixed with visual sensors are sonar [125,126], laser rangefinders [127,128], radio frequency identification (RFID) [128,129], inertial sensors [130,131], GPS [132,133]. However, in the case of GPS sensors, they are based on satellite systems to achieve global positioning. In indoor environments, objects are affected by buildings as well as other signals, causing a significant reduction in their positioning accuracy, implying that GPS may be inappropriate for mobile robot location positioning in indoor obscured environments [134,135].

Sensor data fusion algorithms mainly include improved algorithms based on Kalman filters, particle filters, and neural networks.

3.1. Multi-Sensor Fusion Algorithm Based on Kalman Filter

Kalman filtering methods are mainly used for fusing high-frequency and high-precision dynamic data and have been widely applied in multi-sensor fusion [51,136,137]. When assuming that the system state model and observation model are linear and follow a Gaussian distribution and that the noise is also Gaussian, the Kalman filtering method only requires the mean and variance of the noise to iteratively solve for the state. Kalman filter mainly includes prediction and update steps to realize real-time state estimation of the mobile robot from the previous frame to the current one, as shown in Figure 7. However, the traditional Kalman filter cannot meet the requirements of nonlinear problems. Therefore, researchers have developed improved versions of the Kalman filter, such as the Extended Kalman Filter method (EKF) and the Untraceable Kalman Filter method (UKF), to deal with nonlinear problems.

并且无法为移动机器人提供方向信息。相关传感器的结果比较如表3所示。

表 3. 传感器的特性。

Sensors	Advantages	Disadvantages
IMU	High output frequency, free from environmental interference.	Initial conditions required, error accumulation, accelerometer is susceptible to vibration.
Ultrasonic sensor	Low cost, not influenced by transparent or reflective objects	long-range intensity attenuation, vulnerable to ambient noise.
Infrared sensor	Low cost.	Susceptible to environmental interference (light source, reflection).
LiDAR	High distance measurement accuracy.	Restricted in environments surrounded by transparent or reflective objects.
Vision-based sensor	Rich collection of information	Large computational volume, vulnerable to light.
Radio Frequency Sensor	Environment and obstacles have less effect on positioning.	Not suitable for unknown or unpredictable environments, unable to measure orientation.

3. 多传感器融合传感技术概述

各种传感器可以收集多种形式的测量数据。传感器融合可以更全面地获取感官信息，从而提高任务完成时的定位效率。需要注意的是，每种传感器都有其自身的优势、局限性和潜在场景。值得注意的是，传感器信息融合能够扬长避短，增强移动机器人的定位精度、障碍物识别效率等特性。这种方式可以满足多样化工作环境的要求。例如，视觉传感器提供大量图像数据，这对于大规模环境中同一场景的循环闭环检测和优化非常有价值。该方法能够消除累积的传感器误差。然而，室内环境中的视觉传感器也受到图像信息的限制。它们的缺点是图像处理时间长（视觉SLAM图像实时需要30fps），并且环境信息容易被遮挡。因此，通过其他传感器提高视觉传感器性能也是移动机器人感知的热门话题。对于多传感器，经常与视觉传感器混合的传感器有声纳[125,126]、激光测距仪[127,128]、射频识别（RFID）[128,129]、惯性传感器[130,131]、GPS[132,133]。然而，就GPS传感器而言，它们是基于卫星系统来实现全球定位的。在室内环境中，物体受到建筑物以及其他信号的影响，导致其定位精度显着降低，这意味着GPS可能不适合室内遮挡环境中的移动机器人位置定位[134,135]。

传感器数据融合算法主要包括基于卡尔曼滤波器、粒子滤波器和神经网络的改进算法。

3.1. Multi-Sensor Fusion Algorithm Based on Kalman Filter

卡尔曼滤波方法主要用于融合高频、高精度的动态数据，在多传感器融合中得到了广泛的应用[51,136,137]。当假设系统状态模型和观测模型是线性且服从高斯分布且噪声也是高斯分布时，卡尔曼滤波方法只需要噪声的均值和方差即可迭代求解状态。卡尔曼滤波器主要包括预测和更新步骤，实现移动机器人从前一帧到当前帧的实时状态估计，如图7所示。然而，传统的卡尔曼滤波器无法满足非线性问题的要求。因此，研究人员开发了卡尔曼滤波器的改进版本，例如扩展卡尔曼滤波器方法（EKF）和不可追踪卡尔曼滤波器方法（UKF）来处理非线性问题。

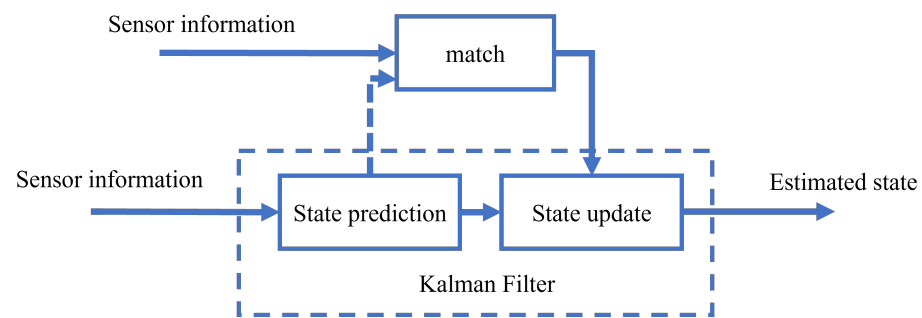


Figure 7. Multi-sensor fusion algorithm based on Kalman filter.

Daniel Magree et al. [138] proposed a modified navigation system by integrating visual SLAM and LiDAR SLAM, thus achieving an EKF-based inertial navigation system. In this vision SLAM approach, the robot's position can be determined by analyzing the environment's image features. The position is estimated and updated using the EKF method. LiDAR SLAM is based on the Monte Carlo method for scan matching. Its results are used to correct the results of visual SLAM and reduce the effect of ambiguous geometry on visual SLAM. Chengguo Zong et al. [139] presented an EKF multi-sensor information fusion method for obstacle avoidance in mobile robots. This method utilizes the previous moment's pose and encoder data to estimate the predicted position. The observed position can be obtained on the basis of observations from ultrasonic sensors, infrared sensors, and an electronic compass. The EKF updates the pose estimation by incorporating the predicted and observed positions, resulting in the optimal estimate of the required pose. LILI MU et al. [140] offered a SLAM approach that fuses LIDAR, an RGBD camera, an encoder, and an IMU. This method utilizes UKF to fuse the data from the four sensors for localization. In this method, the current position of the mobile robot is obtained from the IMU and wheel encoder, and it is input into the UKF for updating the rough position, which is used as the initial position for scanning matching by LIDAR and depth camera point clouds. This speeds up the matching speed of the point cloud and improves the matching accuracy. The optimized pose obtained by point cloud matching is sent to UKF as the measured value. Ping Jiang et al. [141] designed a rank Kalman filtering method for obtaining the robot motion trajectory by utilizing the IMU and LiDAR observations. The proposed method improves the localization accuracy of indoor mobile robots under nonlinear and non-Gaussian noise models. Lin et al. [142] proposed an iterative Kalman filtering scheme by using the error states. It utilizes the IMU information as an information carrier to discretize the continuous motion model and integrate the three sensors to generate more accurate trajectories. Wei Xu et al. [143] proposed a LiDAR–inertial odometry framework that contains a state estimation module and a mapping one. In this way, the IMU and LiDAR data are input into the iterative Kalman filter for optimal estimating position.

3.2. Multi-Sensor Fusion Algorithm Based on Particle Filter

The Kalman filter and its improved algorithms are only capable of handling Gaussian distributions. When dealing with arbitrary distributions, the adoption of Kalman filter-related methods may result in more errors. The particle filter-based approach, known as the Monte Carlo algorithm, is able to deal with the arbitrary distribution of multiple samples [144]. In this method, the probability density function of the indoor AMR's position can be approximated using a set of “samples” or “particles”, and regions with a larger number of particles have a higher probability. Each particle denotes a hypothetical pose of the considered mobile robot, while its weight indicates the extent of the match between the hypothesis and the true state. Typically, larger samples are used for global localization, and smaller samples are used for pose tracking.

When using the Monte Carlo algorithm for global localization, particles must be distributed throughout the entire map due to the unknown initial position. However, this can result in a significant amount of computation and reduced localization efficiency.

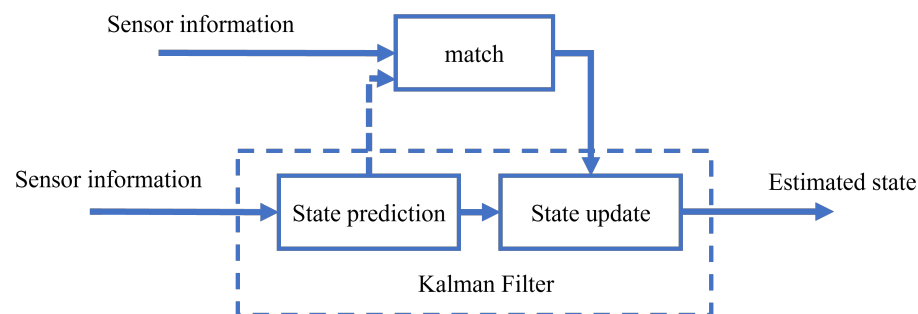


图7. 基于卡尔曼滤波器的多传感器融合算法。

丹尼尔·马格里等人。[138]提出了一种集成视觉SLAM和LiDAR SLAM的改进导航系统，从而实现了基于EKF的惯性导航系统。在这种视觉SLAM方法中，可以通过分析环境的图像特征来确定机器人的位置。使用 EKF 方法估计和更新位置。LiDAR SLAM 基于蒙特卡罗方法进行扫描匹配。其结果用于修正视觉SLAM的结果，减少模糊几何对视觉SLAM的影响。宗成国等。[139]提出了一种用于移动机器人避障的EKF多传感器信息融合方法。该方法利用前一时刻的姿态和编码器数据来估计预测位置。观测位置可以根据超声波传感器、红外传感器和电子罗盘的观测来获得。EKF 通过合并预测位置和观察位置来更新姿态估计，从而得到所需姿态的最佳估计。丽丽穆等人。[140]提供了一种融合 LIDAR、RGBD 相机、编码器和 IMU 的 SLAM 方法。该方法利用 UKF 融合来自四个传感器的数据进行定位。该方法从IMU和车轮编码器获取移动机器人的当前位置，输入UKF更新粗略位置，作为LIDAR和深度相机点云扫描匹配的初始位置。这样加快了点云的匹配速度，提高了匹配精度。将点云匹配得到的优化位姿作为测量值发送给UKF。蒋平等。[141]设计了一种秩卡尔曼滤波方法，利用 IMU 和 LiDAR 观测来获取机器人运动轨迹。该方法提高了室内移动机器人在非线性和非高斯噪声模型下的定位精度。林等人。[142]利用误差状态提出了一种迭代卡尔曼滤波方案。它利用IMU信息作为信息载体，对连续运动模型进行离散化，并集成三个传感器以生成更精确的轨迹。徐伟等人。[143]提出了一种LiDAR惯性里程计框架，其中包含状态估计模块和映射模块。这样，IMU和LiDAR数据被输入到迭代卡尔曼滤波器中以获得最佳估计位置。

3.2. Multi-Sensor Fusion Algorithm Based on Particle Filter

卡尔曼滤波器及其改进算法只能处理高斯分布。当处理任意分布时，采用卡尔曼滤波器相关的方法可能会导致更多的错误。基于粒子滤波器的方法，称为蒙特卡罗算法，能够处理多个样本的任意分布[144]。在该方法中，室内AMR位置的概率密度函数可以使用一组“样本”或“粒子”来近似，并且粒子数量较多的区域具有较高的概率。每个粒子表示所考虑的移动机器人的假设姿势，而其权重表示假设与真实状态之间的匹配程度。通常，较大的样本用于全局定位，较小的样本用于姿态跟踪。

当使用蒙特卡罗算法进行全局定位时，由于初始位置未知，粒子必须分布在整个地图上。然而，这可能会导致大量计算并降低定位效率。

To enhance global localization efficiency, researchers have conducted numerous studies on sensor fusion. This involves rough localization followed by accurate localization, as shown in Figure 8. Song Xu et al. [145] designed a global localization method using the camera and LiDAR. In this method, global localization includes two processes: coarse localization and fine localization. In coarse localization, existing features from a pre-trained convolutional neural network (CNN) are used for determining the candidate positions based on images extracted from a monocular camera. In fine localization, the precise robot position is estimated by adaptive Monte Carlo localization based on LiDAR, where the candidate position is used as a seed for initial random sampling. Gengyu Ge et al. [146] proposed a method for localizing mobile robots by fusing camera and LiDAR data. The method first obtains coarse localization from the camera's text-level semantic information, and then enhances the localization by using the Monte Carlo localization (MCL) method with LiDAR data. Huang Y et al. [147] presented a global localization approach using LiDAR and dual AprilTag for enhancing the global localization of mobile robots. This method utilizes AprilTag-based localization results as a rough localization for the adaptive Monte Carlo global localization algorithm. Based on this rough localization, precisely located particles are generated, which enhances the efficiency, as well as the success rate during global localization.

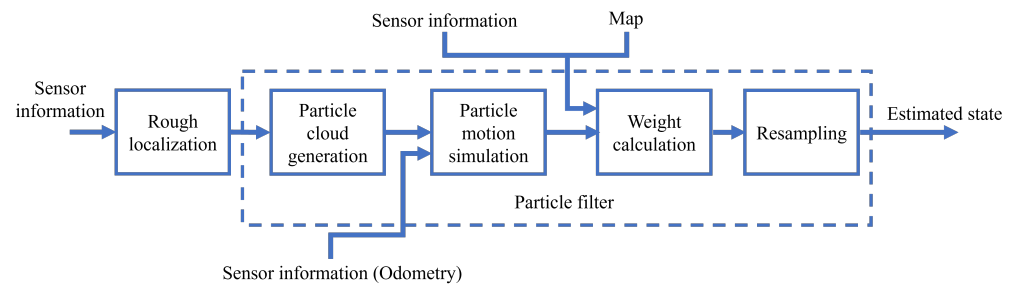


Figure 8. Multi-sensor fusion algorithm based on particle filter.

3.3. Multi-Sensor Fusion Algorithm Based on Neural Network

In recent years, researchers have introduced machine learning technology to mobile robots, which has led to the rapid development of mobile robotic systems [148,149]. Neural networks have also been widely studied and applied in multi-sensor fusion for mobile robots. Neural network-based methods can automatically extract features from data, avoiding the difficulty and uncertainty of manually designing features while having an improved generalization ability and robustness, as shown in Figure 9. Compared to Kalman filter-based methods, neural network-based methods are more capable of handling nonlinear problems and achieving higher accuracy.

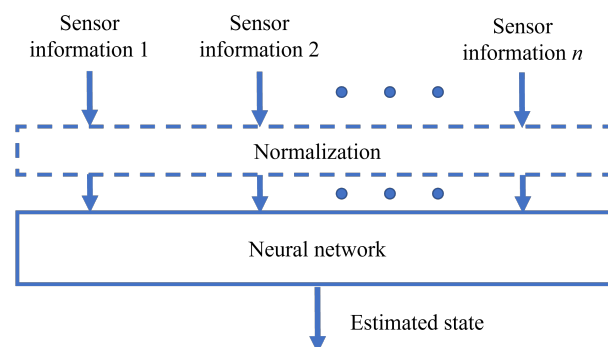
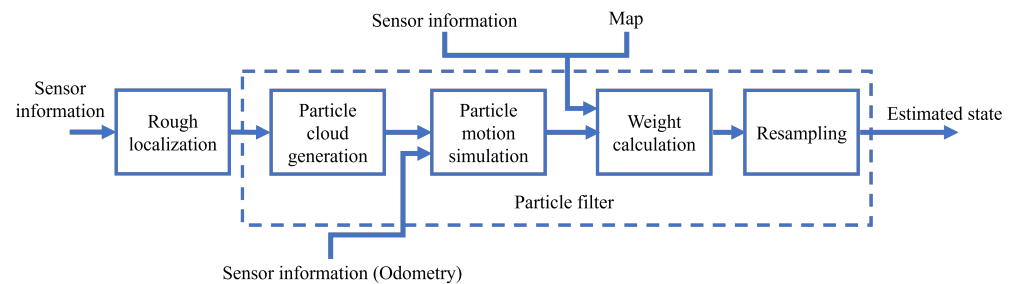


Figure 9. Multi-sensor fusion algorithm based on neural network.

Carlos Eduardo Magrin et al. [150] proposed a hierarchical sensor fusion (HSF) method with an artificial neural network (ANN) for mobile robot self-localization. This method uses

为了提高全球定位效率，研究人员对传感器融合进行了大量研究。这涉及到粗定位和精确定位，如图 8 所示。Song Xu 等人。[145]设计了一种使用相机和激光雷达的全局定位方法。在该方法中，全局定位包括粗定位和精定位两个过程。在粗定位中，预训练的卷积神经网络（CNN）的现有特征用于根据从单目相机提取的图像确定候选位置。在精细定位中，通过基于激光雷达的自适应蒙特卡罗定位来估计精确的机器人位置，其中候选位置被用作初始随机采样的种子。葛庚宇等人。[146]提出了一种通过融合相机和激光雷达数据来定位移动机器人的方法。该方法首先从相机的文本级语义信息中获得粗略定位，然后使用蒙特卡罗定位（MCL）方法和激光雷达数据来增强定位。黄Y等。[147]提出了一种使用 LiDAR 和双 AprilTag 的全球定位方法，以增强移动机器人的全球定位。该方法利用基于 AprilTag 的定位结果作为自适应蒙特卡罗全局定位算法的粗略定位。基于这种粗略定位，生成精确定位的粒子，从而提高了全局定位的效率和成功率。



数字 8. 基于粒子的多传感器融合算法

筛选。

3.3. Multi-Sensor Fusion Algorithm Based on Neural Network

近年来，研究人员将机器学习技术引入移动机器人，带动了移动机器人系统的快速发展[148,149]。神经网络在移动机器人的多传感器融合中也得到了广泛的研究和应用。基于神经网络的方法可以自动从数据中提取特征，避免了手动设计特征的困难和不确定性，同时具有提高的泛化能力和鲁棒性，如图9所示。与基于卡尔曼滤波器的方法相比，基于神经网络的方法更能够处理非线性问题并获得更高的精度。

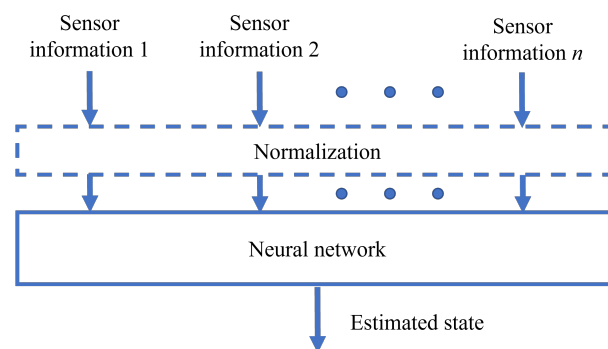


图 9. 基于神经网络的多传感器融合算法

工作。

卡洛斯·爱德华多·马格林等人。[150]提出了一种带有人工神经网络（ANN）的分层传感器融合（HSF）方法，用于移动机器人的自定位。该方法使用

a two-level multilayer perceptron (MLP) to fuse the normalized sensor data of the sonar octagon, digital compass, and wireless network signal strength measure. Li et al. [151] applied neural network fusion of LiDAR and an encoder for mobile robot localization. In this method, LiDAR data and odometer data obtained through an encoder are fed into a three-layer neural network for fusion to improve the mobile robot's positioning accuracy. Chi Li et al. [152] presented a deep learning-based approach to localizing a mobile robot using a 2D LiDAR and an IMU. The method utilizes a recursive convolutional neural network (RCNN) to fuse laser and inertial data and estimate the scan-to-scan attitude. The approach demonstrates greater robustness than traditional geometric-based methods in challenging situations, such as high angular speeds. Jieun Lee et al. [153] provided a deep neural network architecture (FusionLoc) for mobile robot relocalization with a camera and 2D LiDAR. This method contains two feature extraction modules, i.e., a multi-head self-attention (MHSA) module and a regression one. To be more specific, the feature extraction module extracts features from RGB images and LiDAR point clouds, respectively; the MHSA module performs feature fusion; and the regression module realizes position estimation. This approach is capable of achieving improved performance as compared with traditional end-to-end relocalization approaches that only acquire data from one sensor. A fingerprint-assisted localization scheme for mobile robots is explored, which fuses RSSI and magnetometer measurements [154]. However, the accuracy and stability of RSSI-based fingerprint localization methods can be affected by various environmental disturbances. The magnetic field strength can provide information about the level of disturbances. The method uses a fusion of magnetometer measurements and fingerprint data from RSSI for mobile robot localization through multilayer perceptron (MLP) feedforward neural networks. Andres J. Barreto-Cubero et al. [155] fused an ultrasonic sensor and a stereo camera with 2D LiDAR through the artificial neural network to improve obstacle detection in mobile robots by using the capability of ultrasonic sensors to detect glass and the ability to accurately detect objects in the 3D environment of the stereo camera.

3.4. Other Multi-Sensor Fusion Algorithms

The weighted average method is a direct and effective approach that simplifies the data fusion process during algorithm design and ensures real-time calculations. The core idea of the method is how to weight multiple sensors and how to determine the correction method. Wen et al. [156] combine an incremental encoder with a camera to calculate the encoder-based local position and globally update the position through the similarity of the camera to the environment. Jiang et al. [157] achieved the combination of feature point cloud maps and grid maps by establishing joint error co-visual between LiDAR and the monocular camera. Zheng et al. [81] divided the SLAM system into two subsystems, i.e., VIO and LIO. The coupled relationship between the two subsystems is achieved by sharing the point map data between the two to achieve accurate position estimation. However, because the weighted average method does not take into account the relationship between the front and rear states of the robot, its anti-disturbance ability is not excellent in the face of highly dynamic environments. Kosisochukwu Pal Nnoli et al. [43] fused infrared and ultrasonic sensors to detect obstacles. Their approach uses an error-filtering covariance and averaging algorithm to logically fuse distance measurements from a pair of infrared and ultrasonic sensors to track the proximity of environmental obstacles within a 180-degree range in front of a mobile robot.

Graph optimization algorithms in SLAM are also used as a framework for multi-sensor fusion. Ran Liu et al. [158] presented a map-building scheme for UWB, LiDAR, and odometry. In this method, a graph optimization framework with two optimization processes is designed to handle the fusion problem of UWB ranging information, odometry, and LiDAR information. The first optimization process takes the poses of the UWB nodes as vertices in the graph, and the edges of the UWB ranges and odometers are used as constraints, and roughly optimized robot trajectories are obtained. The second one integrates the edges determined based on LiDAR ICP scan matching to obtain the accurate robot trajectory.

两级多层感知器 (MLP), 用于融合声纳八边形、数字罗盘和无线网络信号强度测量的归一化传感器数据。李等人。[151]应用激光雷达和编码器的神经网络融合进行移动机器人定位。该方法将激光雷达数据和通过编码器获得的里程计数据输入三层神经网络进行融合, 以提高移动机器人的定位精度。李驰等人。[152]提出了一种基于深度学习的方法, 使用 2D LiDAR 和 IMU 来定位移动机器人。该方法利用递归卷积神经网络 (RCNN) 融合激光和惯性数据并估计扫描到扫描的姿态。在具有挑战性的情况下, 例如高角速度, 该方法比传统的基于几何的方法表现出更高的鲁棒性。李杰恩等人。[153] 提供了一种深度神经网络架构 (FusionLoc), 用于使用相机和 2D LiDAR 进行移动机器人重定位。该方法包含两个特征提取模块, 即多头自注意力 (MHSA) 模块和回归模块。具体来说, 特征提取模块分别从 RGB 图像和 LiDAR 点云中提取特征; MHSA 模块进行特征融合; 回归模块实现位置估计。与仅从一个传感器获取数据的传统端到端重定位方法相比, 这种方法能够实现更高的性能。探索了一种移动机器人的指纹辅助定位方案, 该方案融合了 RSSI 和磁力计测量[154]。然而, 基于 RSSI 的指纹定位方法的准确性和稳定性会受到各种环境干扰的影响。磁场强度可以提供有关干扰程度的信息。该方法融合磁力计测量值和 RSSI 的指纹数据, 通过多层感知器 (MLP) 前馈神经网络进行移动机器人定位。安德烈斯·J·巴雷托-库韦罗 (Andres J. Barreto-Cubero) 等人。[155]通过人工神经网络将超声波传感器和立体相机与 2D LiDAR 融合在一起, 利用超声波传感器检测玻璃的能力和立体相机在 3D 环境中准确检测物体的能力来改进移动机器人的障碍物检测。

3.4. Other Multi-Sensor Fusion Algorithms

加权平均法是一种直接有效的方法, 可以简化算法设计时的数据融合过程, 保证计算的实时性。该方法的核心思想是如何对多个传感器进行加权以及如何确定校正方法。文等人。[156]将增量编码器与相机结合起来, 计算基于编码器的局部位置, 并通过相机与环境的相似性全局更新位置。江等人。[157]通过在 LiDAR 和单目相机之间建立联合误差共视, 实现了特征点云图和网格图的组合。郑等人。[81]将 SLAM 系统分为两个子系统, 即 VIO 和 LIO。通过共享两者之间的点图数据来实现两个子系统之间的耦合关系, 从而实现精确的位置估计。但由于加权平均法没有考虑机器人前后状态的关系, 因此面对高动态环境时其抗干扰能力并不出色。Kosisochukwu Pal Nnoli 等人。[43]融合红外和超声波传感器来检测障碍物。他们的方法使用误差过滤协方差和平均算法来逻辑地融合一对红外和超声波传感器的距离测量值, 以跟踪移动机器人前方 180 度范围内的环境障碍物的接近程度。

SLAM 中的图优化算法也被用作多传感器融合的框架。刘然等人。[158]提出了一种用于 UWB、LiDAR 和里程计测量的地图构建方案。在该方法中, 设计了具有两个优化过程的图优化框架来处理 UWB 测距信息、里程计和 LiDAR 信息的融合问题。第一个优化过程以 UWB 节点的位姿为图中的顶点, 以 UWB 范围的边和里程计为约束, 得到粗略优化的机器人轨迹。第二个集成基于 LiDAR ICP 扫描匹配确定的边缘以获得准确的机器人轨迹。

Along this line, then, the resulted trajectory can be incorporated into the LiDAR data, yielding the occupancy map of the industrial scenes. The graph optimization process can better deal with the influence of nonlinear systems on the robot. However, the algorithm's high calculation complexity, caused by the need to traverse previous state variables, can make real-time updating difficult during robot navigation. The robot's state is generally optimized when loop closure and relocalization are triggered.

4. Discussions and Future Trends

4.1. Discussions

The perception system of an AMR offers information about the robot's association with the environment, which is essential for successful navigation. To ensure the indoor mobile robot completes the navigation task successfully, it is essential to reasonably select and optimize the sensors and algorithms used in the perception system based on the specific needs of the scenario.

IMU-based trajectory inference methods can only be used for short periods of time and can be used to assist other localization systems. For mobile robot localization, beacon-based positioning systems (including RF sensors, ultrasonic sensors, and infrared sensors) can provide stable and accurate absolute position information. However, they require regular installation and maintenance, and widespread deployment can lead to increased costs. LiDAR, or vision cameras, can be used for accurate and stable positioning. However, their localization relies on environmental information extracted by the sensors, making them susceptible to changes in the environment. For scenarios with frequent changes, beacon-based localization methods are more appropriate.

For the SLAM of a mobile robot, both LiDAR and vision cameras have their own advantages. Laser point cloud data require less computation, are easy to compute, are accurate in ranging, and are less susceptible to large changes in lighting. However, the environmental information collected by LiDAR is not sufficient, and the map constructed using 2D LiDAR only contains distance information about the environment. The rotating mechanism of the device is simple, which negatively impacts the stability of its internal structure over time. Vision SLAM provides a more detailed description of the environment, including texture information, which enhances accuracy and enables the recognition of differences between environments. While laser SLAM has limitations, vision SLAM, equipped with a camera, offers greater stability in terms of internal structure. Mobile robots with only a single camera as a sensor in indoor environments are limited by environmental conditions and lack robustness.

For obstacle detection, vision cameras are effective for recognizing obstacles as they provide rich environmental information. Ultrasonic sensors and LiDAR are precise in measuring the distance and position of obstacles.

4.2. Future Trends

Benefiting from computer science, artificial intelligence, and visual approaches, perception technology will develop rapidly, and the following aspects are worth emphasizing:

- 1: Multi-source Information Fusion.** The study and development of mobile robots have extended their possible applications, along with increased demands for their usage. However, a single sensor may struggle to satisfy the navigation requirements of complex scenes due to its inherent limitations. Along this line, then, multi-sensor fusion becomes a crucial area of research and development for indoor AMR. Multi-sensor fusion can improve the accuracy, efficiency, and stability of localization, map building, and obstacle detection by providing more comprehensive environmental information to the perception system. Multi-sensor fusion provides accurate and comprehensive information to the decision-making system, enabling mobile robots to adapt better to complex environments and complete navigation tasks more efficiently and robustly.

然后，沿着这条线，所得到的轨迹可以合并到激光雷达数据中，从而生成工业场景的占用图。图优化过程可以更好地处理非线性系统对机器人的影响。然而，由于需要遍历先前的状态变量，导致该算法的计算复杂度较高，导致机器人导航过程中的实时更新变得困难。当触发闭环和重定位时，机器人的状态通常会得到优化。

4. 讨论和未来趋势

4.1. Discussions

AMR 的感知系统提供有关机器人与环境关联的信息，这对于成功导航至关重要。为了保证室内移动机器人顺利完成导航任务，需要根据场景的具体需求合理选择和优化感知系统所使用的传感器和算法。

基于 IMU 的轨迹推断方法只能在短时间内使用，可用于辅助其他定位系统。对于移动机器人定位，基于信标的定位系统（包括射频传感器、超声波传感器和红外传感器）可以提供稳定、准确的绝对位置信息。然而，它们需要定期安装和维护，广泛部署可能会导致成本增加。激光雷达（LiDAR）或视觉相机可用于精确稳定的定位。然而，它们的定位依赖于传感器提取的环境信息，这使得它们容易受到环境变化的影响。对于变化频繁的场景，基于信标的定位方法更为合适。

对于移动机器人的SLAM，激光雷达和视觉相机都有各自的优势。激光点云数据需要计算量少，易于计算，测距准确，并且不易受光照大幅变化的影响。然而，激光雷达收集的环境信息并不充分，使用二维激光雷达构建的地图仅包含环境的距离信息。该装置的旋转机构简单，随着时间的推移对其内部结构的稳定性产生负面影响。视觉SLAM提供了更详细的环境描述，包括纹理信息，这提高了准确性并能够识别环境之间的差异。虽然激光SLAM有局限性，但配备摄像头的视觉SLAM在内部结构方面提供了更大的稳定性。在室内环境中仅使用单个摄像头作为传感器的移动机器人受到环境条件的限制且缺乏鲁棒性。

对于障碍物检测，视觉相机可以有效地识别障碍物，因为它们提供了丰富的环境信息。超声波传感器和激光雷达可以精确测量障碍物的距离和位置。

4.2. Future Trends

受益于计算机科学、人工智能和视觉方法，感知技术将快速发展，以下几个方面值得强调：

1: 多源信息融合。移动机器人的研究和开发扩展了其可能的应用范围，同时也增加了对其使用的需求。然而，单个传感器由于其固有的局限性，可能难以满足复杂场景的导航需求。因此，多传感器融合成为室内 AMR 研发的关键领域。多传感器融合可以通过向感知系统提供更全面的环境信息来提高定位、地图构建和障碍物检测的准确性、效率和稳定性。多传感器融合为决策系统提供准确、全面的信息，使移动机器人能够更好地适应复杂环境，更高效、鲁棒地完成导航任务。

- 2: **Flexible and efficient optimization strategies.** When mobile robots work in various complex and dynamic environments, there is a large amount of unknown model noise and roughness in the observation information obtained by sensors. To ensure the efficiency, stability, and accuracy of the AMR system, appropriate optimization strategies should be explored based on the specific environment and dynamic changes. In addition, there are many complex nonlinear operations in the optimization process. To improve computational efficiency when dealing with large-scale or multi-modal data, methods such as parallel computing and distributed computing can be adopted.
- 3: **Integration with neural networks.** Since neural networks are able to enhance the stability of mobile robot systems, the combination of relevant neural networks and mobile robots has attracted significant interest. Additionally, the self-learning capabilities of neural networks can address sensor interference and external environmental factors, thereby improving localization anti-interference ability. However, current neural network-based algorithms still encounter limitations for localizing in various indoor scenarios. When employing the same neural network algorithm, the mobile robot experiences significant errors in varying environments. Further research should explore the combination of additional sensors and the application of more complex neural network structures to meet the navigation requirements in diverse indoor environments. With the emergence of large-scale language models and advancements in chip technology, neural network-based approaches are expected to rapidly expand in the future.

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References

1. Qu, Y.; Yang, M.; Zhang, J.; Xie, W.; Qiang, B.; Chen, J. An Outline of Multi-Sensor Fusion Methods for Mobile Agents Indoor Navigation. *Sensors* **2021**, *21*, 1605. <https://doi.org/10.3390/s21051605>.
2. Bose, D.; Mohan, K.; Cs, M.; Yadav, M.; Saini, D.K. Review of Autonomous Campus and Tour Guiding Robots with Navigation Techniques. *Aust. J. Mech. Eng.* **2022**, *21*, 1580–1590. <https://doi.org/10.1080/14484846.2021.2023266>.
3. Alatisse, M.B.; Hancke, G.P. A Review on Challenges of Autonomous Mobile Robot and Sensor Fusion Methods. *IEEE Access* **2020**, *8*, 39830–39846. <https://doi.org/10.1109/ACCESS.2020.2975643>.
4. Rubio, F.; Valero, F.; Llopis-Albert, C. A review of mobile robots: Concepts, methods, theoretical framework, and applications. *Int. J. Adv. Robot. Syst.* **2019**, *16*, 172988141983959. <https://doi.org/10.1177/1729881419839596>.
5. Kortenkamp, D. Perception for mobile robot navigation: A survey of the state of the art. In Proceedings of the InDual-Use Space Technology Transfer Conference and Exhibition, Houston, TX, USA, 1 May 1994; Volume 2.
6. Obeidat, H.; Shuaib, W.; Obeidat, O.; Abd-Alhameed, R. A Review of Indoor Localization Techniques and Wireless Technologies. *Wirel. Pers. Commun.* **2021**, *119*, 289–327. <https://doi.org/10.1007/s11277-021-08209-5>.
7. Panigrahi, P.K.; Bisoy, S.K. Localization strategies for autonomous mobile robots: A review. *J. King Saud Univ. Comput. Inf. Sci.* **2022**, *34*, 6019–6039. <https://doi.org/10.1016/j.jksuci.2021.02.015>.
8. Huang, J.; Junginger, S.; Liu, H.; Thurrow, K. Indoor Positioning Systems of Mobile Robots: A Review. *Robotics* **2023**, *12*, 47. <https://doi.org/10.3390/robotics12020047>.
9. Kim Geok, T.; Zar Aung, K.; Sandar Aung, M.; Thu Soe, M.; Abdaziz, A.; Pao Liew, C.; Hossain, F.; Tso, C.P.; Yong, W.H. Review of Indoor Positioning: Radio Wave Technology. *Appl. Sci.* **2020**, *11*, 279. <https://doi.org/10.3390/app11010279>.
10. Han, X.; Li, S.; Wang, X.; Zhou, W. Semantic Mapping for Mobile Robots in Indoor Scenes: A Survey. *Information* **2021**, *12*, 92. <https://doi.org/10.3390/info12020092>.
11. Chen, W.; Zhou, C.; Shang, G.; Wang, X.; Li, Z.; Xu, C.; Hu, K. SLAM Overview: From Single Sensor to Heterogeneous Fusion. *Remote Sens.* **2022**, *14*, 6033. <https://doi.org/10.3390/rs14236033>.

2: 灵活高效的优化策略。当移动机器人在各种复杂、动态的环境中工作时, 传感器获得的观测信息中存在大量的未知模型噪声和粗糙度。为保证AMR系统的高效、稳定、准确, 应根据具体环境和动态变化探索合适的优化策略。此外, 优化过程中还存在许多复杂的非线性运算。为了提高处理大规模或多模态数据时的计算效率, 可以采用并行计算、分布式计算等方法。3: 与神经网络集成。由于神经网络能够增强移动机器人系统的稳定性, 相关神经网络与移动机器人的结合引起了人们的极大兴趣。此外, 神经网络的自学习能力可以解决传感器干扰和外部环境因素, 从而提高定位抗干扰能力。然而, 当前基于神经网络的算法在各种室内场景中的定位仍然遇到限制。当采用相同的神经网络算法时, 移动机器人在不同的环境中会遇到显着的错误。进一步的研究应探索附加传感器的组合和更复杂的神经网络结构的应用, 以满足不同室内环境的导航要求。随着大规模语言模型的出现和芯片技术的进步, 基于神经网络的方法有望在未来迅速扩展。

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参考

1. 曲Y.;杨, M. 张, J. 谢W.; 强, B. Chen, J. 移动代理室内导航的多传感器融合方法概述。 *Sensors* 2021, 21, 1605. <https://doi.org/10.3390/s21051605>.
2. 博斯, D.; 莫汉, K.; Cs, M. ; 亚达夫, M.; 赛尼, D.K. 自主校园和带导航技术的导游机器人综述。 *Aust. J. Mech. Eng.* 2022 年, 21, 1580–1590. <https://doi.org/10.1080/14484846.2021.2023266>.
3. 阿拉蒂斯, M.B.; 汉克, G.P. 自主移动机器人和传感器融合方法的挑战综述。 *IEEE Access* 2020, 8, 39830–39846. <https://doi.org/10.1109/ACCESS.2020.2975643>.
4. 卢比奥, F.; 瓦莱罗, F.; Llopis-Albert, C. 移动机器人综述: 概念、方法、理论框架和应用。 *Int. J. Adv. Robot. Syst.* 2019, 16, 17298814 1983959. <https://doi.org/10.1177/1729881419839596>.
5. Kortenkamp, D. 移动机器人导航的感知: 对现有技术的调查。1994年5月1日美国德克萨斯州休斯敦举行的双用途空间技术转让会议和展览的会议记录; 第2卷。
6. Obeidat, H.; 舒艾布, W.; 奥贝达特, O.; Abd-Alhameed, R. 室内定位技术和无线技术综述。 *Wirel. Pers. Commun.* 2021, 119, 289–327. <https://doi.org/10.1007/s11277-021-08209-5>.
7. 帕尼格拉希 (Panigrahi), P.K.; 比索伊, S.K. 自主移动机器人的本地化策略: 综述。 *J. King Saud Univ. Comput. Inf. Sci.* 2022, 34, 6019–6039. <https://doi.org/10.1016/j.jksuci.2021.02.015>.
8. 黄杰; 容金格, S.; 刘, H.; Thurow, K. 移动机器人室内定位系统: 综述。 *Robotics* 2023 年, 12, 47. <https://doi.org/10.3390/robotics12020047>.
9. Kim Geok, T.; 扎尔昂, K.; 桑达尔·昂, M.; Thu Soe, M.; 阿布达齐兹, A.; 保刘, C.; 侯赛因, F.; 曹, C.P.; 勇W.H. 室内定位回顾: 无线电波技术。 *Appl. Sci.* 2020, 11, 279. <https://doi.org/10.3390/app11010279>.
10. 韩X.; 李, S. 王X; Zhou, W. 室内场景中移动机器人的语义映射: 一项调查。 *Information* 2021 年, 12, 92. <https://doi.org/10.3390/info12020092>.
11. 陈W.; 周, C.; 尚, G. 王X; 李, Z. 徐, C.; Hu, K. SLAM 概述: 从单一传感器到异构融合。 *Remote Sens.* 2022, 14, 6033. <https://doi.org/10.3390/rs14236033>.

12. Chen, W.; Shang, G.; Ji, A.; Zhou, C.; Wang, X.; Xu, C.; Li, Z.; Hu, K. An Overview on Visual SLAM: From Tradition to Semantic. *Remote Sens.* **2022**, *14*, 3010. <https://doi.org/10.3390/rs14133010>.
13. Chou, C.; Li, H.; Song, D. Encoder-Camera-Ground Penetrating Radar Sensor Fusion: Bimodal Calibration and Subsurface Mapping. *IEEE Trans. Robot.* **2021**, *37*, 67–81. <https://doi.org/10.1109/TRO.2020.3010640>.
14. Shi, H.; Shi, L.; Xu, M.; Hwang, K.S. End-to-End Navigation Strategy With Deep Reinforcement Learning for Mobile Robots. *IEEE Trans. Ind. Inform.* **2020**, *16*, 2393–2402. <https://doi.org/10.1109/TII.2019.2936167>.
15. Zeng, H.; Song, X.; Jiang, S. Multi-Object Navigation Using Potential Target Position Policy Function. *IEEE Trans. Image Process.* **2023**, *32*, 2608–2619. <https://doi.org/10.1109/TIP.2023.3263110>.
16. Wang, Y.; Li, X.; Zhang, J.; Li, S.; Xu, Z.; Zhou, X. Review of wheeled mobile robot collision avoidance under unknown environment. *Sci. Prog.* **2021**, *104*, 003685042110377. <https://doi.org/10.1177/00368504211037771>.
17. Qin, H.; Shao, S.; Wang, T.; Yu, X.; Jiang, Y.; Cao, Z. Review of Autonomous Path Planning Algorithms for Mobile Robots. *Drones* **2023**, *7*, 211. <https://doi.org/10.3390/drones7030211>.
18. Choi, B.S.; Lee, J.J. Mobile robot localization in indoor environment using RFID and sonar fusion system. In Proceedings of the 2009 IEEE/RSJ International Conference on Intelligent Robots and Systems, St. Louis, MO, USA, 10–15 October 2009; pp. 2039–2044. <https://doi.org/10.1109/IROS.2009.5354104>.
19. Ouyang, M.; Cao, Z.; Guan, P.; Li, Z.; Zhou, C.; Yu, J. Visual-Gyroscope-Wheel Odometry with Ground Plane Constraint for Indoor Robots in Dynamic Environment. *IEEE Sens. Lett.* **2021**, *5*, 1–4. <https://doi.org/10.1109/LSSENS.2021.3057088>.
20. Morar, A.; Moldoveanu, A.; Mocanu, I.; Moldoveanu, F.; Radoi, I.E.; Asavei, V.; Gradinaru, A.; Butean, A. A Comprehensive Survey of Indoor Localization Methods Based on Computer Vision. *Sensors* **2020**, *20*, 2641. <https://doi.org/10.3390/s20092641>.
21. Li, R.; Du, Z.; Zhao, Y.; Liu, S. Design and implementation of mobile robot ultrasonic localization system. In Proceedings of the 2016 Chinese Control and Decision Conference (CCDC), Yinchuan, China, 28–30 May 2016; pp. 5347–5352. <https://doi.org/10.1109/CCDC.2016.7531954>.
22. Chan, T.H.; Hesse, H.; Ho, S.G. LiDAR-Based 3D SLAM for Indoor Mapping. In Proceedings of the 2021 7th International Conference on Control, Automation and Robotics (ICCAR), Singapore, 23–26 April 2021; pp. 285–289. <https://doi.org/10.1109/ICCAR52225.2021.9463503>.
23. Borodacz, K.; Szczepański, C.; Popowski, S. Review and selection of commercially available IMU for a short time inertial navigation. *Aircr. Eng. Aerosp. Technol.* **2022**, *94*, 45–59. <https://doi.org/10.1108/AEAT-12-2020-0308>.
24. Luo, L.; Feng, Z.; Hong-Wei, W.; Long, C. Crawler Robot Indoor Positioning Based on a Combination of Bluetooth and IMU. In Proceedings of the 2022 6th International Conference on Robotics, Control and Automation (ICRCA), Xiamen, China, 26–28 February 2022; pp. 34–39. <https://doi.org/10.1109/ICRCA55033.2022.9829003>.
25. Alatisse, M.; Hancke, G. Pose Estimation of a Mobile Robot Based on Fusion of IMU Data and Vision Data Using an Extended Kalman Filter. *Sensors* **2017**, *17*, 2164. <https://doi.org/10.3390/s17102164>.
26. Zhang, S.; Tan, X.; Wu, Q. Self-Positioning for Mobile Robot Indoor Navigation Based on Wheel Odometry, Inertia Measurement Unit and Ultra Wideband. In Proceedings of the 2021 5th International Conference on Vision, Image and Signal Processing (ICVISP), Kuala Lumpur, Malaysia, 18–20 December 2021; pp. 105–110. <https://doi.org/10.1109/ICVISP54630.2021.00027>.
27. Lee, S.J.; Lee, K.; Song, J.B. Development of advanced grid map building model based on sonar geometric reliability for indoor mobile robot localization. In Proceedings of the 2014 11th International Conference on Ubiquitous Robots and Ambient Intelligence (URAI), Kuala Lumpur, Malaysia, 12–15 November 2014; pp. 292–297. <https://doi.org/10.1109/URAI.2014.7057437>.
28. Liu, H.; Sun, F.; Fang, B.; Zhang, X. Robotic Room-Level Localization Using Multiple Sets of Sonar Measurements. *IEEE Trans. Instrum. Meas.* **2017**, *66*, 2–13. <https://doi.org/10.1109/TIM.2016.2618978>.
29. Liu, Y.; Fan, R.; Yu, B.; Bocus, M.J.; Liu, M.; Ni, H.; Fan, J.; Mao, S. Mobile Robot Localisation and Navigation Using LEGO NXT and Ultrasonic Sensor. In Proceedings of the 2018 IEEE International Conference on Robotics and Biomimetics (ROBIO), Kuala Lumpur, Malaysia, 12–15 December 2018; pp. 1088–1093. <https://doi.org/10.1109/ROBIO.2018.8665350>.
30. Shen, M.; Wang, Y.; Jiang, Y.; Ji, H.; Wang, B.; Huang, Z. A New Positioning Method Based on Multiple Ultrasonic Sensors for Autonomous Mobile Robot. *Sensors* **2019**, *20*, 17. <https://doi.org/10.3390/s20010017>.
31. Hsu, C.C.; Chen, H.C.; Wong, C.C.; Lai, C.Y. Omnidirectional Ultrasonic Localization for Mobile Robots. *Sens. Mater.* **2022**, *34*, 453. <https://doi.org/10.18494/SAM3419>.
32. Yuan, S.; Guo, P.; Han, X.; Luan, F.; Zhang, F.; Liu, T.; Mao, H. DSMT-Based Ultrasonic Detection Model for Estimating Indoor Environment Contour. *IEEE Trans. Instrum. Meas.* **2020**, *69*, 4002–4014. <https://doi.org/10.1109/TIM.2019.2943715>.
33. Takai, H.; Miyake, M.; Okuda, K.; Tachibana, K. A simple obstacle arrangement detection algorithm for indoor mobile robots. In Proceedings of the 2010 2nd International Asia Conference on Informatics in Control, Automation and Robotics (CAR 2010), Wuhan, China, 6–7 March 2010; pp. 110–113. <https://doi.org/10.1109/CAR.2010.5456759>.
34. Jean, J.H.; Wang, J.L. Development of an indoor patrol robot based on ultrasonic and vision data fusion. In Proceedings of the 2013 IEEE International Conference on Mechatronics and Automation, Takamatsu, Japan, 4–7 August 2013; pp. 1234–1238. <https://doi.org/10.1109/ICMA.2013.6618090>.
35. Grami, T.; Sghaier Tlili, A. Indoor Mobile Robot Localization based on a Particle Filter Approach. In Proceedings of the 2019 19th International Conference on Sciences and Techniques of Automatic Control and Computer Engineering (STA), Sousse, Tunisia, 24–26 March 2019; pp. 47–52. <https://doi.org/10.1109/STA.2019.8717214>.

- 12.陈W.; 尚, G. 吉, A.; 周, C.; 王X.; 徐, C.; 李, Z. Hu, K. 视觉 SLAM 概述: 从传统到语义。 *Remote Sens.* 2022, 14, 3010。 <https://doi.org/10.3390/rs14133010>.
- 13.Chou, C.; 李, H. Song, D. 编码器-相机-探地雷达传感器融合: 双模态校准和地下测绘。 *IEEE Trans. Robot.* 2021, 37, 67–81。 <https://doi.org/10.1109/TRO.2020.3010640>.
- 14.石H.; 施L.; 徐, M. 黄K.S.具有深度强化学习的移动机器人端到端导航策略。 *IEEE Trans. Ind. Inform.* 2020, 16, 2393–2402。 <https://doi.org/10.1109/TII.2019.2936167>.
- 15.曾 H.; 宋 X.; Jiang, S. 使用潜在目标位置策略函数的多目标导航。 *IEEE Trans. Image Process.* 2023, 32, 2608–2619。 <https://doi.org/10.1109/TIP.2023.3263110>.
- 16.王Y.;李, X. 张, J. 李, S. 徐, Z. 未知环境下轮式移动机器人防撞综述。 *Sci. Prog.* 2021 年, 104, 003685042 110377。 <https://doi.org/10.1177/00368504211037771>.
- 17.秦H.; 邵, S. 王, T. 于X. 蒋, Y. 曹, Z. 移动机器人自主路径规划算法综述。 *Drones* 2023 年, 7, 211。 <https://doi.org/10.3390/drones7030211>.
18. Choi, 理学学士; 李, J.J.使用 RFID 和声纳融合系统在室内环境中进行移动机器人定位。 2009 年 IEEE/RSJ 国际智能机器人和系统会议论文集, 美国密苏里州圣路易斯, 2009 年 10 月 10 日至 15 日; 第 2039–2044 页。 <https://doi.org/10.1109/IROS.2009.5354104>.
- 19.欧阳M.; 曹, Z. 关, P. 李, Z. 周, C.; Yu, J. 动态环境中室内机器人具有地平线约束的视觉陀螺仪里程计。 *IEEE Sens. Lett.* 2021 年, 5, 1–4。 <https://doi.org/10.1109/LENS.2021.3057088>.
- 20.莫拉尔, A.; 摩尔多瓦努, A.; 莫卡努, I.; 摩尔多瓦努, F.; 拉多伊, I.E.; 阿萨维, V.; 格拉迪纳鲁, A.; Butean, A. 基于计算机视觉的室内定位方法的综合综述。 *Sensors* 2020, 20, 2641。 <https://doi.org/10.3390/s20092641>.
- 21.李, R.; 杜, Z. 赵, Y. 刘S.移动机器人超声定位系统的设计与实现。 2016 年中国控制与决策会议 (CCDC) 论文集, 中国银川, 2016 年 5 月 28–30 日; 第 5347–5352 页。 <https://doi.org/10.1109/CCDC.2016.7531954>.
- 22.陈T.H.; 黑塞, H.; Ho, S.G. 基于 LiDAR 的 3D SLAM 用于室内测绘。 2021 年第七届国际控制、自动化和机器人会议 (ICCAR) 会议记录, 新加坡, 2021 年 4 月 23–26 日; 第 285–289 页。 <https://doi.org/10.1109/ICCAR522.2021.9463503>.
- 23.博罗达茨, K.; Szczepański, C.; Popowski, S. 用于短时惯性导航的商用 IMU 的审查和选择。 *Aircr. Eng. Aerosp. Technol.* 2022, 94, 45–59。 <https://doi.org/10.1108/AEAT-12-2020-0308>.
- 24.罗L.; 冯, Z.; 洪伟, W.; Long, C. 基于蓝牙和IMU组合的履带机器人室内定位。 2022 年第六届机器人、控制与自动化国际会议 (ICRCA) 论文集, 中国厦门, 2022 年 2 月 2 6–28 日; 第 34–39 页。 <https://doi.org/10.1109/ICRCA55033.2022.9829003>.
- 25.阿拉蒂斯, M.; Hancke, G. 使用扩展卡尔曼滤波器基于 IMU 数据和视觉数据融合的移动机器人姿态估计。 *Sensors* 2017 年, 17, 2164。 <https://doi.org/10.3390/s17102164>.
- 26.张S.; 谭, X. 吴 Q. 基于车轮里程计、惯性测量单元和超宽带的移动机器人室内导航自定位。 2021 年第五届视觉、图像和信号处理国际会议 (ICVISP) 论文集, 马来西亚吉隆坡, 2021 年 12 月 18–20 日; 第 105–110 页。 <https://doi.org/10.1109/ICVISP54630.2021.00027>.
- 27.李, S.J.; 韭葱.; Song, J.B. 基于声纳几何可靠性的先进网格地图构建模型的开发, 用于室内移动机器人定位。 2014 年第 11 届普适机器人和环境智能国际会议 (URAI) 会议记录, 马来西亚吉隆坡, 2014 年 11 月 12–15 日; 第 292–297 页。 <https://doi.org/10.1109/URAI.2014.7057437>.
- 28.刘H.; 孙F.; 方, B.; 张, X. 使用多组声纳测量的机器人房间级定位。 *IEEE Trans. Instrum. Meas.* 2017 年, 66, 2–13。 <https://doi.org/10.1109/TIM.2016.2618978>.
- 29.刘, Y.; 范, R. 于, B.; 博库斯, M.J.; 刘, M.; NIH.; 范, J. Mao, S. 使用乐高 NXT 和超声波传感器的移动机器人定位和导航。 2018 年 IEEE 国际机器人与仿生学会议 (ROBIO) 会议记录, 马来西亚吉隆坡, 2018 年 12 月 12 日至 15 日; 第 1088–1093 页。 <https://doi.org/10.1109/ROBIO.2018.8665350>.
- 30.沉, M.; 王, Y. 蒋, Y. 吉, H. 王, B.; 黄Z. 一种基于多个超声波传感器的自主移动机器人定位新方法。 *Sensors* 2019, 20, 17。 <https://doi.org/10.3390/s20010017>.
- 31.许C.C.; 陈H.C.; 黄, C.C.; 赖春云移动机器人的全向超声波定位。 *Sens. Mater.* 2022, 34, 453。 <https://doi.org/10.18494/SAM3419>.
- 32.袁S.; 郭, P. 韩X.; 栾, F.; 张, F. 刘, T. Mao, H. 基于 DSMT 的 Estima 超声波检测模型

36. Derkach, M.; Matiuk, D.; Skarga-Bandurova, I. Obstacle Avoidance Algorithm for Small Autonomous Mobile Robot Equipped with Ultrasonic Sensors. In Proceedings of the 2020 IEEE 11th International Conference on Dependable Systems, Services and Technologies (DESSERT), Kyiv, Ukraine, 14–18 May 2020; pp. 236–241. <https://doi.org/10.1109/DESSERT50317.2020.9125019>.
37. Brassart, E.; Pegard, C.; Mouaddib, M. Localization using infrared beacons. *Robotica* **2000**, *18*, 153–161. <https://doi.org/10.1017/S0263574799001927>.
38. Krejsa, J.; Vechet, S. Infrared Beacons based Localization of Mobile Robot. *Electron. Electr. Eng.* **2012**, *117*, 17–22. <https://doi.org/10.5755/j01.eee.117.1.1046>.
39. Li, T.H.; Chang, S.J.; Tong, W. Fuzzy Target Tracking Control of Autonomous Mobile Robots by Using Infrared Sensors. *IEEE Trans. Fuzzy Syst.* **2004**, *12*, 491–501. <https://doi.org/10.1109/TFUZZ.2004.832526>.
40. Juang, J.; Yang, Y. Fuzzy Sensor Fusion and Curve Approximation for indoor Map Building. In Proceedings of the 2015 International Conference on Artificial Intelligence and Industrial Engineering, Phuket, Thailand, 26–27 July 2015. <https://doi.org/10.2991/aiie-15.2015.148>.
41. Oultiligh, A.; Ayad, H.; Pozna, C.; Mogan, G.; ELbouzekraoui, M.; Elkari, B. Obstacle Avoidance using Fuzzy Controller for Unicycle Robot. In Proceedings of the 2020 International Conference on Control, Automation and Diagnosis (ICCAD), Paris, France, 7–9 October 2020; pp. 1–6. <https://doi.org/10.1109/ICCAD49821.2020.9260553>.
42. Solano, D.M.; Grande, R.E.; Bonilla, M.N.I. PID Control and Fuzzy Logic System to the Obstacle Avoidance in an Autonomous Robot. In Proceedings of the 2021 18th International Conference on Electrical Engineering, Computing Science and Automatic Control (CCE), Mexico City, Mexico, 10–12 November 2021; pp. 1–6. <https://doi.org/10.1109/CCE53527.2021.9633079>.
43. Nnoli, K.P.; Benyeogor, M.S.; Bolu, J.I.; Olakanmi, O.O. Edge-Based Infrared-Ultrasonic Anti-Collision Radar System for Robotic Navigation: *Applications of Cost-effective Bisensory System for Obstacle Detection, Tracking, and Avoidance. In Proceedings of the 2022 IEEE International Symposium on Technologies for Homeland Security (HST), Boston, MA, USA, 14–15 November 2022; pp. 1–7. <https://doi.org/10.1109/HST56032.2022.10024985>.
44. Zhao, J.; Fang, J.; Wang, S.; Wang, K.; Liu, C.; Han, T. Obstacle Avoidance of Multi-Sensor Intelligent Robot Based on Road Sign Detection. *Sensors* **2021**, *21*, 6777. <https://doi.org/10.3390/s21206777>.
45. Habich, T.L.; Stuede, M.; Labbe, M.; Spindeldreier, S. Have I been here before? Learning to Close the Loop with LiDAR Data in Graph-Based SLAM. In Proceedings of the 2021 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM), Delft, The Netherlands, 12–16 July 2021; pp. 504–510. <https://doi.org/10.1109/AIM46487.2021.9517565>.
46. Meng, J.; Wan, L.; Wang, S.; Jiang, L.; Li, G.; Wu, L.; Xie, Y. Efficient and Reliable LiDAR-Based Global Localization of Mobile Robots Using Multiscale/Resolution Maps. *IEEE Trans. Instrum. Meas.* **2021**, *70*, 1–15. <https://doi.org/10.1109/TIM.2021.3093933>.
47. Hähnel, D.; Burgard, W.; Thrun, S. Learning compact 3D models of indoor and outdoor environments with a mobile robot. *Robot. Auton. Syst.* **2003**, *44*, 15–27. [https://doi.org/10.1016/S0921-8890\(03\)00007-1](https://doi.org/10.1016/S0921-8890(03)00007-1).
48. Wang, X.; Marcotte, R.J.; Olson, E. GLFP: Global Localization from a Floor Plan. In Proceedings of the 2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Macau, China, 3–8 November 2019; pp. 1627–1632. <https://doi.org/10.1109/IROS40897.2019.8968061>.
49. Kuang, H.; Chen, X.; Guadagnino, T.; Zimmerman, N.; Behley, J.; Stachniss, C. IR-MCL: Implicit Representation-Based Online Global Localization. *IEEE Robot. Autom. Lett.* **2023**, *8*, 1627–1634. <https://doi.org/10.1109/LRA.2023.3239318>.
50. Kim, J.; Chung, W. Localization of a Mobile Robot Using a Laser Range Finder in a Glass-Walled Environment. *IEEE Trans. Ind. Electron.* **2016**, *63*, 3616–3627. <https://doi.org/10.1109/TIE.2016.2523460>.
51. Grisetti, G.; Stachniss, C.; Burgard, W. Improved Techniques for Grid Mapping With Rao-Blackwellized Particle Filters. *IEEE Trans. Robot.* **2007**, *23*, 34–46. <https://doi.org/10.1109/TRO.2006.889486>.
52. Kohlbrecher, S.; Von Stryk, O.; Meyer, J.; Klingauf, U. A flexible and scalable SLAM system with full 3D motion estimation. In Proceedings of the 2011 IEEE International Symposium on Safety, Security, and Rescue Robotics, Kyoto, Japan, 1–5 November 2011; pp. 155–160. <https://doi.org/10.1109/SSRR.2011.6106777>.
53. Konolige, K.; Grisetti, G.; Kümmerle, R.; Burgard, W.; Limketkai, B.; Vincent, R. Efficient Sparse Pose Adjustment for 2D mapping. In Proceedings of the 2010 IEEE/RSJ International Conference on Intelligent Robots and Systems, Taipei, Taiwan, 18–22 October 2010; pp. 22–29. <https://doi.org/10.1109/IROS.2010.5649043>.
54. Hess, W.; Kohler, D.; Rapp, H.; Andor, D. Real-time loop closure in 2D LIDAR SLAM. In Proceedings of the 2016 IEEE International Conference on Robotics and Automation (ICRA), Stockholm, Sweden, 16–21 May 2016; pp. 1271–1278. <https://doi.org/10.1109/ICRA.2016.7487258>.
55. Xiang, Y.; Yang, X.Q.; Yang, W.W.; Miao, W.H. Localization and Mapping Algorithm for the Indoor Mobile Robot Based on LIDAR. *IOP Conf. Ser. Mater. Sci. Eng.* **2020**, *831*, 012021. <https://doi.org/10.1088/1757-899X/831/1/012021>.
56. Wang, H.; Wang, C.; Xie, L. Intensity-SLAM: Intensity Assisted Localization and Mapping for Large Scale Environment. *IEEE Robot. Autom. Lett.* **2021**, *6*, 1715–1721. <https://doi.org/10.1109/LRA.2021.3059567>.
57. Tian, Y.; Liu, X.; Li, L.; Wang, W. Intensity-Assisted ICP for Fast Registration of 2D-LIDAR. *Sensors* **2019**, *19*, 2124. <https://doi.org/10.3390/s19092124>.
58. Hu, W.; Zhang, K.; Shao, L.; Lin, Q.; Hua, Y.; Qin, J. Clustering Denoising of 2D LiDAR Scanning in Indoor Environment Based on Keyframe Extraction. *Sensors* **2022**, *23*, 18. <https://doi.org/10.3390/s23010018>.
59. Jiang, S.; Wang, S.; Yi, Z.; Zhang, M.; Lv, X. Autonomous Navigation System of Greenhouse Mobile Robot Based on 3D Lidar and 2D Lidar SLAM. *Front. Plant Sci.* **2022**, *13*, 815218. <https://doi.org/10.3389/fpls.2022.815218>.

- 36.德卡赫, M.; 马蒂克, D.; Skarga-Bandurova, I. 配备超声波传感器的小型自主移动机器人的避障算法。2020年IEEE第11届可靠系统、服务和技术国际会议 (DESSERT) 会议记录, 乌克兰基辅, 2020年5月14日至18日; 第236–241页。https://doi.org/10.1109/DESSERT50317.2020.9125019.
- 37.布拉萨尔特, E.; 佩加德, C.; Mouaddib, M. 使用红外信标进行定位。Robotica 2000, 18, 153–161。https://doi.org/10.1017/S0263574799001927.
- 38.克雷萨, J.; Vechet, S. 基于红外信标的移动机器人定位。Electron. Electr. Eng. 2012, 117, 17–22。https://doi.org/10.5755/j01.eee.117.1.1046.
- 39.李T.H.; 张, S.J.; Tong, W. 使用红外传感器的自主移动机器人的模糊目标跟踪控制。IEEE Trans. Fuzzy Syst. 2004, 12, 491–501。https://doi.org/10.1109/TFUZZ.2004.832526.
40. Juang, J.; Yang, Y. 室内地图构建的模糊传感器融合和曲线逼近。2015年人工智能与工业工程国际会议论文集, 泰国普吉岛, 2015年7月26–27日。https://doi.org/10.2991/aiie-15.2015.148.
41. 奥蒂利, A.; 阿亚德, H.; 波兹纳, C.; 莫根, G.; ELbouzekraoui, M.; Elkari, B. 使用独轮车机器人模糊控制器进行避障。2020年国际控制、自动化和诊断会议 (ICCAD) 会议记录, 法国巴黎, 2020年10月7–9日; 第1–6页。https://doi.org/10.1109/ICCAD49821.2020.9260553.
42. 索拉诺, D.M.; 格兰德, R.E.; 博尼拉, M.N.I. PID控制和模糊逻辑系统在自主机器人中的避障。2021年第18届电气工程、计算科学和自动控制国际会议 (CCE) 会议记录, 2021年11月10日至12日, 墨西哥墨西哥城; 第1–6页。https://doi.org/10.1109/CCE53527.2021.9633079.
43. Nnoli, K.P.; 本尼奥戈尔, 硕士; 博卢, J.I.; 奥坎坎米, O.O. 用于机器人导航的基于边缘的红外超声波防撞雷达系统: *经济高效的双传感系统在障碍物检测、跟踪和规避中的应用。2022年IEEE国际国土安全技术研讨会 (HST) 会议记录, 2022年11月14日至15日, 美国马萨诸塞州波士顿; 第1–7页。https://doi.org/10.1109/HST56032.2022.10024985.
44. 赵杰; 方, J. 王, S. 王, K. 刘, C.; Han, T. 基于路标检测的多传感器智能机器人避障。Sensors 2021, 21, 6777。https://doi.org/10.3390/s21206777.
45. 哈比奇, T.L.; 斯图德, M.; 拉贝, M.; Spindeldreier, S. 我以前来过这里吗? 学习在基于图的SLAM中使用LiDAR数据闭环循环。2021年IEEE/ASME国际先进智能机电一体化会议 (AIM) 会议记录, 荷兰代尔夫特, 2021年7月12–16日; 第504–510页。https://doi.org/10.1109/AIM46487.2021.9517565.
46. 孟J.; 万, L.; 王, S. 蒋L.; 李, G. 吴L.; Xie, Y. 使用多尺度/分辨率地图进行基于LiDAR的移动机器人高效可靠的全球定位。IEEE Trans. Instrum. Meas. 2021, 70, 1–15。https://doi.org/10.1109/TIM.2021.3093933.
47. 哈内尔, D.; 布加德, W.; Thrun, S. 使用移动机器人学习室内和室外环境的紧凑3D模型。Robot. Auton. Syst. 2003, 44, 15–27。https://doi.org/10.1016/S0921-8890(03)00007-1.
48. 王X.; 马科特, R.J.; Olson, E. GLFP: 从平面图进行全球本地化。2019年IEEE/RSJ国际智能机器人与系统会议 (IROS) 论文集, 中国澳门, 2019年11月3–8日; 第1627–1632页。https://doi.org/10.1109/IROS40897.2019.8968061.
49. 匡, H.; 陈X.; 瓜达尼诺, T.; 齐默尔曼, N.; 贝利, J.; Stachniss, C. IR-MCL: 基于隐式表示的在线全局本地化。IEEE Robot. Autom. Lett. 2023, 8, 1627–1634。https://doi.org/10.1109/LRA.2023.3239318.
50. 金, J.; Chung, W. 在玻璃墙环境中使用激光测距仪定位移动机器人。IEEE Trans. Ind. Electron. 2016, 63, 3616–3627。https://doi.org/10.1109/TIE.2016.2523460.
51. 格里塞蒂, G.; 斯塔尼斯, C.; Burgard, W. 使用 Rao-Blackwellized 粒子滤波器改进网格映射技术。IEEE Trans. Robot. 2007, 23, 34–46。https://doi.org/10.1109/TRO.2006.889486.
52. Kohlbrecher, S.; 冯·斯特莱克, O.; 迈耶, J.; Klingauf, U. 具有完整3D运动估计功能的灵活且可扩展的SLAM系统。2011年IEEE国际安全、安保和救援机器人研讨会论文集, 日本京都, 2011年11月1–5日; 第155–160页。https://doi.org/10.1109/SSRR.2011.6106777.
53. 科诺利格, K.; 格里塞蒂, G.; 库梅尔, R.; 布加德, W.; Limketkai, B.; Vincent, R. 用于2D映射的高效稀疏姿势调整。2010年IEEE/RSJ国际智能机器人和系统会议论文集, 台湾台北, 2010年10月18–22日; 第22–29页。https://doi.org/10.1109/IROS.2010.5649043.
54. 赫斯, W.; 科勒, D.; 拉普, H.; Andor, D. 2D LIDAR SLAM中的实时闭环。2016年IEEE国际机器人与自动化会议 (ICRA) 会议记录, 瑞典斯德哥尔摩, 2016年16–21日

60. Guan, W.; Huang, L.; Wen, S.; Yan, Z.; Liang, W.; Yang, C.; Liu, Z. Robot Localization and Navigation Using Visible Light Positioning and SLAM Fusion. *J. Light. Technol.* **2021**, *39*, 7040–7051. <https://doi.org/10.1109/JLT.2021.3113358>.
61. Lee, J.; Tsubouchi, T.; Yamamoto, K.; Egawa, S. People Tracking Using a Robot in Motion with Laser Range Finder. In Proceedings of the 2006 IEEE/RSJ International Conference on Intelligent Robots and Systems, Beijing, China, 9–15 October 2006; pp. 2936–2942. <https://doi.org/10.1109/IROS.2006.282147>.
62. Mozos, O.M.; Kurazume, R.; Hasegawa, T. Multi-Part People Detection Using 2D Range Data. *Int. J. Soc. Robot.* **2010**, *2*, 31–40. <https://doi.org/10.1007/s12369-009-0041-3>.
63. Guerrero-Higueras, Á.M.; Álvarez Aparicio, C.; Calvo Olivera, M.C.; Rodríguez-Lera, F.J.; Fernández-Llamas, C.; Rico, F.M.; Matellán, V. Tracking People in a Mobile Robot From 2D LIDAR Scans Using Full Convolutional Neural Networks for Security in Cluttered Environments. *Front. Neurobot.* **2019**, *12*, 85. <https://doi.org/10.3389/fnbot.2018.00085>.
64. Yan, Z.; Duckett, T.; Bellotto, N. Online learning for 3D LiDAR-based human detection: Experimental analysis of point cloud clustering and classification methods. *Auton. Robot.* **2020**, *44*, 147–164. <https://doi.org/10.1007/s10514-019-09883-y>.
65. Li, Z.; Wang, C.; Yan, W.; Ji, Y.; Zou, A.; Lai, J. Research on obstacle detection and location of indoor robot based on LIDAR. In Proceedings of the 9th International Symposium on Advanced Optical Manufacturing and Testing Technologies: Optical Test, Measurement Technology, and Equipment, Chengdu, China, 26–29 June 2019; p. 31. <https://doi.org/10.1117/12.2504952>.
66. Henry, P.; Krainin, M.; Herbst, E.; Ren, X.; Fox, D. RGB-D Mapping: Using Depth Cameras for Dense 3D Modeling of Indoor Environments. In *Experimental Robotics*; Khatib, O., Kumar, V., Sukhatme, G., Eds.; Springer: Berlin/Heidelberg, Germany, 2014; Volume 79, pp. 477–491. https://doi.org/10.1007/978-3-642-28572-1_33.
67. Wimbauer, F.; Yang, N.; Von Stumberg, L.; Zeller, N.; Cremers, D. MonoRec: Semi-Supervised Dense Reconstruction in Dynamic Environments from a Single Moving Camera. In Proceedings of the 2021 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Nashville, TN, USA, 20–25 June 2021; pp. 6108–6118. <https://doi.org/10.1109/CVPR46437.2021.00605>.
68. Campos, C.; Elvira, R.; Rodriguez, J.J.G.; Montiel, J.M.M.; Tardos, J.D. ORB-SLAM3: An Accurate Open-Source Library for Visual, Visual-Inertial, and Multimap SLAM. *IEEE Trans. Robot.* **2021**, *37*, 1874–1890. <https://doi.org/10.1109/TRO.2021.3075644>.
69. Li, Y.; Liu, X.; Dong, W.; Zhou, H.; Bao, H.; Zhang, G.; Zhang, Y.; Cui, Z. DELTAR: Depth Estimation from a Light-weight ToF Sensor and RGB Image. *arXiv* **2022**, arXiv:2209.13362.
70. Davison, A.J.; Reid, I.D.; Molton, N.D.; Stasse, O. MonoSLAM: Real-Time Single Camera SLAM. *IEEE Trans. Pattern Anal. Mach. Intell.* **2007**, *29*, 1052–1067. <https://doi.org/10.1109/TPAMI.2007.1049>.
71. Konolige, K.; Agrawal, M. FrameSLAM: From Bundle Adjustment to Real-Time Visual Mapping. *IEEE Trans. Robot.* **2008**, *24*, 1066–1077. <https://doi.org/10.1109/TRO.2008.2004832>.
72. Newcombe, R.A.; Fitzgibbon, A.; Izadi, S.; Hilliges, O.; Molyneaux, D.; Kim, D.; Davison, A.J.; Kohi, P.; Shotton, J.; Hodges, S. KinectFusion: Real-time dense surface mapping and tracking. In Proceedings of the 2011 10th IEEE International Symposium on Mixed and Augmented Reality, Basel, Switzerland, 26–29 October 2011; pp. 127–136. <https://doi.org/10.1109/ISMAR.2011.6092378>.
73. Rebecq, H.; Horstschaefer, T.; Gallego, G.; Scaramuzza, D. EVO: A Geometric Approach to Event-Based 6-DOF Parallel Tracking and Mapping in Real Time. *IEEE Robot. Autom. Lett.* **2017**, *2*, 593–600. <https://doi.org/10.1109/LRA.2016.2645143>.
74. Bajpayee, A.; Techet, A.H.; Singh, H. Real-Time Light Field Processing for Autonomous Robotics. In Proceedings of the 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Madrid, Spain, 1–5 October 2018; pp. 4218–4225. <https://doi.org/10.1109/IROS.2018.8594477>.
75. Liu, L.; Song, X.; Wang, M.; Liu, Y.; Zhang, L. Self-supervised Monocular Depth Estimation for All Day Images using Domain Separation. *arXiv* **2021**, arXiv:2108.07628.
76. Bescos, B.; Cadena, C.; Neira, J. Empty Cities: A Dynamic-Object-Invariant Space for Visual SLAM. *IEEE Trans. Robot.* **2021**, *37*, 433–451. <https://doi.org/10.1109/TRO.2020.3031267>.
77. Tourani, A.; Bavle, H.; Sanchez-Lopez, J.L.; Voos, H. Visual SLAM: What Are the Current Trends and What to Expect? *Sensors* **2022**, *22*, 9297. <https://doi.org/10.3390/s22239297>.
78. Zhan, H.; Weerasekera, C.S.; Bian, J.W.; Reid, I. Visual Odometry Revisited: What Should Be Learnt? In Proceedings of the 2020 IEEE International Conference on Robotics and Automation (ICRA), Paris, France, 31 May–31 August 2020; pp. 4203–4210. <https://doi.org/10.1109/ICRA40945.2020.9197374>.
79. Zhou, Y.; Gallego, G.; Shen, S. Event-Based Stereo Visual Odometry. *IEEE Trans. Robot.* **2021**, *37*, 1433–1450. <https://doi.org/10.1109/TRO.2021.3062252>.
80. Zhu, Z.; Peng, S.; Larsson, V.; Xu, W.; Bao, H.; Cui, Z.; Oswald, M.R.; Pollefeys, M. NICE-SLAM: Neural Implicit Scalable Encoding for SLAM. In Proceedings of the 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), New Orleans, LA, USA, 18–24 June 2022; pp. 12776–12786. <https://doi.org/10.1109/CVPR52688.2022.01245>.
81. Zheng, C.; Zhu, Q.; Xu, W.; Liu, X.; Guo, Q.; Zhang, F. FAST-LIVO: Fast and Tightly-coupled Sparse-Direct LiDAR-Inertial-Visual Odometry. In Proceedings of the 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Kyoto, Japan, 23–27 October 2022; pp. 4003–4009. <https://doi.org/10.1109/IROS47612.2022.9981107>.
82. Wu, M.; Wang, S.; Meng, J.; Xie, Y. AprilTag-Aided Pose Estimation Optimization of Landmark VSLAM. In Proceedings of the 2023 42nd Chinese Control Conference (CCC), Tianjin, China, 24–26 July 2023; pp. 4556–4561. <https://doi.org/10.23919/CCC58697.2023.10241127>.

60. 关W.; 黄L.; 温, S. 严, Z. 梁, W. 杨, C.; Liu, Z. 使用可见光定位和 SLAM 融合的机器人定位和导航。 *J. Light. Technol.* 2021, 39, 7040–7051. <https://doi.org/10.1109/JLT.2021.3113358>.
61. 李, J.; 坪内, T.; 山本, K.; Ekawa, S. 使用带有激光测距仪的运动机器人进行人员跟踪。2006 年 IEEE/RSJ 国际智能机器人与系统会议论文集, 中国北京, 2006 年 10 月 9-15 日; 第 2936–2942 页。 <https://doi.org/10.1109/IROS.2006.282147>.
62. 莫索斯, O.M.; 库拉祖姆, R.; Hasekawa, T. 使用 2D 范围数据进行多部分人员检测。 *Int. J. Soc. Robot.* 2010, 2, 31–40. <https://doi.org/10.1007/s12369-009-0041-3>.
63. 格雷罗-伊格拉斯, Á.M.; 阿尔瓦雷斯-阿帕里西奥, C.; 卡尔沃-奥利维拉, M.C.; 罗德里格斯-莱拉, F.J.; 费尔南德斯-拉马斯, C.; Rico, F.M.; Matellán, V. 使用全卷积神经网络通过 2D LIDAR 扫描跟踪移动机器人中的人员, 以确保杂乱环境中的安全。 *Front. Neurobot.* 2019, 12, 85. <https://doi.org/10.3389/fnbot.2018.00085>.
64. Yan, Z.; 达卡特, T.; Bellotto, N. 基于 3D LiDAR 的人体检测在线学习: 点云聚类 and 分类方法的实验分析。 *Auton. Robot.* 2020, 44, 147–164. <https://doi.org/10.1007/s10514-019-09883-y>.
65. 李Z.; 王, C.; 严, W. 吉, Y. 邹, A.; 基于激光雷达的室内机器人障碍物检测与定位研究。第九届先进光学制造和测试技术国际研讨会论文集: 光学测试、测量技术和设备, 中国成都, 2019 年 6 月 26-29 日; p. 31. <https://doi.org/10.1117/12.2504952>.
66. 亨利, P.; 克莱宁, M.; 赫布斯特, E.; 任, X. Fox, D. RGB-D 映射: 使用深度相机对室内环境进行密集 3D 建模。在 *Experimental Robotics* 中; Khatib, O., Kumar, V., Sukhatme, G., 编辑; 施普林格: 德国柏林/海德堡, 2014 年; 第 79 卷, 第 477–491 页。 https://doi.org/10.1007/978-3-642-28572-1_33.
67. 温鲍尔, F.; 杨, N. 冯·斯坦伯格, L.; 泽勒, N.; Cremers, D. MonoRec: 通过单个移动摄像机在动态环境中进行半监督密集重建。2021 年 IEEE/CVF 计算机视觉和模式识别会议 (CVPR) 会议记录, 美国田纳西州纳什维尔, 2021 年 6 月 20-25 日; 第 6108–6118 页。 <https://doi.org/10.1109/CVPR46437.2021.00605>.
68. 坎波斯, C.; 埃尔维拉, R.; 罗德里格斯, J.J.G.; 蒙蒂尔, J.M.M.; Tardos, J.D. ORB-SLAM3: 用于视觉、视觉惯性和多地图 SLAM 的精确开源库。 *IEEE Trans. Robot.* 2021, 37, 1874–1890. <https://doi.org/10.1109/TRO.2021.3075644>.
69. 李, Y.; 刘X. 董, W. 周, H.; 鲍, H.; 张, G. 张, Y. Cui, Z. DELTAR: 利用轻量级 ToF 传感器和 RGB 图像进行深度估计。 *arXiv* 2022 年, arXiv: 2209.13362.
70. 戴维森, A.J.; 里德, I.D.; 北达科他州莫尔顿; Stasse, O. MonoSLAM: 实时单相机 SLAM。 *IEEE Trans. Pattern Anal. Mach. Intell.* 2007, 29, 1052–1067. <https://doi.org/10.1109/TPAMI.2007.1049>.
71. 科诺利格, K.; Agrawal, M. FrameSLAM: 从捆绑调整到实时视觉映射。 *IEEE Trans. Robot.* 2008, 24, 1066–1077. <https://doi.org/10.1109/TRO.2008.2004832>.
72. 纽科姆, RA; 菲茨吉本, A.; 伊扎迪, S.; 希利格斯, O.; 莫利诺, D.; 金, D. 戴维森, A.J.; 科希, P.; 肖顿, J.; Hodges, S. KinectFusion: 实时密集表面测绘和跟踪。2011 年第 10 届 IEEE 国际混合与增强现实研讨会论文集, 瑞士巴塞尔, 2011 年 10 月 26-29 日, 第 127–136 页。 <https://doi.org/10.1109/ISMAR.2011.6092378>.
73. Rebecq, H.; 霍斯特舍费尔, T.; 加列戈, G.; Scaramuzza, D. EVO: 基于事件的 6 自由度并行实时跟踪和绘图的几何方法。 *IEEE Robot. Autom. Lett.* 2017, 2, 593–600. <https://doi.org/10.1109/LRA.2016.2645143>.
74. 巴杰帕伊, A.; 泰切特, A.H.; Singh, H. 自主机器人实时光场处理。2018 年 IEEE/RSJ 国际智能机器人和系统会议 (IROS) 会议记录, 西班牙马德里, 2018 年 10 月 1-5 日; 第 4218–4225 页。 <https://doi.org/10.1109/IROS.2018.8594477>.
75. 刘L.; 宋X.; 王, M. 刘, Y. 张, L. 使用域分离对全天图像进行自监督单目深度估计。 *arXiv* 2021 年, arXiv: 2108.07628.
76. 贝斯科斯, B.; 卡德纳, C.; Neira, J. 空城: 视觉 SLAM 的动态对象不变空间。 *IEEE Trans. Robot.* 2021, 37, 433–451. <https://doi.org/10.1109/TRO.2020.3031267>.
77. 图拉尼, A.; 巴夫勒, H.; 桑切斯-洛佩兹, J.L.; Voos, H. 视觉 SLAM: 当前趋势是什么以及预期是什么? *Sensors* 2022 年, 22, 9297. <https://doi.org/10.3390/s22239297>.
78. 詹H.; 韦拉塞克拉, C.S.; 卞, J.W.; Reid, I. 视觉里程计重温: 应该学习什么? 2020 年 IEEE 国际机器人与自动化会议 (ICRA) 会议记录, 法国巴黎, 2020 年 5 月 31 日至 8 月 31 日; 第 4203–4210 页。 <https://doi.org/10.1109/ICRA40945.2020.9197374>.
79. 周, Y.; 加列戈, G.; Shen, S. 基于事件的立体视觉里程计。 *IEEE Trans. Robot.* 2021, 37, 1433–1450. <https://doi.org/10.1109/TRO.2021.3062252>.
80. 朱Z.; 彭, S. 拉尔森, V.; 徐, W. 鲍, H.; 崔, Z. 奥斯瓦尔德, M

83. Schreiber, M.; Belagiannis, V.; Glaser, C.; Dietmayer, K. Dynamic Occupancy Grid Mapping with Recurrent Neural Networks. In Proceedings of the 2021 IEEE International Conference on Robotics and Automation (ICRA), Xi'an, China, 30 May–5 June 2021; pp. 6717–6724. <https://doi.org/10.1109/ICRA48506.2021.9561375>.
84. Eisoldt, M.; Flottmann, M.; Gaal, J.; Buschermöhle, P.; Hinderink, S.; Hillmann, M.; Nitschmann, A.; Hoffmann, P.; Wiemann, T.; Porrmann, M. HATSDF SLAM—Hardware-accelerated TSDF SLAM for Reconfigurable SoCs. In Proceedings of the 2021 European Conference on Mobile Robots (ECMR), Bonn, Germany, 31 August–3 September 2021; pp. 1–7. <https://doi.org/10.1109/ECMR50962.2021.9568815>.
85. Rosinol, A.; Abate, M.; Chang, Y.; Carlone, L. Kimera: An Open-Source Library for Real-Time Metric-Semantic Localization and Mapping. In Proceedings of the 2020 IEEE International Conference on Robotics and Automation (ICRA), Paris, France, 31 May–31 August 2020; pp. 1689–1696. <https://doi.org/10.1109/ICRA40945.2020.9196885>.
86. Vijayanarasimhan, S.; Ricco, S.; Schmid, C.; Sukthankar, R.; Fragkiadaki, K. SfM-Net: Learning of Structure and Motion from Video. *arXiv* **2017**, arXiv:1704.07804.
87. Cao, A.Q.; De Charette, R. MonoScene: Monocular 3D Semantic Scene Completion. In Proceedings of the 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), New Orleans, LA, USA, 18–24 June 2022; pp. 3981–3991. <https://doi.org/10.1109/CVPR52688.2022.00396>.
88. Kim, H.; Moon, J.; Lee, B. RGB-to-TSDF: Direct TSDF Prediction from a Single RGB Image for Dense 3D Reconstruction. In Proceedings of the 2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Macau, China, 3–8 November 2019; pp. 6714–6720. <https://doi.org/10.1109/IROS40897.2019.8968566>.
89. Yu, C.; Liu, Z.; Liu, X.J.; Xie, F.; Yang, Y.; Wei, Q.; Fei, Q. DS-SLAM: A Semantic Visual SLAM towards Dynamic Environments. In Proceedings of the 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Madrid, Spain, 1–5 October 2018; pp. 1168–1174. <https://doi.org/10.1109/IROS.2018.8593691>.
90. Kulhanek, J.; Derner, E.; De Bruin, T.; Babuska, R. Vision-based Navigation Using Deep Reinforcement Learning. In Proceedings of the 2019 European Conference on Mobile Robots (ECMR), Prague, Czech Republic, 4–6 September 2019; pp. 1–8. <https://doi.org/10.1109/ECMR.2019.8870964>.
91. Xiao, W.; Yuan, L.; He, L.; Ran, T.; Zhang, J.; Cui, J. Multigoal Visual Navigation With Collision Avoidance via Deep Reinforcement Learning. *IEEE Trans. Instrum. Meas.* **2022**, *71*, 1–9. <https://doi.org/10.1109/TIM.2022.3158384>.
92. Xie, L.; Markham, A.; Trigoni, N. SnapNav: Learning Mapless Visual Navigation with Sparse Directional Guidance and Visual Reference. In Proceedings of the 2020 IEEE International Conference on Robotics and Automation (ICRA), Paris, France, 31 May–31 August 2020; pp. 1682–1688. <https://doi.org/10.1109/ICRA40945.2020.9197523>.
93. Tai, L.; Li, S.; Liu, M. A deep-network solution towards model-less obstacle avoidance. In Proceedings of the 2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Daejeon, Republic of Korea, 9–14 October 2016; pp. 2759–2764. <https://doi.org/10.1109/IROS.2016.7759428>.
94. De Cristóforis, P.; Nitsche, M.; Krajník, T.; Pire, T.; Mejail, M. Hybrid vision-based navigation for mobile robots in mixed indoor/outdoor environments. *Pattern Recognit. Lett.* **2015**, *53*, 118–128. <https://doi.org/10.1016/j.patrec.2014.10.010>.
95. Biswas, J.; Veloso, M.M. WiFi Localization and Navigation for Autonomous Indoor Mobile Robots. In Proceedings of the 2010 IEEE International Conference on Robotics and Automation, Anchorage, AK, USA, 3–7 May 2010; pp. 4379–4384. <https://doi.org/10.1109/ROBOT.2010.5509842>.
96. Schwarz, D.; Schwarz, M.; Stücker, J.; Behnke, S. Cosero, Find My Keys! Object Localization and Retrieval Using Bluetooth Low Energy Tags. In *RoboCup 2014: Robot World Cup XVIII*; Bianchi, R.A.C., Akin, H.L., Ramamoorthy, S., Sugiura, K., Eds.; Series Title: Lecture Notes in Computer Science; Springer International Publishing: Cham, Germany, 2015; Volume 8992, pp. 195–206. https://doi.org/10.1007/978-3-319-18615-3_16.
97. Loganathan, A.; Ahmad, N.S. Self-Adaptive Filtering Approach for Improved Indoor Localization of a Mobile Node with Zigbee-Based RSSI and Odometry. *Sensors* **2019**, *19*, 4748. <https://doi.org/10.3390/s19214748>.
98. Xin, J.; Xie, G.; Yan, B.; Shan, M.; Li, P.; Gao, K. Multimobile Robot Cooperative Localization Using Ultrawideband Sensor and GPU Acceleration. *IEEE Trans. Autom. Sci. Eng.* **2022**, *19*, 2699–2710. <https://doi.org/10.1109/TASE.2021.3117949>.
99. Yang, B.; Yang, E. A Survey on Radio Frequency based Precise Localisation Technology for UAV in GPS-denied Environment. *J. Intell. Robot. Syst.* **2021**, *103*, 38. <https://doi.org/10.1007/s10846-021-01500-4>.
100. Shen, J.; Junyang, S.; Molisch, A.F.; Salmi, J. Accurate Passive Location Estimation Using TOA Measurements. *IEEE Trans. Wirel. Commun.* **2012**, *11*, 2182–2192. <https://doi.org/10.1109/TWC.2012.040412.110697>.
101. Guvenç, I.; Chong, C.C. A Survey on TOA Based Wireless Localization and NLOS Mitigation Techniques. *IEEE Commun. Surv. Tutorials* **2009**, *11*, 107–124. <https://doi.org/10.1109/SURV.2009.090308>.
102. Ho, K.C. Bias Reduction for an Explicit Solution of Source Localization Using TDOA. *IEEE Trans. Signal Process.* **2012**, *60*, 2101–2114. <https://doi.org/10.1109/TSP.2012.2187283>.
103. Zafari, F.; Gkelias, A.; Leung, K.K. A Survey of Indoor Localization Systems and Technologies. *IEEE Commun. Surv. Tutorials* **2019**, *21*, 2568–2599. <https://doi.org/10.1109/COMST.2019.2911558>.
104. Wang, Y.; Ho, K.C. An Asymptotically Efficient Estimator in Closed-Form for 3-D AOA Localization Using a Sensor Network. *IEEE Trans. Wirel. Commun.* **2015**, *14*, 6524–6535. <https://doi.org/10.1109/TWC.2015.2456057>.

- 83.施赖伯, M.; 贝拉吉安尼斯, V.; 格拉泽, C.; Dietmayer, K. 使用循环神经网络进行动态占用网格映射。2021 年 IEEE 国际机器人与自动化会议 (ICRA) 论文集, 中国西安, 2021 年 5 月 30 日至 6 月 5 日; 第 6717–6724 页。https://doi.org/10.1109/ICRA48506.2021.9561375。
- 84.艾索尔特, M.; 弗洛特曼, M.; 盖尔, J.; 布舍莫勒, P.; 亨德林克, S.; 希尔曼, M.; 尼奇曼, A.; 霍夫曼, P.; 维曼, T.; Porrmann, M. HATSDF SLAM — 用于可重新配置 SoC 的硬件加速 TSDF SLAM。2021 年欧洲移动机器人会议 (ECMR) 会议记录, 德国波恩, 2021 年 8 月 31 日至 9 月 3 日; 第 1-7 页。https://doi.org/10.1109/ECMR50962.2021.9568815。
- 85.松香醇, A.; 阿巴特, M.; 张, Y. Carlone, L. Kimera: 用于实时度量语义定位和映射的开源库。2020 年 IEEE 国际机器人与自动化会议 (ICRA) 会议记录, 法国巴黎, 2020 年 5 月 31 日至 8 月 31 日; 第 1689–1696 页。https://doi.org/10.1109/ICRA40945.2020.9196885。
- 86.Vijayanarasimhan, S.; 里科, S.; 施密德, C.; 苏坎坦卡, R.; Fragkiadaki, K. SfM-Net: 从视频中学习结构和运动。arXiv 2017 年, arXiv: 1704.07804。
- 87.曹A.Q.; De Charette, R. MonoScene: 单目 3D 语义场景完成。2022 年 IEEE/CVF 计算机视觉和模式识别会议 (CVPR) 会议记录, 美国路易斯安那州新奥尔良, 2022 年 6 月 18-24 日; 第 3981–3991 页。https://doi.org/10.1109/CVPR52688.2022.00396。
- 88.金, H.; 穆恩, J.; Lee, B. RGB 到 TSDF: 从单个 RGB 图像直接进行 TSDF 预测以进行密集 3D 重建。2019 年 IEEE/RSJ 国际智能机器人与系统会议 (IROS) 论文集, 中国澳门, 2019 年 11 月 3-8 日; 第 6714–6720 页。https://doi.org/10.1109/IROS40897.2019.8968566。
- 89.Yu, C.; 刘, Z.; 刘XJ; 谢F.; 杨, Y. 魏, Q. Fei, Q. DS-SLAM: 针对动态环境的语义视觉 SLAM。2018 年 IEEE/RSJ 国际智能机器人和系统会议 (IROS) 会议记录, 西班牙马德里, 2018 年 10 月 1-5 日; 第 1168–1174 页。https://doi.org/10.1109/IROS.2018.8593691。
- 90.库尔哈内克, J.; 德尔纳, E.; 德布鲁因, T.; Babuska, R. 使用深度强化学习的基于视觉的导航。2019 年欧洲移动机器人会议 (ECMR) 会议记录, 捷克共和国布拉格, 2019 年 9 月 4-6 日; 第 1-8 页。https://doi.org/10.1109/ECMR.2019.8870964。
- 91.肖W.; 袁L.; 他, L.; 兰, T. 张, J. Cui, J. 通过深度强化学习实现避免碰撞的多目标视觉导航。IEEE Trans. Instrum. Meas. 2022, 71, 1-9。https://doi.org/10.1109/TIM.2022.3158384。
- 92.谢L.; 马卡姆, A.; Trgoni, N. SnapNav: 通过稀疏方向指导和视觉参考学习无地图视觉导航。2020 年 IEEE 国际机器人与自动化会议 (ICRA) 会议记录, 法国巴黎, 2020 年 5 月 31 日至 8 月 31 日; 第 1682–1688 页。https://doi.org/10.1109/ICRA40945.2020.9197523。
- 93.Tai, L.; 李, S. Liu, M. 一种无模型避障的深度网络解决方案。2016 年 IEEE/RSJ 国际智能机器人和系统会议 (IROS) 会议记录, 2016 年 10 月 9-14 日, 韩国大田; 第 2759–2764 页。https://doi.org/10.1109/IROS.2016.7759428。
- 94.德·克里斯托福里斯, P.; 尼采, M.; 克拉伊尼克, T.; 皮雷, T.; Mejail, M. 用于室内/室外混合环境中移动机器人的混合视觉导航。Pattern Recognit. Lett. 2015, 53, 118–128。https://doi.org/10.1016/j.patrec.2014.10.010。
- 95.比斯瓦斯, J.; 维罗索, M.M. 自主室内移动机器人的 WiFi 定位和导航。2010 年 IEEE 国际机器人与自动化会议论文集, 美国阿拉斯加州安克雷奇, 2010 年 5 月 3-7 日; 第 4379–4384 页。https://doi.org/10.1109/ROBOT.2010.5509842。
- 96.施瓦茨, D.; 施瓦茨, M.; 施图克勒, J.; Behnke, S. Cosero, 找到我的钥匙! 使用蓝牙低功耗标签进行对象定位和检索。在 RoboCup 2014: Robot World Cup XVIII 中; Bianchi, R.A.C., Akin, H.L., Ramamoorthy, S., Sugiura, K., 编辑; 系列名称: 计算机科学讲义; 施普林格国际出版社: Cham, 德国, 2015 年; 第 8992 卷, 第 195–206 页。https://doi.org/10.1007/978-3-319-18615-3_16。
- 97.洛加纳森, A.; 艾哈迈德, N.S. 利用基于 Zigbee 的 RSSI 和里程计改进移动节点室内定位的自适应滤波方法。Sensors 2019, 19, 4748。https://doi.org/10.3390/s19214748。
- 98.辛, J.; 谢G.; 严, B.; 单, M. 李, P. 高, K. 使用超宽带传感器和 GPU 加速的多移动机器人协作定位。IEEE Trans. Autom. Sci. Eng. 2022, 19, 2699–2710。https://doi.org/10.1109/TASE.2021.3117949。
- 99.杨, B.; Yang, E. GPS 拒绝环境中基于射频的无人机精确定位技术综述。J. Intell. Robot. Syst. 2021 年, 103, 38。https://doi.org/10.1007/s10846-021-01500-4。
- 100.沉J.; 俊阳, S.; 莫里施, A.F.; Salmi, J. 使用 TOA 测量进行准确的无源位置估计。IEEE Trans. Wirel. Commun. 2012, 11, 2182–2192。https://doi.org/10.1109/TWC.2012.040412.110697。
- 101.Guvenç, I.; 冲, C.C.

105. Mazuelas, S.; Bahillo, A.; Lorenzo, R.M.; Fernandez, P.; Lago, F.A.; Garcia, E.; Blas, J.; Abril, E.J. Robust Indoor Positioning Provided by Real-Time RSSI Values in Unmodified WLAN Networks. *IEEE J. Sel. Top. Signal Process.* **2009**, *3*, 821–831. <https://doi.org/10.1109/JSTSP.2009.2029191>.
106. He, S.; Chan, S.H.G. Wi-Fi Fingerprint-Based Indoor Positioning: Recent Advances and Comparisons. *IEEE Commun. Surv. Tutorials* **2016**, *18*, 466–490. <https://doi.org/10.1109/COMST.2015.2464084>.
107. Davidson, P.; Piche, R. A Survey of Selected Indoor Positioning Methods for Smartphones. *IEEE Commun. Surv. Tutorials* **2017**, *19*, 1347–1370. <https://doi.org/10.1109/COMST.2016.2637663>.
108. Ye, H.; Peng, J. Robot Indoor Positioning and Navigation Based on Improved WiFi Location Fingerprint Positioning Algorithm. *Wirel. Commun. Mob. Comput.* **2022**, *2022*, 8274455. <https://doi.org/DOI10.1155/2022/8274455>.
109. Zhang, L.; Chen, Z.; Cui, W.; Li, B.; Chen, C.; Cao, Z.; Gao, K. WiFi-Based Indoor Robot Positioning Using Deep Fuzzy Forests. *IEEE Internet Things J.* **2020**, *7*, 10773–10781. <https://doi.org/10.1109/JIOT.2020.2986685>.
110. Zhou, G.; Xu, S.; Zhang, S.; Wang, Y.; Xiang, C. Multi-Floor Indoor Localization Based on Multi-Modal Sensors. *Sensors* **2022**, *22*, 4162. <https://doi.org/10.3390/s22114162>.
111. Zhang, L.; Zhang, S.; Leng, C.T. A Study on the Location System Based on Zigbee for Mobile Robot. *Appl. Mech. Mater.* **2014**, *651–653*, 612–615. <https://doi.org/10.4028/www.scientific.net/AMM.651-653.612>.
112. Wang, Z.; Liu, M.; Zhang, Y. Mobile localization in complex indoor environment based on ZigBee wireless network. *J. Phys.* **2019**, *1314*, 012214. <https://doi.org/10.1088/1742-6596/1314/1/012214>.
113. Raghavan, A.N.; Ananthapadmanaban, H.; Sivamurugan, M.S.; Ravindran, B. Accurate mobile robot localization in indoor environments using bluetooth. In Proceedings of the 2010 IEEE International Conference on Robotics and Automation, Anchorage, AK, USA, 3–7 May 2010; pp. 4391–4396. <https://doi.org/10.1109/ROBOT.2010.5509232>.
114. Yamami, Y.; Tang, S. AoA Estimation for High Accuracy BLE Positioning. In Proceedings of the IEEE Consumer Communications and Networking Conference, Las Vegas, NV, USA, 8–11 January 2023. <https://doi.org/10.1109/CCNC51644.2023.10059866>.
115. Weinmann, K.; Simske, S. Design of Bluetooth 5.1 Angle of Arrival Homing Controller for Autonomous Mobile Robot. *Robotics* **2023**, *12*, 115. <https://doi.org/10.3390/robotics12040115>.
116. Alexandr, A.; Anton, D.; Mikhail, M.; Ilya, K. Comparative Analysis of Indoor Positioning Methods Based on the Wireless Sensor Network of Bluetooth Low Energy Beacons. In Proceedings of the 2020 International Conference Engineering and Telecommunication (En & T), Dolgoprudny, Russia, 25–26 November 2020. <https://doi.org/10.1109/EnT50437.2020.9431286>.
117. Shi, D.; Mi, H.; Collins, E.G.; Wu, J. An Indoor Low-Cost and High-Accuracy Localization Approach for AGVs. *IEEE Access* **2020**, *8*, 50085–50090. <https://doi.org/10.1109/ACCESS.2020.2980364>.
118. Leng, J.; Ma, G.; Zhu, J.; Ma, H. Improved TDOA Two-Stage UWB Localization Algorithm For Indoor Mobile Robot. In Proceedings of the IEEE International Conference on Recent Advances in Systems Science and Engineering (IEEE RASSE 2021), Shanghai, China, 12–14 December 2021. <https://doi.org/10.1109/RASSE53195.2021.9686768>.
119. Li, P.; Xu, Y.; Shen, T.; Bi, S. INS/UWB integrated AGV localization employing Kalman filter for indoor LOS/NLOS mixed environment. In Proceedings of the 2019 International Conference on Advanced Mechatronic Systems (ICAMEchS), Kusatsu, Shiga, Japan, 26–28 August 2019; pp. 294–298. <https://doi.org/10.1109/ICAMEchS.2019.8861620>.
120. Xu, T.; Zhang, H.; Zhou, X.; Yuan, X.; Tan, X.; Zhang, J.; Zhong, H. A Weight Adaptive Kalman Filter Localization Method Based on UWB and Odometry. In Proceedings of the 2022 International Conference on Advanced Robotics and Mechatronics (ICARM), Guilin, China, 9–11 July 2022; pp. 956–961. <https://doi.org/10.1109/ICARM54641.2022.9959604>.
121. Zhou, H.; Yao, Z.; Zhang, Z.; Liu, P.; Lu, M. An Online Multi-Robot SLAM System Based on Lidar/UWB Fusion. *IEEE Sens. J.* **2022**, *22*, 2530–2542. <https://doi.org/10.1109/JSEN.2021.3136929>.
122. Wu, H.; Tao, B.; Gong, Z.; Yin, Z.; Ding, H. A Standalone RFID-Based Mobile Robot Navigation Method Using Single Passive Tag. *IEEE Trans. Autom. Sci. Eng.* **2021**, *18*, 1529–1537. <https://doi.org/10.1109/TASE.2020.3008187>.
123. Wang, J.; Takahashi, Y. Particle Smoother-Based Landmark Mapping for the SLAM Method of an Indoor Mobile Robot with a Non-Gaussian Detection Model. *J. Sens.* **2019**, *2019*, 3717298. <https://doi.org/10.1155/2019/3717298>.
124. Shamsfakhr, F.; Macii, D.; Fontanelli, D.; Motroni, A.; Nepa, P.; Palopoli, L.; Buffi, A. RFID-based robot localisation: An unconstrained optimisation problem by exploiting RSSI. In Proceedings of the 2022 IEEE International Instrumentation and Measurement Technology Conference (I2MTC), Ottawa, ON, Canada, 16–19 May 2022; pp. 1–6. <https://doi.org/10.1109/I2MTC48687.2022.9806604>.
125. Wijk, O.; Christensen, H. Localization and navigation of a mobile robot using natural point landmarks extracted from sonar data. *Robot. Auton. Syst.* **2000**, *31*, 31–42. [https://doi.org/10.1016/S0921-8890\(99\)00085-8](https://doi.org/10.1016/S0921-8890(99)00085-8).
126. Lee, J.S.; Nam, S.Y.; Chung, W.K. Robust RBPF-SLAM for Indoor Mobile Robots Using Sonar Sensors in Non-Static Environments. *Adv. Robot.* **2011**, *25*, 1227–1248. <https://doi.org/10.1163/016918611X574696>.
127. Arras, K.; Tomatis, N. Improving robustness and precision in mobile robot localization by using laser range finding and monocular vision. In Proceedings of the 1999 Third European Workshop on Advanced Mobile Robots (Eurobot'99). Proceedings (Cat. No.99EX355), Zurich, Switzerland, 6–8 September 1999; pp. 177–185. <https://doi.org/10.1109/EURBOT.1999.827638>.
128. Lingemann, K.; Nüchter, A.; Hertzberg, J.; Surmann, H. High-speed laser localization for mobile robots. *Robot. Auton. Syst.* **2005**, *51*, 275–296. <https://doi.org/10.1016/j.robot.2005.02.004>.
129. Choi, B.S.; Lee, J.W.; Lee, J.J.; Park, K.T. A Hierarchical Algorithm for Indoor Mobile Robot Localization Using RFID Sensor Fusion. *IEEE Trans. Ind. Electron.* **2011**, *58*, 2226–2235. <https://doi.org/10.1109/TIE.2011.2109330>.

- 105.马苏埃拉斯, S.; 巴希罗, A.; 洛伦佐, R.M.; 费尔南德斯, P.; 拉戈, F.A.; 加西亚, E.; 布拉斯, J.; 阿布里尔, E.J.在未修改的 WLAN 网络中通过实时 RSSI 值提供稳健的室内定位。 *IEEE J. Sel. Top. Signal Process.* 2009, 3, 821–831。 <https://doi.org/10.1109/JSTSP.2009.2029191>。
- 106.他, S.; 陈S.H.G.基于 Wi-Fi 指纹的室内定位: 最新进展和比较。 *IEEE Commun. Surv. Tutorials* 2016, 18, 466–490。 <https://doi.org/10.1109/COMST.2015.2464084>。
- 107.戴维森, P.; Piche, R. 智能手机室内定位方法的调查。 *IEEE Commun. Surv. Tutorials* 2017, 19, 1347–1370。 <https://doi.org/10.1109/COMST.2016.2637663>。
- 108.叶, H.; 彭杰. 基于改进WiFi定位指纹定位算法的机器人室内定位导航。 *Wirel. Commun. Mob. Comput.* 2022 年, 2022, 8274455。 <https://doi.org/DOI10.1155/2022/8274455>。
- 109.张L.; 陈, Z.; 崔, W. 李, B. 陈, C.; 曹, Z. 高, K. 使用深度模糊森林的基于 WiFi 的室内机器人定位。 *IEEE Internet Things J.* 2020, 7, 10773–10781。 <https://doi.org/10.1109/JIOT.2020.2986685>。
- 110.周, G.; 徐, S. 张, S. 王, Y. Xiang, C. 基于多模态传感器的多层室内定位。 *Sensors* 2022 年, 22, 4162。 <https://doi.org/10.3390/s22114162>。
- 111.张L.; 张, S. 冷, C.T. 基于Zigbee的移动机器人定位系统研究。 *Appl. Mech. Mater.* 2014 年, 651–653, 612–615。 <https://doi.org/10.4028/www.scientific.net/AMM.651-653.612>。
- 112.王Z.; 刘, M.; 基于ZigBee无线网络的复杂室内环境移动定位。 *J. Phys.* 2019, 1314, 012214。 <https://doi.org/10.1088/1742-6596/1314/1/012214>。
- 113.拉格万, A.N.; 阿南塔帕德马纳班, H.; 西瓦穆鲁甘, 硕士; Ravindran, B. 使用蓝牙在室内环境中精确移动机器人定位。 2010 年 IEEE 国际机器人与自动化会议论文集, 美国阿拉斯加州安克雷奇, 2010 年 5 月 3-7 日; 第 4391–4396 页。 <https://doi.org/10.1109/ROBOT.2010.5509232>。
- 114.山美, Y.; Tang, S. 高精度 BLE 定位的 AoA 估计。 IEEE 消费者通信和网络会议记录, 美国内华达州拉斯维加斯, 2023 年 1 月 8 日至 11 日。 <https://doi.org/10.1109/CCNC51644.2023.10059866>。
- 115.Weinmann, K.; Simske, S. 用于自主移动机器人的蓝牙 5.1 到达角归航控制器设计。 *Robotics* 2023, 12, 115。 <https://doi.org/10.3390/robotics12040115>。
- 116.亚历山大, A.; 安东, D.; 米哈伊尔, M.; Ilya, K. 基于蓝牙低功耗信标无线传感器网络的室内定位方法的比较分析。 2020 年工程与电信国际会议 (En & T) 会议记录, 俄罗斯多尔戈普鲁德内, 2020 年 11 月 25-26 日。 <https://doi.org/10.1109/EnT50437.2020.9431286>。
- 117.石D.; 米, H. 柯林斯, E.G.; Wu, J. AGV 室内低成本高精度定位方法。 *IEEE Access* 2020, 8, 50085–50090。 <https://doi.org/10.1109/ACCESS.2020.2980364>。
- 118.冷, J.; 马, G. 朱, J. Ma, H. 室内移动机器人改进的 TDOA 两级 UWB 定位算法。 IEEE 系统科学与工程最新进展国际会议 (IEEE RASSE 2021) 论文集, 中国上海, 2021 年 12 月 12-14 日。 <https://doi.org/10.1109/RASSE53195.2021.9686768>。
- 119.李, P.; 徐, Y. 沉, T. Bi, S. INS/UWB 集成 AGV 定位采用卡尔曼滤波器, 适用于室内 LOS/NLOS 混合环境。 2019 年先进机电系统国际会议 (ICAMechS) 论文集, 日本滋贺县草津, 2019 年 8 月 26 日至 28 日; 第 294–298 页。 <https://doi.org/10.1109/ICAMechS.2019.8861620>。
- 120.徐T.; 张, H. 周X; 袁X.; 谭, X. 张, J. 钟浩. 一种基于UWB和里程计的权重自适应卡尔曼滤波器定位方法。 2022 年先进机器人与机电一体化国际会议 (ICARM) 论文集, 中国桂林, 2022 年 7 月 9-11 日; 第 956–961 页。 <https://doi.org/10.1109/ICARM54641.2022.9959604>。
- 121.周H.; 姚, Z.; 张, Z. 刘, P. 陆明. 基于激光雷达/UWB融合的在线多机器人SLAM系统。 *IEEE Sens. J.* 2022, 22, 2530–2542。 <https://doi.org/10.1109/JSEN.2021.3136929>。
- 122.吴H.; 陶, B.; 龚, Z. 尹, Z.; Ding, H. 使用单个无源标签的基于 RFID 的独立移动机器人导航方法。 *IEEE Trans. Autom. Sci. Eng.* 2021, 18, 1529–1537。 <https://doi.org/10.1109/TASE.2020.3008187>。
- 123.王杰; Takahashi, Y. 基于粒子平滑器的非高斯检测模型室内移动机器人 SLAM 方法的地标映射。 *J. Sens.* 2019, 2019, 3717298。 <https://doi.org/10.1155/2019/3717298>。
- 124.沙姆斯法赫尔, F.; 马西, D.; 丰塔内利, D.; 莫特罗尼, A.; 尼帕, P.; 帕洛波利, L.; Buffi, A. 基于 RFID 的机器人定位: 利用 RSSI 的无约束优化问题。 2022 年 IEEE 国际仪器和测量技术会议 (I2MTC) 会议记录, 加拿大安大略省渥太华, 2022 年 5 月 16-19 日; 第 1-6 页。 <https://doi.org/10.1109/I2MTC48687.2022.9806604>。
- 125.韦克, O.; Christensen, H. 使用从其中提取的自然点地标对移动机器人进行定位和导航

130. Cho, B.S.; Moon, W.s.; Seo, W.J.; Baek, K.R. A dead reckoning localization system for mobile robots using inertial sensors and wheel revolution encoding. *J. Mech. Sci. Technol.* **2011**, *25*, 2907–2917. <https://doi.org/10.1007/s12206-011-0805-1>.
131. Myung, H.; Lee, H.K.; Choi, K.; Bang, S. Mobile robot localization with gyroscope and constrained Kalman filter. *Int. J. Control Autom. Syst.* **2010**, *8*, 667–676. <https://doi.org/10.1007/s12555-010-0321-6>.
132. Zhang, X.; Xian, B.; Zhao, B.; Zhang, Y. Autonomous Flight Control of a Nano Quadrotor Helicopter in a GPS-Denied Environment Using On-Board Vision. *IEEE Trans. Ind. Electron.* **2015**, *62*, 6392–6403. <https://doi.org/10.1109/TIE.2015.2420036>.
133. López, E.; García, S.; Barea, R.; Bergasa, L.; Molinos, E.; Arroyo, R.; Romera, E.; Pardo, S. A Multi-Sensorial Simultaneous Localization and Mapping (SLAM) System for Low-Cost Micro Aerial Vehicles in GPS-Denied Environments. *Sensors* **2017**, *17*, 802. <https://doi.org/10.3390/s17040802>.
134. Bachrach, A.; Prentice, S.; He, R.; Henry, P.; Huang, A.S.; Krainin, M.; Maturana, D.; Fox, D.; Roy, N. Estimation, planning, and mapping for autonomous flight using an RGB-D camera in GPS-denied environments. *Int. J. Robot. Res.* **2012**, *31*, 1320–1343. <https://doi.org/10.1177/0278364912455256>.
135. Chowdhary, G.; Johnson, E.N.; Magree, D.; Wu, A.; Shein, A. GPS-denied Indoor and Outdoor Monocular Vision Aided Navigation and Control of Unmanned Aircraft: J. Field Robotics. *J. Field Robot.* **2013**, *30*, 415–438. <https://doi.org/10.1002/rob.21454>.
136. Yang, Z.; Shen, S. Monocular Visual-Inertial State Estimation With Online Initialization and Camera-IMU Extrinsic Calibration. *IEEE Trans. Autom. Sci. Eng.* **2017**, *14*, 39–51. <https://doi.org/10.1109/TASE.2016.2550621>.
137. Huh, S.; Shim, D.H.; Jonghyuk Kim. Integrated navigation system using camera and gimbaled laser scanner for indoor and outdoor autonomous flight of UAVs. In Proceedings of the 2013 IEEE/RSJ International Conference on Intelligent Robots and Systems, Tokyo, Japan, 3–7 November 2013; pp. 3158–3163. <https://doi.org/10.1109/IROS.2013.6696805>.
138. Magree, D.; Johnson, E.N. Combined laser and vision-aided inertial navigation for an indoor unmanned aerial vehicle. In Proceedings of the 2014 American Control Conference, Portland, OR, USA, 4–6 June 2014; pp. 1900–1905. <https://doi.org/10.1109/ACC.2014.6858995>.
139. Zong, C.; Ji, Z.; Yu, Y.; Shi, H. Research on Obstacle Avoidance Method for Mobile Robot Based on Multisensor Information Fusion. *Sens. Mater.* **2020**, *32*, 1159. <https://doi.org/10.18494/SAM.2020.2540>.
140. Mu, L.; Yao, P.; Zheng, Y.; Chen, K.; Wang, F.; Qi, N. Research on SLAM Algorithm of Mobile Robot Based on the Fusion of 2D LiDAR and Depth Camera. *IEEE Access* **2020**, *8*, 157628–157642. <https://doi.org/10.1109/ACCESS.2020.3019659>.
141. Jiang, P.; Chen, L.; Guo, H.; Yu, M.; Xiong, J. Novel indoor positioning algorithm based on Lidar/inertial measurement unit integrated system. *Int. J. Adv. Robot. Syst.* **2021**, *18*, 172988142199992. <https://doi.org/10.1177/1729881421999923>.
142. Lin, J.; Zheng, C.; Xu, W.; Zhang, F. R2LIVE: A Robust, Real-time, LiDAR-Inertial-Visual tightly-coupled state Estimator and mapping. *arXiv* **2021**, arXiv:2102.12400.
143. Xu, W.; Cai, Y.; He, D.; Lin, J.; Zhang, F. FAST-LIO2: Fast Direct LiDAR-Inertial Odometry. *IEEE Trans. Robot.* **2022**, *38*, 2053–2073. <https://doi.org/10.1109/TRO.2022.3141876>.
144. Alhamdi, E.; Hedjar, R. Comparative Study of Two Localization Approaches for Mobile Robots in an Indoor Environment. *J. Robot.* **2022**, *2022*, 1999082. <https://doi.org/10.1155/2022/1999082>.
145. Xu, S.; Chou, W.; Dong, H. A Robust Indoor Localization System Integrating Visual Localization Aided by CNN-Based Image Retrieval with Monte Carlo Localization. *Sensors* **2019**, *19*, 249. <https://doi.org/10.3390/s19020249>.
146. Ge, G.; Zhang, Y.; Wang, W.; Jiang, Q.; Hu, L.; Wang, Y. Text-MCL: Autonomous Mobile Robot Localization in Similar Environment Using Text-Level Semantic Information. *Machines* **2022**, *10*, 169. <https://doi.org/10.3390/machines10030169>.
147. Huang, Y.H.; Lin, C.T. Indoor Localization Method for a Mobile Robot Using LiDAR and a Dual AprilTag. *Electronics* **2023**, *12*, 1023. <https://doi.org/10.3390/electronics12041023>.
148. Kendall, A.; Grimes, M.; Cipolla, R. PoseNet: A Convolutional Network for Real-Time 6-DOF Camera Relocalization. In Proceedings of the 2015 IEEE International Conference on Computer Vision (ICCV), Santiago, Chile, 7–13 December 2015; pp. 2938–2946. <https://doi.org/10.1109/ICCV.2015.336>.
149. Chen, C.; Wang, B.; Lu, C.X.; Trigoni, N.; Markham, A. A Survey on Deep Learning for Localization and Mapping: Towards the Age of Spatial Machine Intelligence. *arXiv* **2020**, arXiv:2006.12567.
150. Magrin, C.E.; Todt, E. Multi-Sensor Fusion Method Based on Artificial Neural Network for Mobile Robot Self-Localization. In Proceedings of the 2019 Latin American Robotics Symposium (LARS), 2019 Brazilian Symposium on Robotics (SBR) and 2019 Workshop on Robotics in Education (WRE), Rio Grande, Brazil, 23–25 October 2019; pp. 138–143. <https://doi.org/10.1109/LARS-SBR-WRE48964.2019.00032>.
151. Li, H.; Mao, Y.; You, W.; Ye, B.; Zhou, X. A neural network approach to indoor mobile robot localization. In Proceedings of the 2020 19th International Symposium on Distributed Computing and Applications for Business Engineering and Science (DCABES), Xuzhou, China, 16–19 October 2020; pp. 66–69. <https://doi.org/10.1109/DCABES50732.2020.00026>.
152. Li, C.; Wang, S.; Zhuang, Y.; Yan, F. Deep Sensor Fusion Between 2D Laser Scanner and IMU for Mobile Robot Localization. *IEEE Sens. J.* **2021**, *21*, 8501–8509. <https://doi.org/10.1109/JSEN.2019.2910826>.
153. Lee, J.; Lee, H.; Oh, J. FusionLoc: Camera-2D LiDAR Fusion Using Multi-Head Self-Attention for End-to-End Serving Robot Relocalization. *IEEE Access* **2023**, *11*, 75121–75133. <https://doi.org/10.1109/ACCESS.2023.3297202>.
154. Sarcevic, P.; Csik, D.; Odry, A. Indoor 2D Positioning Method for Mobile Robots Based on the Fusion of RSSI and Magnetometer Fingerprints. *Sensors* **2023**, *23*, 1855. <https://doi.org/10.3390/s23041855>.

130. Cho, 理学学士; 穆恩, W.S.; Seo, W.J.; 白, K.R.使用惯性传感器和车轮转数编码的移动机器人的航位推算定位系统。 *J. Mech. Sci. Technol.* 2011, 25, 2907–2917. <https://doi.org/10.1007/s12206-011-0805-1>.
131. 明, H.; 李, 香港; 崔, K. Bang, S. 使用陀螺仪和约束卡尔曼滤波器的移动机器人定位。 *Int. J. Control Autom. Syst.* 2010, 8, 667–676. <https://doi.org/10.1007/s12555-010-0321-6>.
132. 张X.; 西安, B.; 赵, B. 张, Y. 使用机载视觉在无GPS的环境中对纳米四旋翼直升机进行自主飞行控制。 *IEEE Trans. Ind. Electron.* 2015, 62, 6392–6403. <https://doi.org/10.1109/TIE.2015.2420036>.
133. 洛佩斯, E.; 加西亚, S.; 巴里亚, R.; 贝尔加萨, L.; 莫利诺斯, E.; 阿罗约, R.; 罗梅拉, E.; Pardo, S. 用于GPS拒绝环境中的低成本微型飞行器的多传感器同步定位和绘图(SLAM)系统。 *Sensors* 2017, 17, 802. <https://doi.org/10.3390/s17040802>.
134. 巴克拉赫, A.; 普伦蒂斯, S.; 她.; 亨利, P.; 黄, A.S.; 克莱宁, M.; 马图拉纳, D.; 福克斯, D.; Roy, N. 在GPS拒绝的环境中使用RGB-D相机进行自主飞行的估计、规划和绘图。 *Int. J. Robot. Res.* 2012, 31, 1320–1343. <https://doi.org/10.1177/0278364912455256>.
135. 乔杜里, G.; 约翰逊, E.N.; 马格里, D.; 吴, A. Shein, A. GPS拒绝的室内和室外单目视觉辅助无人机导航和控制: J. Field Robotics. *J. Field Robot.* 2013, 30, 415–438. <https://doi.org/10.1002/rob.21454>.
136. 杨, Z.; Shen, S. 单目视觉惯性状态估计与在线初始化和相机IMU外部校准。 *IEEE Trans. Autom. Sci. Eng.* 2017年, 14, 39–51. <https://doi.org/10.1109/TASE.2016.2550621>.
137. 哈, S.; 希姆, D.H.; 金钟赫. 使用摄像头和万向激光扫描仪的集成导航系统, 用于无人机的室内和室外自主飞行。 2013年IEEE/RSJ国际智能机器人和系统会议论文集, 日本东京, 2013年11月3–7日; 第3158–3163页。 <https://doi.org/10.1109/IROS.2013.6696805>.
138. 马格里, D.; 约翰逊, E.N. 用于室内无人机的激光和视觉辅助惯性导航组合。 2014年美国控制会议论文集, 美国俄勒冈州波特兰, 2014年6月4–6日; 第1900–1905页。 <https://doi.org/10.1109/ACC.2014.6858995>.
139. 宗, C.; 吉, Z. 于, Y. 石浩. 基于多传感器信息融合的移动机器人避障方法研究。 *Sens. Mater.* 2020, 32, 1159. <https://doi.org/10.18494/SAM.2020.2540>.
140. 穆L.; 姚, P. 郑, Y. 陈, K. 王, F. 齐娜. 基于二维激光雷达和深度相机融合的移动机器人SLAM算法研究。 *IEEE Access* 2020, 8, 157628–157642. <https://doi.org/10.1109/ACCESS.2020.3019659>.
141. 蒋, P.; 陈L.; 郭, H. 于, M. 熊健. 基于激光雷达/惯性测量单元集成系统的新型室内定位算法。 *Int. J. Adv. Robot. Syst.* 2021年, 18, 172988142199992. <https://doi.org/10.1177/1729881421999923>.
142. 林, J.; 郑, C.; 徐, W. 张, F. R2LIVE: 鲁棒、实时、LiDAR-惯性-视觉紧耦合状态估计器和映射。 *arXiv* 2021年, arXiv: 2102.12400.
143. 徐W.; 蔡, Y. 他, D. 林, J. 张, F. FAST-LIO2: 快速直接激光雷达惯性里程计。 *IEEE Trans. Robot.* 2022年, 38, 2053–2073年。 <https://doi.org/10.1109/TRO.2022.3141876>.
144. 阿尔哈姆迪, E.; Hedjar, R. 室内环境中移动机器人两种定位方法的比较研究。 *J. Robot.* 2022年, 2022, 1999082. <https://doi.org/10.1155/2022/1999082>.
145. 徐S.; 周, W.; Dong, H. 一种鲁棒的室内定位系统, 集成了基于CNN的图像检索和蒙特卡罗定位辅助的视觉定位。 *Sensors* 2019, 19, 249. <https://doi.org/10.3390/s19020249>.
146. 葛, G.; 张, Y. 王, W. 姜Q. 胡, L. Wang, Y. Text-MCL: 使用文本级语义信息在类似环境中进行自主移动机器人定位。 *Machines* 2022年, 10, 169. <https://doi.org/10.3390/machines10030169>.
147. 黄YH; 林正堂使用激光雷达和双AprilTag的移动机器人的室内定位方法。 *Electronics* 2023, 12, 1023. <https://doi.org/10.3390/electronics12041023>.
148. 肯德尔, A.; 格莱姆斯, M.; Cipolla, R. PoseNet: 用于实时6-DOF相机重定位的卷积网络。 2015年IEEE国际计算机视觉会议(ICCV)会议记录, 智利圣地亚哥, 2015年12月7–13日; 第2938–2946页。 <https://doi.org/10.1109/ICCV.2015.336>.
149. 陈, C.; 王, B.; 卢, C.X.; 特里戈尼, N.; Markham, A. 定位和绘图深度学习调查: 迈向空间机器智能时代。 *arXiv* 2020年, arXiv: 2006.12567.
150. 马格林, CE; Todt, E. 基于人工神经网络的移动机器人自定位多传感器融合方法。 2019年拉丁美洲机器人研讨会(LARS)、2019年巴西机器人研讨会(SBR)和2019年教育机器人研讨会(WRE)论文集, 巴西里奥格兰德, 2019年10月23–25日; 第138–143页。 <https://doi.org/10.1109/LARS-SBR-WRE48964.2019.00032>.
- 151.

155. Barreto-Cubero, A.J.; Gómez-Espinosa, A.; Escobedo Cabello, J.A.; Cuan-Urquizo, E.; Cruz-Ramírez, S.R. Sensor Data Fusion for a Mobile Robot Using Neural Networks. *Sensors* **2021**, *22*, 305. <https://doi.org/10.3390/s22010305>.
156. Lv, W.; Kang, Y.; Qin, J. FVO: Floor vision aided odometry. *Sci. China Inf. Sci.* **2019**, *62*, 12202. <https://doi.org/10.1007/s11432-017-9306-x>.
157. Jiang, G.; Yin, L.; Jin, S.; Tian, C.; Ma, X.; Ou, Y. A Simultaneous Localization and Mapping (SLAM) Framework for 2.5D Map Building Based on Low-Cost LiDAR and Vision Fusion. *Appl. Sci.* **2019**, *9*, 2105. <https://doi.org/10.3390/app9102105>.
158. Liu, R.; He, Y.; Yuen, C.; Lau, B.P.L.; Ali, R.; Fu, W.; Cao, Z. Cost-Effective Mapping of Mobile Robot Based on the Fusion of UWB and Short-Range 2-D LiDAR. *IEEE/ASME Trans. Mechatron.* **2022**, *27*, 1321–1331. <https://doi.org/10.1109/TMECH.2021.3087957>.

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155.巴雷托-库韦罗, A.J.; 戈麦斯-埃斯皮诺萨, A.; 埃斯科贝多-卡贝洛 (Escobedo Cabello), J.A.; 库安-乌尔基佐, E.; 克鲁兹-拉米雷斯, S.R.使用神经网络的移动机器人的传感器数据融合。 *Sensors* 2021, 22, 305。 <https://doi.org/10.3390/s22010305>。 156.Lv, W.; 康, Y. 秦 J. FVO: 地面视觉辅助里程计。 *Sci. China Inf. Sci.* 2019, 62, 12202。 <https://doi.org/10.1007/s11432-017-9306-x>。 157.江, G.; 尹, L.; 金, S. 田, C.; 最大限度。; Ou, Y. 基于低成本 LiDAR 和视觉融合的 2.5D 地图构建同步定位与建图 (SLAM) 框架。 *Appl. Sci.* 2019, 9, 2105。 <https://doi.org/10.3390/app9102105>。 158.刘, R.; 嘿。; 袁, C.; 刘, B.P.L.; 阿里, R. 傅, W. 曹志。基于 UWB 和短距离 2-D LiDAR 融合的移动机器人成本效益测绘。 *IEEE/ASME Trans. Mechatron.* 2022, 27, 1321–1331。 <https://doi.org/10.1109/TMECH.2021.3087957>。

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